

Modulation of aperiodic EEG activity provides sensitive index of cognitive state changes during working memory task

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eLife Assessment

This **valuable** work explores the timely idea that aperiodic activity in human electrophysiology recordings is dynamically modulated in response to task events in a manner that may be relevant for behavioral performance. Moreover, the authors present **solid** evidence that, in some circumstances, these aperiodic changes might be misinterpreted as oscillatory changes. While many aspects of the manuscript were intriguing, there was a sense that some of the interpretations were overstated - for instance the claim that aperiodic activity distorts interpretations of theta specifically, versus having a more nuanced impact on the time-frequency representation. Softening some of the language may further improve the manuscript.

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Abstract

To better understand cognitive processes, it is valuable to consider both periodic (oscillatory) and aperiodic electrophysiological brain activity. In this study, we aim to clarify how the periodic and aperiodic electrophysiological components respectively reflect the cognitive processes involved in working memory. Fifty-seven participants performed an n-back task while their brain activity was recorded using EEG. The analysis of both components of the EEG spectrum during task performance indicates that both periodic and aperiodic activities exhibit distinct task-related spatiotemporal dynamics that are closely related to cognitive demands. The results suggest that a substantial portion of the changes often attributed to theta oscillations in working memory tasks may be influenced by shifts in the spectral slope of aperiodic activity. This finding indicates that the modulation of aperiodic activity, associated with cognitive control processes, could provide a more sensitive index of cognitive state changes than previously recognised. To further confirm our findings, we also used these analysis methods in an item-recognition task, which showed similar patterns of periodic and aperiodic activity. These observations challenge the conventional understanding of low-

frequency oscillations in cognitive processing and raise concerns about the routine practice of EEG baseline correction in time-frequency analysis, as it may obscure significant modulations in continuous working memory tasks. Consequently, the inclusion of aperiodic activity as a fundamental component of EEG analysis is likely to be critical for the accurate representation of the neural basis of cognition.

Introduction

Working memory is a crucial cognitive ability that enables goal-directed behaviour and higher cognitive function. It involves the maintenance and manipulation of information in the absence of sensory stimuli (Baddeley, 2012 [↗](#)) and depends on a complex set of cognitive processes and neural mechanisms (Wager and Smith, 2003 [↗](#)). Due to its central role in cognition, working memory has been extensively studied using electroencephalography (EEG). This method allows for time-frequency analysis of brain activity (Cohen, 2014 [↗](#)), often focusing on predefined frequency bands such as theta (4–8 Hz), alpha (6–12 Hz), and beta (13–30 Hz). One widely used paradigm for investigating working memory via time-frequency analysis is the n-back task (Owen et al., 2005 [↗](#); Gevins et al., 1997 [↗](#)), which integrates key working memory processes such as encoding, maintenance, updating, and recall (Pesonen et al., 2007 [↗](#); Chen and Huang, 2016 [↗](#)).

Beyond band-specific oscillations, EEG power spectra exhibit a broadband decrease in power with increasing frequency, following a $1/f$ scaling pattern (where f is frequency). This pattern is widely thought to reflect non-oscillatory (aperiodic) neural activity that lacks distinct spectral peaks. Because aperiodic activity spans a continuous frequency range it can obscure changes in periodic (oscillatory) activity, complicating interpretation, unless appropriately separated or accounted for (Cohen, 2014 [↗](#); Donoghue et al., 2020b [↗](#)).

Given that aperiodic activity is sometimes regarded as background noise, a common approach in task-based EEG studies is to apply baseline correction to emphasize task-related oscillatory changes. Baseline correction aims to normalise the data by removing this background activity, often using decibel or z-score normalisation. These methods typically assume stable aperiodic activity across time (Gyurkovics et al., 2021 [↗](#); Merkin et al., 2023 [↗](#); Donoghue et al., 2020b [↗](#)), however, in continuous paradigms like the n-back task, where no neutral pre-stimulus interval exists, this assumption may not hold. Moreover, recent studies show that ignoring task-related aperiodic fluctuations can distort oscillatory measurements and mask meaningful physiological and behavioural information (Gyurkovics et al., 2021 [↗](#); Donoghue et al., 2020a [↗](#); Gyurkovics et al., 2022 [↗](#); Virtue-Griffiths et al., 2022 [↗](#); Kałamała et al., 2024 [↗](#); Akbarian et al., 2023b [↗](#)).

Computational models suggest that these aperiodic fluctuations reflect the balance between excitation and inhibition (E:I) in the brain (Gao et al., 2017 [↗](#); Lim and Goldman, 2013 [↗](#)), essential for efficient cognitive processing (Ahmad et al., 2022 [↗](#)). Aperiodic activity is often approximated as linear in log-log space (Donoghue et al., 2020b [↗](#); Barry and De Blasio, 2021 [↗](#)), and while sometimes referred to as the ‘aperiodic exponent,’ we use the term ‘slope’ here, except where methodological descriptions require exponent notation. A steeper aperiodic slope implies higher low-frequency power and lower high-frequency power, potentially indicating stronger cognitive engagement (Gao et al., 2017 [↗](#); Akbarian et al., 2023a [↗](#)), whereas a flatter slope can reflect more uniform power across frequencies (Gyurkovics et al., 2022 [↗](#); Kałamała et al., 2024 [↗](#)). However, some tasks show flatter slopes even under increased engagement (Waschke et al., 2021 [↗](#); Cunningham et al., 2023 [↗](#)). Moreover, task-related changes in overall power, typically interpreted as oscillatory dynamics, may be significantly co-modulated by the aperiodic slope shift (Akbarian et al., 2023b [↗](#); Gyurkovics et al., 2022 [↗](#); Virtue-Griffiths et al., 2022 [↗](#)).

Given these findings, it is methodologically imperative to carefully parameterise both periodic and aperiodic components of an EEG before interpreting time-frequency analysis results. Common approaches to spectral or time-frequency analysis often assume that distinct peaks in the power spectrum reflect true oscillatory activity once the $1/f$ background is removed (Donoghue et al., 2020b [a](#)). However, the presence of power in a particular frequency band is not necessarily indicative of oscillatory dynamics, as aperiodic activity can constitute a significant proportion of observed power spectra (Bullock et al., 2003 [b](#); Donoghue et al., 2020a [b](#)). Moreover, time-frequency representations (TFRs) typically highlight temporal fluctuations in power (Cohen, 2014 [b](#)), but do not inherently differentiate between periodic and aperiodic contributions. If part of the TFR signal arises from aperiodic shifts, baseline correction may confound rather than clarify the nature of observed effects.

In this study, we examined how aperiodic and periodic EEG components each contribute to working memory demands in the n-back task, addressing potential biases introduced by baseline correction. Furthermore, we focused on disentangling how aperiodic slope is influenced by memory load, stimulus type, or task modality. We based our analyses on a dataset of middle-aged to older adults, half of whom reported subjective cognitive complaints; nonetheless, behavioural outcomes and electrophysiological measures did not differ between groups. To assess the generalisability of our findings, we applied the same analysis to two additional datasets which allowed us to test whether similar aperiodic-periodic dynamics emerge across different working memory contexts.

Our study builds on previous research by using high temporal resolution data to investigate the aperiodic slope across two modalities (visuospatial/verbal). In contrast to n-back studies that use discrete time windows (Akbarian et al., 2023b [b](#); Kałamała et al., 2024 [b](#)), we analyse the activity at each time point, capturing shortlived slope fluctuations that could reveal crucial insights into continuous working memory processes. While a similar approach has been used to track temporal dynamics in sleep and resting state (e.g., Wilson et al., 2022 [b](#); Ameen et al., 2024 [b](#)), we are not aware of any study using time-resolved spectral parameterisation in task-based paradigms. While some evidence suggests modality-specific aperiodic variations (Waschke et al., 2021 [b](#)), we propose that the n-back task primarily recruits domain-general processes (Owen et al., 2005 [b](#); Gazzaley and Nobre, 2012 [b](#); Olesen et al., 2004 [b](#)), leading to similar slope modulations across both verbal and visuospatial modalities (Gyurkovics et al., 2022 [b](#)). We therefore predict that the aperiodic slope would be modulated by the processing demands of the n-back task, varying with load $\times$ and stimulus type but not between visuospatial and verbal conditions.

By explicitly separating aperiodic and periodic components and tracking their dynamics over time, our study offers a novel perspective on the relationship between baseline correction and aperiodic activity in continuous tasks.

Results

We analysed the EEG data from the visuospatial and verbal n-back tasks. We transformed the data into the time-frequency domain by convolving it with superlets (Moca et al., 2021 [b](#)).

Subsequently, we applied the FOOOF (Fitting Oscillations and One-Over-F) algorithm (Donoghue et al., 2020b [b](#)) to decompose the activity into periodic and aperiodic components (**Figure 2** [b](#)). We estimated two aperiodic parameters using FOOOF: the offset and the exponent. While the FOOOF framework refers to this parameter as the exponent, we use the term ‘aperiodic slope’ throughout the manuscript for consistency with broader literature on spectral power scaling. Note that the exponent is numerically positive, whereas the slope is its negative counterpart (i. e., a steeper (more negative) slope in log-log space corresponds to a higher (positive) exponent value).

Figure 1.

Schematic representation of the n-back task.

The n-back working memory task was conducted in two distinct modalities: visuospatial and verbal. In the 2-back condition, participants had to identify whether the current stimulus matched the one presented two steps previously. In the visuospatial modality, the target was a spatial location, whereas in the verbal modality, the target was a letter. In the 0-back condition participants' task was to respond to a predefined target, with the type of target corresponding to the task modality. Target stimuli are highlighted in orange.

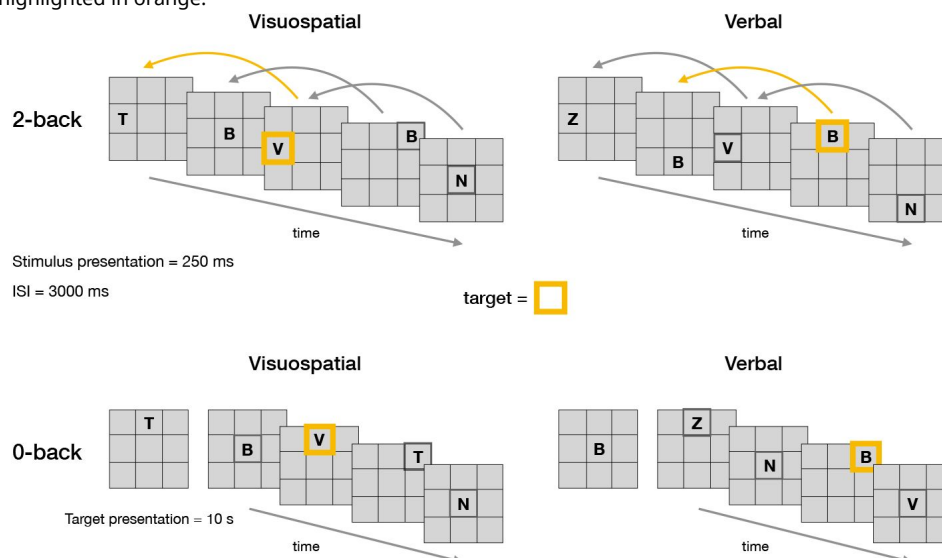
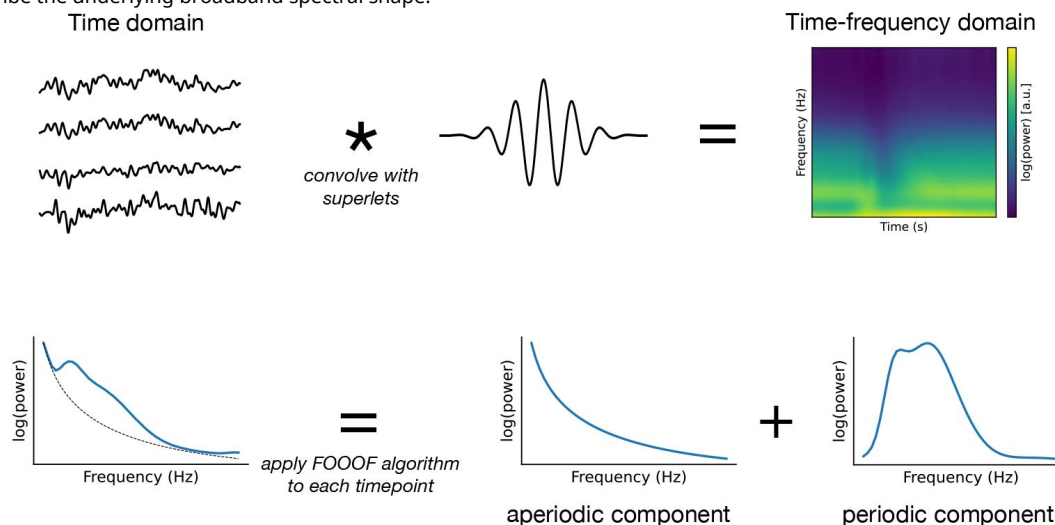


Figure 2.

Schematic of the analysis.

The time domain data were first transformed into the time-frequency domain by convolution with superlets (Moca et al., 2021). Next, the periodic and aperiodic components of the power spectrum density were estimated for each time point using FOOOF (Fitting Oscillations and One-Over-F) algorithm (Donoghue et al., 2020b). The aperiodic component was characterised by the aperiodic slope (the negative counterpart of the exponent parameter) and the offset, which together describe the underlying broadband spectral shape.



Changes in aperiodic activity appear as low-frequency oscillations in baseline-corrected time-frequency plots

As the first step of the analysis, we compared periodic and aperiodic activity with baseline-corrected time-frequency activity. In the baseline-corrected time-frequency plot, we observed a decrease in alpha and beta power up to 1 second post-stimulus, while low-frequency power increased from 0 seconds post-stimulus ([Figure 3](#), [Figure S18](#)). The observed pattern was minimally affected by the choice of baseline ([Figure S2](#)). Specifically, whether we employed subtractive or divisive baseline normalisation ([Gyurkovics et al., 2021](#)), the qualitative pattern of alpha, beta, and low-frequency (delta and theta) changes remained consistent.

Note that although no prominent oscillatory peak in the theta range was observed at the group level, and some of this activity could potentially fall within the delta range, similar low-frequency patterns have often been referred to as ‘theta’ in previous work, even in the absence of a clear spectral peak (e.g., [Palomäki et al., 2012](#); [Pesonen et al., 2007](#); [Rossi et al., 2023](#)). To acknowledge both perspectives, we use the more neutral term ‘low-frequency activity’ throughout the paper, while noting its potential overlap with activity commonly described as frontal theta.

Notably, when we examined periodic activity, we only observed changes in alpha and beta frequency bands, while increases in the low-frequency range were absent ([Figure 6](#)). This suggests that low-frequency changes may, at least in part, reflect aperiodic activity rather than oscillatory processes. To quantitatively assess this observation, we directly compared the FOOOF and baseline-corrected time-frequency decompositions by performing correlations for each channel-frequency-time point ([Figure 4](#), [Figure S3](#)). We observed strong positive correlations between periodic power and baseline-corrected power in the alpha and beta ranges across all channels, but only weak correlations in the low-frequency range. In contrast, both the aperiodic exponent and offset parameters showed strong positive correlations with baseline-corrected power in the theta range. In addition, the exponent showed strong negative correlations with activity above 35 Hz, suggesting that higher frequency power decreases as the aperiodic slope steepens.

Rhythmicity analysis reveals aperiodic parameters are largely independent of oscillatory activity

A possible concern is that the FOOOF decomposition might not be sufficiently sensitive to peaks at the edges of the frequency spectrum, potentially affecting the interpretation of the results. To address this issue, we conducted simulations demonstrating that when periodic components include peaks at low frequencies (around 3 Hz), they may go undetected, thereby inflating estimates of aperiodic activity (see the Supplement for details).

To overcome this limitation, we employed an alternative metric to assess oscillatory activity: the recently developed measure of rhythmicity, the **phase autocorrelation function (pACF)** ([Myrov et al., 2024](#)). The pACF quantifies the temporal stability of phase dynamics and captures oscillatory activity independent of amplitude fluctuations. Consistent results between FOOOF and pACF would strengthen the conclusion that low-frequency oscillations were truly absent from the recordings, rather than simply undetectable by FOOOF. As expected, the observed rhythmicity patterns were largely consistent with periodic power, particularly in the alpha frequency range ([Figure 5](#), [Figure S4](#), [Figure S5](#)). In contrast, correlations between aperiodic parameters and pACF were weak in comparison, supporting the notion that the aperiodic parameter estimates remain largely independent of oscillatory processes.

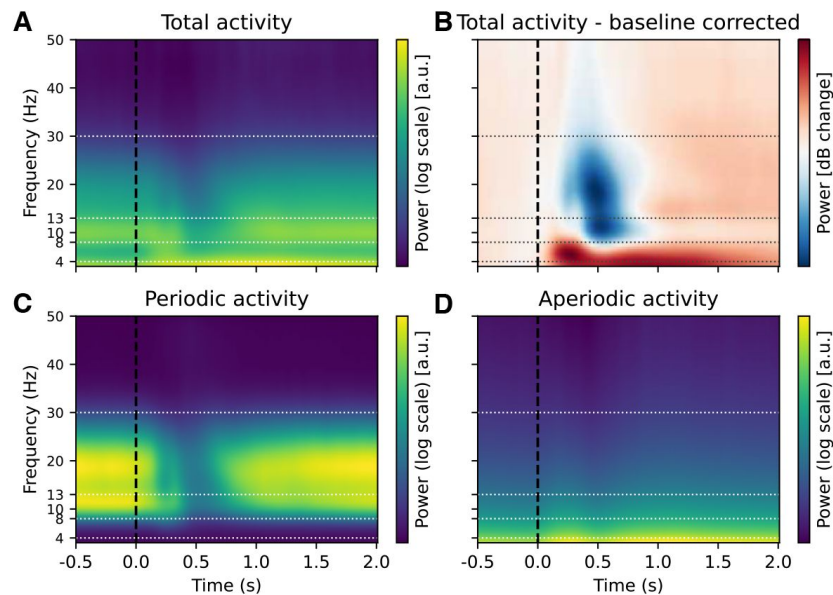


Figure 3.

Effect of baseline correction and FOOOF decomposition on time-frequency decomposition in the n-back task.

Illustration of the effect of baseline correction and FOOOF decomposition on EEG time-frequency analysis. **(A)** “Raw” total power from time-frequency decomposition is difficult to interpret due to the $1/f$ power scaling of EEG power spectra (here representing average data across subjects, channels and conditions). In panel **B** we applied baseline correction (decibel conversion) using a pre-stimulus interval of -0.5 to -0.2 seconds for comparison. The baseline correction showed a significant decrease in alpha and beta power from 0 to 1 second and a concomitant increase in low-frequency power lasting up to 2 seconds. The observed changes were very similar across different choices of baseline correction (**Figure S2**). However, it’s unclear from the baseline-corrected data whether the observed changes in the low-frequency range reflect oscillatory or aperiodic contributions. To disentangle these components, we decomposed the time-frequency signal into periodic and aperiodic contributions using spectral parameterisation. **(C)** The periodic component includes only the parameterised oscillatory peaks, reconstructed from Gaussian fits to the power spectrum (see Methods for details). **(D)** The aperiodic component reflects the $1/f$ -like background activity. This decomposition suggests that changes in power in the low-frequency (delta and theta) range may largely reflect changes in aperiodic activity. See **Figure S1** for the corresponding figure with a logarithmic y-axis.

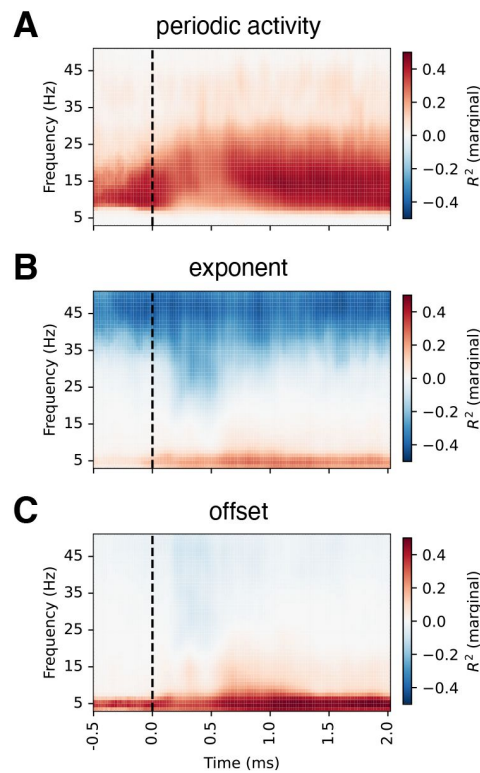


Figure 4.

Correlations between baseline-corrected time-frequency and FOOOF-decomposed EEG activity

For each FOOOF parameter, we estimated its similarity to baseline-corrected time-frequency EEG activity at each channel-frequency-timepoint using a linear mixed model. We observed strong correlations between periodic activity and baseline-corrected time-frequency activity in the alpha and beta ranges. In contrast, the exponent and offset showed strong correlations in the low-frequency range, with the exponent also showing negative correlations in the gamma range. These results suggest that changes in aperiodic activity appear as low-frequency oscillations in baseline-corrected time-frequency plots. Note that while R^2 is strictly non-negative, we assigned its sign based on the beta co-efficient from the fixed effect in the model to facilitate interpretation. The values presented are averaged across all channels; see [Figure S3](#) for topographical distributions.

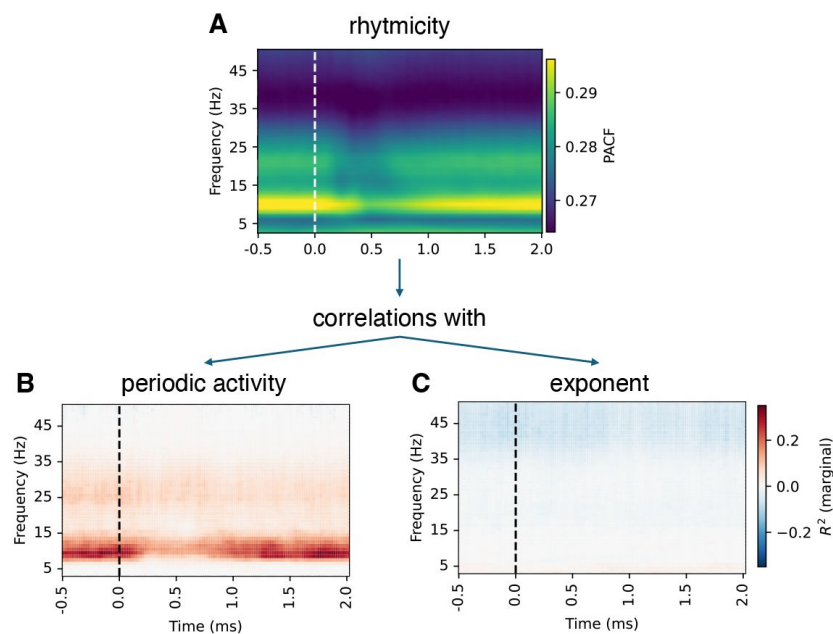


Figure 5.

Phase autocorrelation function and its correlations with FOOOF parameters.

Phase autocorrelation function (pACF) (Myrov et al., 2024) is an alternative measure of oscillations that is sensitive to rhythmicity rather than the amplitude. Patterns of pACF (A) were largely consistent with periodic activity (B), particularly in the alpha range, supporting the notion that pACF and periodic activity reflect the same underlying processes. In contrast, the correlations with exponent and offset were low (only results for exponent are shown as results for exponent and offset were essentially the same). See Figure S4 and Figure S5 for detailed figures.

Periodic activity changes with working memory load, stimulus type and modality

Next, we examined the differences in periodic activity as a function of task conditions. We observed a decrease in the alpha and beta bands of periodic activity in the 2-back compared to the 0-back condition over the entire trial period and over the entire scalp (**Figure 6**, **Figure 7**). Furthermore, we found a greater decrease in alpha and beta power in response to targets compared to non-targets. In addition, we observed a correlation between beta power and reaction times in central channels. This correlation was positive in the early period (up to 0.5 seconds post-stimulus) and negative in the late period (1–2 seconds post-stimulus). Finally, we observed an interaction between modality and stimulus type in the time period from 0.5 to 1 second post-stimulus, reflecting a greater difference between target and non-target stimuli in the verbal task in the alpha and beta bands compared to the visuospatial task (**Figure S7**).

Aperiodic activity has frontal and parietal components

We observed two distinct components (peaks) in the time course of the exponent (**Figure 8**). The first component peaked at around 0.3 s after stimulus onset and was most pronounced in frontal channels, whereas the second component peaked at around 0.7 s post-stimulus and was most pronounced in parietal channels. The exponent exhibited significant differences between stimulus types (**Figure 9**). Specifically, the exponent was larger for targets at approximately 0.3 seconds and for non-targets at approximately 0.7 seconds post-stimulus. We observed no other differences between conditions. We found associations with reaction times at approximately 1.4 seconds post-stimulus in the parieto-occipital channels.

Similar to the exponent, offset also showed a pattern of activity with two components (see **Figure S8**). In addition, offset was sensitive to differences in load (lower offset for 2-back), and there was also an interaction between load and stimulus type (see **Figure S11**).

Aperiodic activity does not reflect event-related potentials

To ensure that the aperiodic activity identified in our study did not simply reflect event-related potentials (ERPs), we removed evoked activity from the data by subtracting ERPs before performing the time-frequency analysis. We then used linear mixed model to assess the association between the ERPs and aperiodic activity for each channel-time point. We observed R^2 values ranging from 0.1 to 0.2 around stimulus onset, but otherwise correlations were negligible (**Figure S14**). This was observed throughout the entire time period examined. In addition, a visual comparison of the topographies of stimulus- and response-locked ERPs with those of aperiodic activity (**Figure S13**) revealed no clear similarities, except for a central ERP component observed around 0.25–1 second after the response, which resembled the fronto-central aperiodic component. However, while this component in aperiodic activity showed large differences between targets and non-targets, in ERPs the component differed more between 0- and 2-back conditions.

We also repeated the analysis without subtracting the ERPs (**Figure S15**, **Figure S16**, **Figure S17**), and the results were largely similar. Nevertheless, when ERP was not subtracted we observed a more pronounced increase in both aperiodic and periodic power between 0 and 0.5 seconds following stimulus presentation compared to the case with subtracted ERPs.

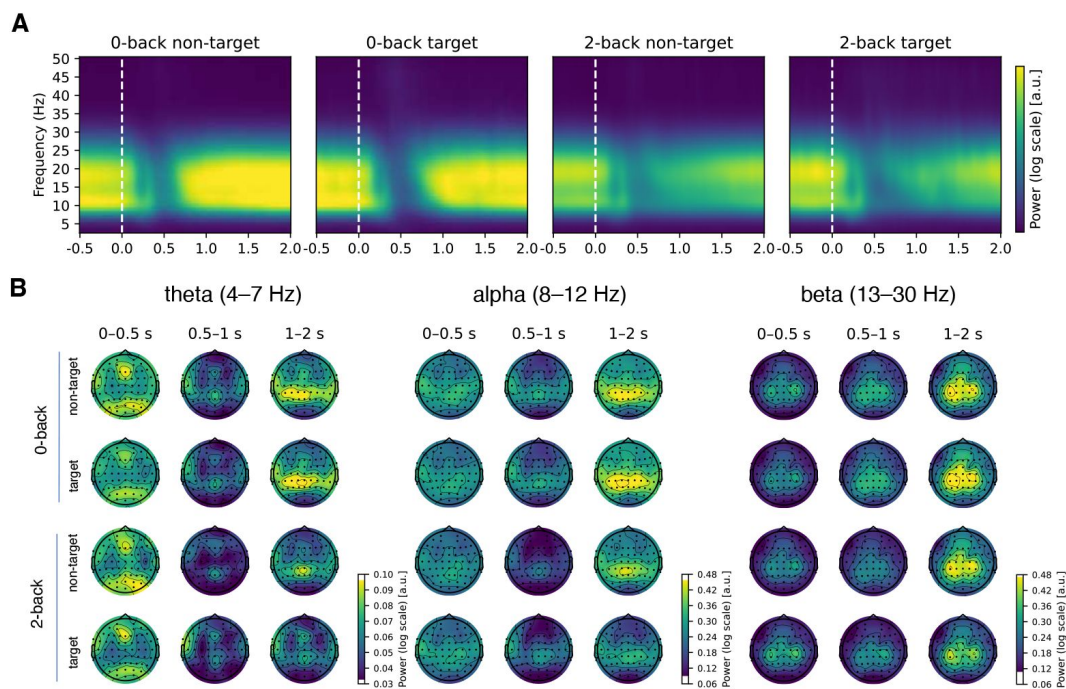


Figure 6.

Changes in periodic (oscillatory) activity as a function of time.

(A) The inspection of periodic activity revealed strong activity in the alpha and beta frequency bands, with a sharp decrease at around 0.5 seconds post-stimulus. The power was stronger in the 0-back condition, compared to the 2-back condition (see also [Figure 7](#)). (B) Early beta activity was most prominent at occipital and frontal channels. Note that colour scale ranges in panel B differ between frequency ranges.

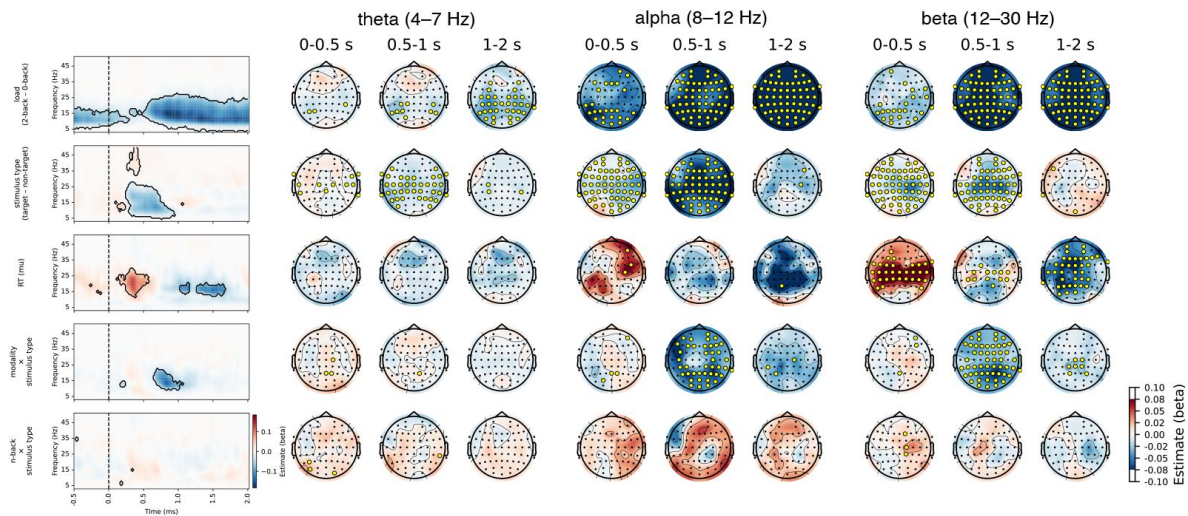


Figure 7.

Results of the linear mixed model analysis of periodic activity for comparison between conditions.

Significant values are highlighted on the heatmap and marked with yellow circles on the topographies. Only factors with significant differences are shown. A p-value was interpreted as significant if it was significant in at least 3 channels or 3 time points (see Methods for details). Note that there are small discrepancies between significant values on heatmaps and topographies due to averaging across time or channels. A significant difference was observed between the 2-back and 0-back conditions, with a reduction in activity in the 2-back condition, particularly in the alpha and beta frequency bands, starting at 0.5 seconds post-stimulus. The differences between stimulus types were most evident from 0.3 to 1 second post-stimulus, with decreased activity for targets compared to non-targets across the whole scalp. Additionally, a smaller effect of the modality $\times$ stimulus type interaction was observed from 0.6 to 1 second post-stimulus (see also [Figure S7](#) for detailed visualisation of the interaction). Associations with reaction times were significant in the beta band across central channels. These associations exhibited a positive correlation in the early phase (0 to 0.5 seconds) and a negative correlation in the later phase (1 to 1.5 seconds).

Figure 8.

Changes in aperiodic activity (exponent, interpreted as aperiodic slope) as a function of time.

(A) We averaged the time course of the aperiodic activity (exponent) over all channels and observed two components (peaks) of the aperiodic activity. The first component peaked around 0.3 s post-stimulus in the frontal channels. The second component peaked at 0.7 s, was stronger in parietal channels and differed between non-target and target conditions (see [Figure 9](#)). The time course of the offset parameter was comparable, although the separation between the frontal and parietal components was more pronounced (see [Figure S8](#)). The vertical lines in panel A represent mean reaction time modes. The shaded areas represent 95% Cousineau-Morey within-subjects confidence intervals, adjusted so that non-overlapping intervals correspond to statistically significant differences ([Cousineau, 2017](#)).

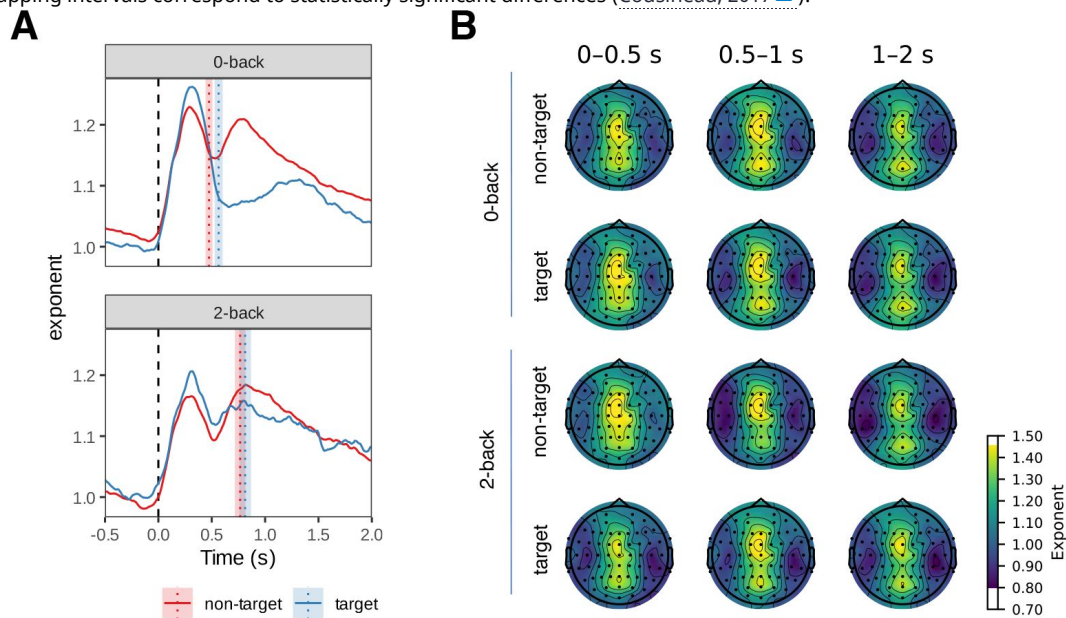
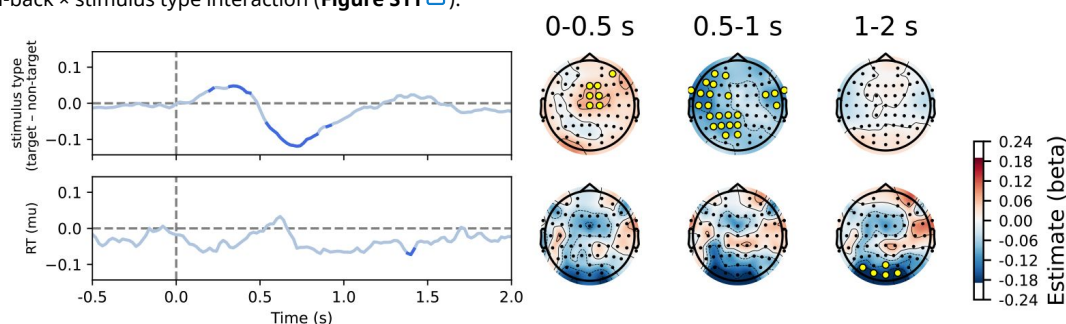


Figure 9.

Results of the linear mixed model analysis of aperiodic exponent (interpreted as slope) for comparison between conditions.

Significant values are shown in blue on line plots and marked with yellow circles on topographies. Only factors with significant differences are shown. The only significant differences between conditions were observed between target and non-target conditions, where the exponent was higher for targets in the early phase (0 to 0.5 s post-stimulus) and for non-targets in the middle phase (0.5 to 1 s post-stimulus). There was also a small association with reaction times in occipital channels around 1.5 seconds post-stimulus. Results were similar for the offset parameter, with an additional effect of load and n-back $\times$ stimulus type interaction ([Figure S11](#)).



Replication confirms main findings

To ensure the generalisability of the findings, we repeated all analyses using two additional datasets. First we analysed a publicly available dataset from a previously published study by Nakuci et al. (2023) [\[1\]](#), in which participants performed a verbal n-back task with three load levels (0-, 1-, 2-back). Next, we analysed the data from an item-recognition task.

In both analyses, we were able to replicate the main findings previously reported, namely the task-related modulation of periodic alpha and beta activity and the task-related changes in aperiodic activity. In the baseline-corrected data, we observed a decrease in alpha and beta power accompanied by an increase in low-frequency power (**Figure S18B** [\[1\]](#), **Figure 10B** [\[1\]](#)). The latter was not evident in the grand average periodic activity (**Figure S18C** [\[1\]](#), **Figure 10C** [\[1\]](#), **Figure S40** [\[1\]](#)) and was only detected in two participants in the n-back task (**Figure S23** [\[1\]](#)). In the item-recognition task, peak in the theta range was present in approximately one-third of the participants (**Figure S39** [\[1\]](#)). In half of these cases, the peaks were at 7–8 Hz, which could be considered low alpha peaks rather than theta oscillations.

Notably, the correlation patterns between baseline-corrected time-frequency data and FOOOF parameters, as well as between FOOOF parameters and pACF, were consistent across all three datasets. Specifically, aperiodic parameters were associated with low-frequency (theta) baseline-corrected activity, whereas periodic activity correlated with alpha and beta baseline-corrected activity and pACF (**Figure S21** [\[1\]](#), **Figure S20** [\[1\]](#), **Figure S22** [\[1\]](#), **Figure S34** [\[1\]](#), **Figure S33** [\[1\]](#), **Figure S35** [\[1\]](#)).

Regarding aperiodic activity, we observed frontal and parietal/occipital components in both tasks. In the itemrecognition task, these components were clearly observable only for the offset parameter, whereas for the exponent, the two components were not clearly distinguishable, likely due to lower spatial resolution (32 channels). In both tasks, aperiodic activity was modulated by task demands, with an increase in the exponent following stimuli presentation and response (**Figure S27** [\[1\]](#), **Figure S37** [\[1\]](#)).

Discussion

This study provides a detailed analysis of the relationship between periodic and aperiodic neural activity during the n-back task. EEG data recorded from 57 participants performing both 0-back and 2-back tasks in visuospatial and verbal modalities were analysed to distinguish these two components within the EEG frequency spectrum. The analysis shows the presence of periodic activity in the alpha/beta frequency range prior to stimulus onset, which modulates according to task demands. Furthermore, the results suggest that low-frequency power may largely reflect the effect of baseline correction on aperiodic activity, which also exhibits its own task-related dynamics. This insight is crucial for the interpretation of EEG results, especially for continuous tasks such as the n-back, which require sustained working memory engagement.

In addition to the interplay between the periodic and aperiodic EEG activity, the study expands on the findings of stimulus-induced aperiodic slope shift by identifying two neural components within the aperiodic activity. These aperiodic components are distinguished by their frontal and posterior topographic locations. Additionally, we observed task-dependent differences, including a slightly different slope modulation in non-target compared to target trials. However, we did not find any differences between modalities. Collectively, these findings enhance our understanding of the neural dynamics involved in cognitive processes during a continuous working memory task.

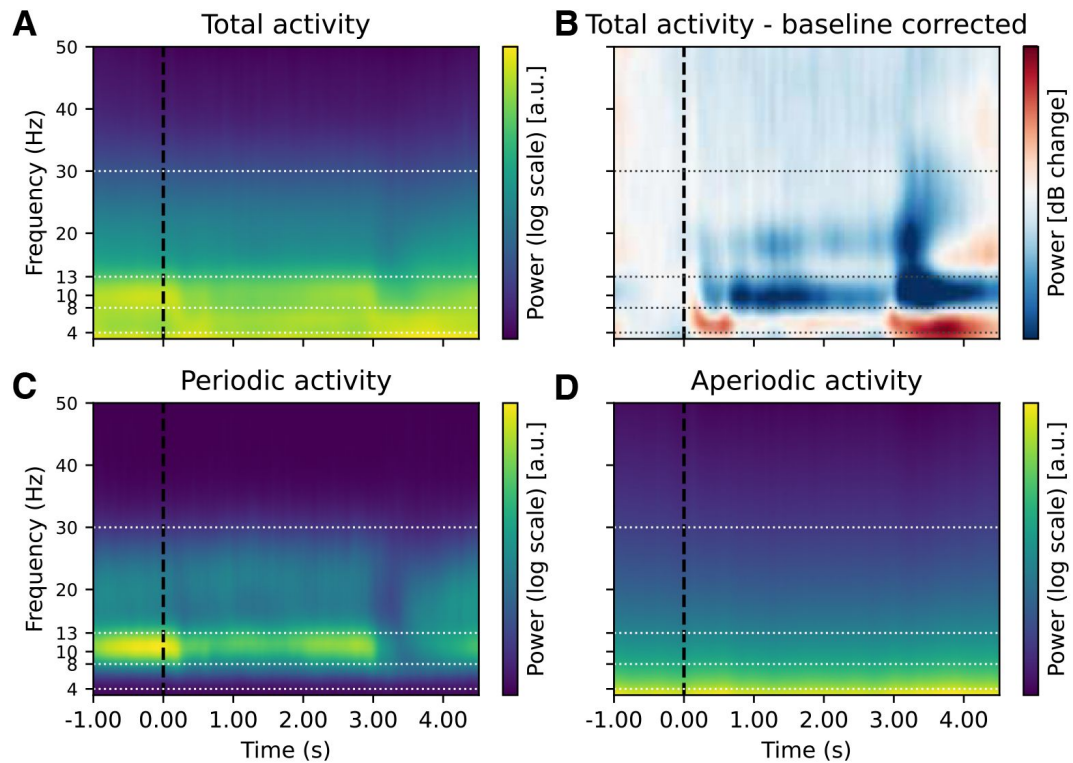


Figure 10.

The effect of baseline correction and FOOOF decomposition on time-frequency decomposition in the item-recognition task.

Similar to the n-back task (Figure 3, Figure S18), a decrease in alpha power is observed following instruction and stimuli presentation (up to 0.8 seconds) and continues throughout the retention period, up to 2.8 seconds post-stimulus (B). This is followed by a decrease in alpha and beta power during probe presentation, which is likely indicative of a motor response. Simultaneously, there is an increase in low-frequency power, which is most pronounced during stimulus presentation (up to 0.8 seconds) and again after probe presentation (after 2.8 seconds). The FOOOF decomposition indicates that a substantial portion of low-frequency activity could be attributed to the aperiodic component (C, D) (see also Figure S33). The data shown represent the group average over all conditions at electrode Fz, where low-frequency activity was most pronounced (see also Figure S40). Horizontal lines indicate the boundaries of the frequency ranges. See Figure S32 for the corresponding figure with a logarithmic y-axis.

Disentangling periodic and aperiodic components in EEG activity: avoiding baseline correction

The analysis of the EEG data shows that when baseline correction is applied to the total EEG activity, changes in alpha, beta and low-frequency activity are observed following stimulus presentation compared to the prestimulus interval (**Figure 3B**). In particular, there is a significant increase in power in the low-frequency range after the stimulus, while a decrease in power in the alpha/beta range is observed for some time after the stimulus presentation. These findings are consistent with previous n-back studies reporting that the presentation of both target and non-target stimuli elicits long-lasting increases in low-frequency power and decreases in alpha/beta power at stimulus onset (Palomäki et al., 2012; Krause et al., 2010; Coleman et al., 2023).

Omitting baseline correction and decomposing the total EEG activity into periodic and aperiodic components provides a more nuanced insight into the dynamics of the spectra, showing that alpha and beta activities remain oscillatory while low-frequency (delta and theta) activity “disappears” from the periodic component. These findings suggest that the apparent power in the low-frequency range may, at least in part, reflect changes in the aperiodic component, which could contribute to the post-stimulus increase in low-frequency power observed under baseline correction in time-frequency analysis of the full EEG spectrum. Consistent with this interpretation, we found that aperiodic parameters correlated with baseline-corrected low-frequency activity. Given the current understanding of aperiodic activity, specifically based on the findings of poststimulus slope steepening in the context of increased cognitive demands (Akbarian et al., 2023b; Gyurkovics et al., 2022; Virtue-Griffiths et al., 2022), these results suggest that changes in power within the low-frequency range may be influenced by aperiodic slope modulation rather than a distinct increase in oscillatory low-frequency activity. At the same time, alpha and beta activity remain present as part of the periodic component throughout the task.

In line with this view, the analyses show that low-frequency activities do not necessarily exhibit an oscillatory characteristic during the n-back task (see **Figure 3C, D**, **Figure 6**). Following a detailed visual inspection at the individual level (**Figure S6**), we found that only two participants exhibited evident theta peak, while only one participant displayed pronounced theta and alpha peaks. The majority of participants exhibited alpha peaks, while only two participants displayed a theta peak with no substantial alpha activity. This pattern points to a low-frequency activity that may support the cognitive demands of the n-back task, yet it appears highly variable across individuals, and potentially within individuals, depending on task requirements. A recent study of visual working memory reached a similar conclusion, reporting a robust alpha peak in 90% of participants but inconsistent or absent theta peaks in 60% of participants (Van Engen et al., 2024).

A potential concern, also raised by reviewers of the original version of this paper, is that FOOOF decomposition may miss low-frequency oscillations near the edges of the spectrum. Indeed, simulations we performed showed that in scenarios with low-frequency periodic components, FOOOF may fail to identify these components, leading to inflated estimates of aperiodic parameters (see the Supplement for details). To address this issue, we performed control analyses using the recently introduced measure of rhythmicity, the **phase autocorrelation function (pACF)** (Myrov et al., 2024), which quantifies oscillatory activity by assessing the temporal stability of phase independent of amplitude. This metric showed no evidence of strong low-frequency periodic activity in our dataset, providing further evidence that the observed aperiodic slope reflects genuine non-oscillatory dynamics, rather than an overlooked theta peak. Furthermore, weak correlations between aperiodic parameters and pACF reinforce the independence of these aperiodic features from classical oscillations.

The obtained results show that classical baseline correction can remove continuous oscillatory activity that is present both during baseline and after stimulus onset, because it treats all baseline signals as ‘background’ to be removed without distinguishing between transient and continuous oscillations. While this is consistent with the intended purpose of baseline correction—to highlight changes relative to ongoing activity—it may also lead to unintended consequences, such as misinterpreting aperiodic activity as an increase in post-stimulus theta oscillations. Importantly, we confirmed that these observations hold regardless of whether baseline normalisation was performed using a subtractive or divisive approach, suggesting that the conclusions are not driven by any single normalisation strategy.

The modulation of periodic activity is concentrated in the alpha and beta bands

When decomposing the EEG signal into periodic and aperiodic components we still observed a significant attenuation in the power of alpha and beta oscillations after stimulus presentation (see **Figure 3B, C** and **Figure 6A**). This is in line with existing literature that documents the decrease of post-stimulus alpha and beta (Krause et al., 2010; Coleman et al., 2023). Moreover, the results support previously reported increased suppressive response that is elicited by target stimuli compared to non-target stimuli (Palomäki et al., 2012; Peng et al., 2015), especially in the alpha range (**Figure S7**).

Our findings support existing research (Pesonen et al., 2007) by demonstrating that reductions in alpha and beta powers are more pronounced and have longer duration under increased task demands, as is typical of higher loads in n-back tasks (see **Figure 7**). Consistent with seminal work (Gevins et al., 1997; Pesonen et al., 2007; Krause et al., 2010), the results of this study show that working memory load mainly affects alpha/beta power in the parietal-occipital electrodes (see **Figure 6B**). In n-back tasks, diminished alpha power is thought to indicate increased engagement in working memory operations (Haegens et al., 2014). The functional significance of beta oscillations remains a topic of debate, with studies often linking centrally located beta oscillations to the coordination of motor responses (e.g. Pavlov and Kotchoubey, 2022; Pfurtscheller et al., 1998; Kaiser et al., 2001). However, recent research indicates that they also play a crucial role in working memory, particularly in supporting the maintenance of memorised information and preventing distractions (Schmidt et al., 2019; HajiHosseini et al., 2020). Additionally, beta oscillations are believed to facilitate the manipulation of stored information, dynamically adjusting to meet task demands and to serve a gating function by inhibiting irrelevant inputs, thus enhancing cognitive efficiency (ElShafei et al., 2022; Schmidt et al., 2019).

The strong correlation between alpha, lower beta activity and reaction times in central regions, as shown in **Figure 7**, suggests a shared relationship between these measures and shows that beta activity, typically associated with motor function, is most pronounced in central electrodes, although no direct temporal relationship with reaction times was observed. We also observed alpha/beta activities in the parietal electrodes, which support current theories that suggest a collaborative function for alpha and beta oscillations in supporting advanced cognitive functions such as working memory (Scharinger et al., 2017; Coleman et al., 2023; Hanslmayr et al., 2016).

Aperiodic activity dynamics and their cognitive implications

The analysis of aperiodic activity at each time point within the peristimulus interval builds upon prior research that has documented stimulus-induced changes in the aperiodic component, specifically the poststimulus steepening of the aperiodic slope (Gyurkovics et al., 2022; Virtue-Griffiths et al., 2022; Kałamała et al., 2024). The results of our study indicate the presence of two components in the dynamics of the aperiodic slope during the n-back task, differentiated both temporally and spatially.

In terms of temporality, the first component appears early in the task cycle, whereas the second component appears at a later stage (see **Figure 8A** [↗](#)). The presence of these components differs between target and non-target stimuli, with a more pronounced difference for the second component (see **Figure 9** [↗](#)). While non-target stimuli consistently exhibit a clear second component, in the case of target stimuli, the slope returns to its baseline steepness more quickly, with a significantly reduced or non-existent second slope steepening. This raises the question of whether the two components represent distinct cognitive operations inherent to the n-back task.

We speculate that the second component, which emerges more prominently following non-target stimuli, may be indicative of an elevated state of vigilance. Since it is rare for one target stimulus to be immediately succeeded by another, participants may become accustomed to a period of “rest” following a target stimulus, given the low anticipation for another. The increased aperiodic slope following non-target stimuli may reflect the level of alertness for potential targets, whereas after actual target stimuli, vigilance may diminish more rapidly, accounting for the absence of a second slope increase.

The two aperiodic components also exhibit distinct spatial manifestations. The first component is predominantly observed on the frontal electrodes, whereas the second component is also present in the parietal channels. This topographical distinction further supports the hypothesis that these two components reflect different cognitive processes. The activation in the parietal channels may be related to anticipatory mechanisms for the next stimulus, whereas the activation in the frontal channels may be involved in the decision-making process regarding the classification of the previous stimulus as a target.

Overall, the results strongly suggest that solely examining total EEG power or oscillatory dynamics, without distinctly analysing the aperiodic slope, would overlook a subtle yet potentially significant cognitive processing feature. This feature may be related to distinct working memory processes, which warrants further research.

Importantly, to exclude the possibility that aperiodic activity simply reflects the event-related potentials (ERPs; see **Figure S12** [↗](#)), ERPs were subtracted before performing the time-frequency decomposition. We observed low correlations between the ERPs and aperiodic activity (**Figure S14** [↗](#)) at the stimulus onset, but no correlations after 0.5 seconds, suggesting the task demands within the n-back paradigm manifest in aperiodic modulations that are distinct from other stimulus-locked ERPs. While it is possible that response-related ERPs contributed to the second aperiodic component, the differing patterns between conditions in ERPs and aperiodic activity, along with the weak association between reaction times and aperiodic activity, suggest that ERPs and aperiodic activity capture distinct aspects of neural processing rather than reflecting the same underlying phenomenon.

While contrasting response-related ERPs with aperiodic components can help address potential confounds, we believe that ERPs are not inherently separate from aperiodic or periodic activity. Instead, ERPs may reflect underlying changes in aperiodic and periodic activity. Therefore, different approaches to studying EEG activity should be seen as providing complementary rather than competing perspectives.

These findings are consistent with previous research that has consistently observed systematic, stimulus-induced changes in the aperiodic component that are distinct from concurrent stimulus-evoked EEG activity (Gyurkovics et al., 2022 [↗](#); Kałamała et al., 2024 [↗](#); Virtue-Griffiths et al., 2022 [↗](#)). We thus underscore the importance of acknowledging the unique characteristics of aperiodic activity, particularly in the context of the n-back task.

Distinct functions of periodic and aperiodic EEG components during cognitive engagement

The findings on periodic and aperiodic EEG activities during the n-back task suggest a complex pattern of cognitive engagement. Following stimulus presentation, the analysis shows an almost immediate aperiodic slope increase in the fronto-central channels (see **Figure 8**), followed by a brief alpha/beta attenuation in the posterior channels (see **Figure 6**). Both of these features have been associated with inhibitory processes that temporarily disrupt ongoing excitatory activity to allow the brain to engage with the presented stimulus (Gyurkovics et al., 2022; Kałamała et al., 2024; Gevins et al., 1997; Pesonen et al., 2007). The results show that both the alpha/beta suppression and the aperiodic change in slope are more pronounced after target stimuli compared to non-target stimuli (see **Figure 7** and **Figure 9**). These observations may indicate greater cognitive engagement associated with decision-making processes following the onset of the target stimulus.

Further analysis of the data shows a significant difference between the 0-back and 2-back conditions in the periodic component (**Figure 7**), indicating more sustained alpha/beta suppression in the 2-back condition, whereas the aperiodic component did not show such spectral slope differences. Sustained suppression of periodic alpha and beta activity during the 2-back condition may reflect ongoing active engagement in the form of information encoding, retention, and updating. This finding suggests that the periodic and aperiodic components may reflect distinct cognitive processes. The characteristics of the second component in the aperiodic slope after non-target stimuli (see **Figure 8**), with no obvious co-occurring processes in the periodic component, further support this idea. It is important to analyse both components to ensure that the processes they reflect are not overlooked.

The results of the study show a clear distinction between the periodic and aperiodic components in relation to reaction times. These results support the association between central beta activity and motor planning (e.g. Eggermont, 2021; Pfurtscheller et al., 1998), as evidenced by its correlation with reaction times in the analysis of the periodic component (**Figure 7**). In contrast, the analysis indicates a relatively weak correlation between the aperiodic slope and reaction times (**Figure 9**). This observation differs from the most commonly reported patterns (e.g. Voytek et al., 2015; Thuwal et al., 2021; Akbarian et al., 2023b), but the evidence remains inconclusive, as some studies have reported contrasting results (e.g. Monchy et al., 2023). The discrepancy highlights the need for further research to understand how different factors influence the manifestation of the aperiodic component as a reflection of cognitive processes.

Prior studies have consistently reported sustained elevations in theta power during the n-back task (e.g. Palomäki et al., 2012; Krause et al., 2010; Pesonen et al., 2007). These increases, particularly in the frontal cortex, have been linked to various cognitive functions such as attention, information encoding, and cognitive control (Roux and Uhlhaas, 2014; Rossi et al., 2023; Cavanagh and Frank, 2014). While power in the theta range can indeed be oscillatory, as demonstrated in some studies, the sustained low-frequency activity observed in the analysis of scalp EEG data in this study appears to be largely influenced by aperiodic components, with post-stimulus increases potentially attributable to shifts in spectral slope rather than exclusively to oscillatory changes. Because low-frequency activity can reflect either oscillatory or aperiodic processes, distinguishing between them is key to avoiding misinterpretation. Nevertheless, previously reported patterns of theta modulation are consistent with the notion that post-stimulus changes in aperiodic activity may reflect enhanced inhibitory control commensurate with task demands (Voytek et al., 2015; Gao et al., 2017).

Main findings replicated in additional datasets

Analyses of control data from a previously published study employing n-back task (Nakuci et al., 2023 [DOI](#)) confirmed our initial results, indicating that similar task-related modulations in alpha and beta periodic activities, as well as aperiodic activity dynamics, can be divided into frontal and parietal components. However, although both aperiodic components were observed in the control dataset, their temporal differentiation was less pronounced (Figure S27 [DOI](#), Figure S29 [DOI](#)). Notably, in this comparative dataset, the aperiodic slope was only related to task load, with a steeper slope observed for the 2-back level, and no significant differences based on stimulus type (Figure S28 [DOI](#)). Furthermore, in the comparative dataset, there was an association with reaction times in the parietal channels, more so than in our original dataset.

To test whether these findings generalise across different task paradigms, we applied the same methods to an item-recognition task, which showed similar low-frequency dynamics despite having distinct mechanisms of memory maintenance and a non-continuous structure (Figure 10 [DOI](#)). Analyses of the original n-back dataset, the control n-back dataset, and the item-recognition dataset all point to the same core pattern: low-frequency power in baseline-corrected data correlates with aperiodic activity, while periodic activity is associated with alpha/beta power in both baseline-corrected data and pACF. This cross-task consistency underscores the robustness of the interpretation that periodic and aperiodic components should be carefully distinguished when studying working memory.

Theoretical and methodological implications

This study highlights the importance of refined EEG analysis methods that take into account the dynamic nature of both periodic and aperiodic components during cognitive tasks. The findings suggest a need for reconsidering baseline correction techniques, particularly in continuous paradigms where traditional baseline subtraction can obscure meaningful neural dynamics by potentially misrepresenting aperiodic activity as low-frequency (theta) oscillations (Gyurkovics et al., 2021 [DOI](#)). Such misrepresentation is particularly relevant under sustained cognitive demand, as corrections intended to normalise data may inadvertently lead to misleading or artefactual interpretations of neural activity. Further investigation is warranted to explore how changes in the aperiodic slope might underlie the commonly observed increases in frontal low-frequency power, which have traditionally been documented as theta oscillations. In order to further validate this interpretation, future research should extend to a wider range of working memory and cognitive control tasks.

The aperiodic slope dynamics observed during the n-back task in this study align with the E:I balance framework (Gao et al., 2017 [DOI](#); Vogels and Abbott, 2009 [DOI](#); Lim and Goldman, 2013 [DOI](#)). Specifically, the results suggest that stimulus-induced shifts in the slope likely reflect a proportional modulation of ongoing excitatory activities tailored to meet processing demands (Gyurkovics et al., 2022 [DOI](#); Voytek et al., 2015 [DOI](#); Virtue-Griffiths et al., 2022 [DOI](#)). These adjustments in the post-stimulus aperiodic component may signify enhanced inhibitory control, which has implications for cognitive control capacities previously associated with changes in the frontal theta activity (e.g. Scharinger et al., 2017 [DOI](#); Cavanagh and Frank, 2014 [DOI](#); Ratcliffe et al., 2022 [DOI](#)). The findings provide support for the hypothesis that modifications in aperiodic activity following cognitive load could serve as a reliable index of cognitive control mechanisms. Future studies should consider aperiodic slope modulations as a potential biomarker for cognitive control, investigating how these measures correlate with traditional indicators and whether they can predict cognitive performance across different task contexts.

To better understand the functional role of the observed second component in the aperiodic activity (second slope steepening) following non-target trials, a study design incorporating a larger dataset with an increased number of target-followed-by-target sequences is essential. This would

enable a detailed comparison of aperiodic components in scenarios where a target is succeeded by another target versus when it is followed by a non-target, potentially revealing how attentional mechanisms are reset or sustained under different conditions.

Furthermore, investigating the potential relationship between aperiodic slope dynamics and theta oscillations in tasks requiring enhanced cognitive control could provide deeper insights into the neural basis of cognitive flexibility and decision-making. It is also imperative to consider individual variations in theta dynamics and their relationship with aperiodic slope modulation, as this could highlight potential biomarkers for cognitive efficiency or susceptibility to cognitive fatigue.

Finally, the findings highlight a potential issue for EEG-based cross-frequency coupling analyses, particularly phase-amplitude coupling (Tort et al., 2010 [DOI](#); Van Der Meij et al., 2012 [DOI](#); Tseng et al., 2019 [DOI](#)). Numerous studies have demonstrated gamma amplitude modulation by the phase of theta (e.g. Park et al., 2013 [DOI](#); Köster et al., 2014 [DOI](#); Goodman et al., 2018 [DOI](#)), posited to organise information in working memory (e.g. Rajji et al., 2016 [DOI](#); Brooks et al., 2020 [DOI](#); McGill and Kieffaber, 2024 [DOI](#); Goodman et al., 2018). However, the results of this study suggest that scalp-recorded low-frequency fluctuations may be predominantly influenced by aperiodic shifts rather than true oscillations. Consequently, it may be unwarranted to assume an inherently oscillatory theta band when investigating theta-gamma coupling. In future cross-frequency coupling studies, it is therefore crucial to confirm that theta power reflects genuine oscillations, rather than aperiodic activity, before drawing conclusions about cross-frequency interactions.

Strengths and limitations

The main strength of the study is the replication of key findings by performing the same analyses on two independent samples. Replication of the results in an independent sample of younger individuals performing a standard verbal n-back task extends the generalisability of the findings beyond the original cohort of older adults. Furthermore, replication of these results in the item-recognition task demonstrates the robustness of conclusions across different task paradigms.

Furthermore, we used pACF, an alternative measure of oscillations, to verify the robustness of the approach, thereby ensuring that low-frequency oscillatory activity was not simply missed by FOOOF near the edges of the spectrum.

There are a number of limitations that need to be considered. First, the task difficulty in the 2-back condition may have been too high. Some participants had to be excluded from the analysis due to poor performance. This exclusion may have reduced the range of cognitive abilities in the sample of older adults, potentially inflating them.

In addition, the higher level of difficulty in the tasks may have influenced the observed differences between the 0-back and 2-back conditions, potentially affecting the magnitude and direction of the reported neural modulations. For the sample of older adults, a simplified 2-back task using a single modality may have been more appropriate. The single modality (verbal, with letters displayed in the centre of the screen) may help to reduce task complexity and align better with the cognitive abilities of older participants. In addition, the dual-modality task imposed time constraints that limited the number of trials for each condition, potentially leading to participant fatigue and reduced data quality.

Conclusions

The study highlights the critical importance of distinguishing between the periodic and aperiodic components of the EEG spectrum in order to ensure accurate interpretation of changes traditionally attributed to oscillatory activity. Using the n-back task, we have demonstrated that the aperiodic component, which is often dismissed as mere background noise, reflects important

features of cognitive processing. The findings also challenge the routine practice of baseline correction, which can obscure crucial modulations of aperiodic activity, particularly in continuous cognitive tasks. It is important to note that a substantial proportion of the low-frequency activity commonly observed in scalp EEG may in fact be attributable to shifts in the aperiodic slope. It is therefore crucial to independently verify whether the observed low-frequency signals are genuinely oscillatory or primarily aperiodic. This is of particular importance for accurately deciphering the neural underpinnings of working memory and cognitive control, and highlights the need for a refined approach in the analysis and interpretation of EEG data.

Methods

Participants

We analysed EEG recordings from fifty-seven volunteers (nine males) aged 55–78 years ($M = 67.4$, $SD = 6.4$), who initially participated in a study on subjective cognitive complaints. All participants had normal or corrected-to-normal vision and were free from neurological or psychiatric illness or cognitive decline, as assessed by neuropsychological testing prior to study entry. Of these participants, twenty-seven reported subjective cognitive complaints, while thirty did not. The current study focused exclusively on within-subject comparisons and did not investigate subjective cognitive complaints or between-group differences. We excluded data from 10 participants from all analyses due to poor quality.

The study protocol was approved by the Medical Ethics Committee of the Republic of Slovenia (protocol number 0120-128/2019/9; 21.08.2019). All study participants gave written informed consent prior to the initiation of any study-related procedures.

Experimental task and procedure

The participants were instructed to press the left button when the letter or position matched the letter/position n -back trials before (i.e. on target trials), and the right button otherwise (on non-target trials). Some of the stimuli were lures, i.e. stimuli that would have been considered targets in the 1-back condition but appeared during the 2-back condition.

The participants completed the n -back task in two modalities (visuospatial and verbal) with two load conditions (0- and 2-back) (**Figure 1** [↗](#)). During the task, participants were presented with white capital letters on a grey background one at a time in one of nine possible positions, in a continuous sequence. Each letter was presented on the screen for 250 ms, followed by a 3000 ms interval during which participants were required to respond. In the verbal task conditions, participants were instructed to memorise the presented letter, whereas in the visuospatial task conditions, participants were asked to memorise the position of the presented letter. The order of modality was counterbalanced across participants.

Each modality always began with the 0-back load condition in which the target stimulus was presented for 10 seconds at the beginning of the block and stayed the same throughout the whole block. The 0-back load condition was followed by a 2-back load condition of the same modality, where participants had to remember each stimulus as it appeared and discern whether it matched the one displayed two trials prior. The 0-back and 2-back load conditions each comprised three 5-minute blocks of continuously presented stimuli. For each modality, the 0-back load condition comprised a total of 201 stimuli, 39 of which were targets, and the 2-back load condition comprised a total of 240 stimuli, 39 of which were targets, and 30 were lures. All stimuli were presented in a pseudorandomised order.

The task began only after participants indicated their readiness by pressing a designated key, after reviewing the instructions and clarifying any doubts with the experimenter. Before each condition, participants completed a practice session consisting of a short block to familiarise themselves with the task demands. The session included a variation of the main task in which participants practised identifying both the target letter and its location, as well as responding to the 2-back challenge for both letters and positions. During the main task, response accuracy and reaction times were recorded by keystroke on the Cedrus response pad (Cedrus Corporation, San Pedro, CA, USA, model RB-540). Participants were instructed to favour accurate responses over fast responses. The participants' responses were monitored, and only trials with correct responses were included in the EEG analysis.

The stimuli were presented in Helvetica font on a 24-inch LCD screen with a refresh rate of 120 Hz. The stimuli, arranged in a 3×3 matrix, spanned approximately 7.22 degrees of both vertical and horizontal visual angles. The task was programmed and executed using PsychoPy 3 (Peirce et al., 2022 [↗](#)).

EEG data acquisition

We recorded EEG activity using either a BrainAmp MR-plus or ActiChamp amplifier (Brain Products GmbH, Germany). We acquired the recordings with a 64-channel ActiCap plus a reference channel, configured according to the extended 10-10 international system layout. To accommodate all participants, we used two slightly different electrode layouts, differing only in the position of two electrodes. Due to inconsistencies in electrode placement between subjects, we excluded channels AFz, Iz, PO9, and PO10 from the analysis, resulting in a dataset of 63 electrodes (including the reference channel). We grounded the electrode setup at AFz, with FCz serving as the reference. During the EEG recording, we band-pass filtered the signal between 0.1 and 250 (or 1000) Hz with a slope of 12dB/octave and digitised it at a sampling rate of 500, 1000, or 2500 Hz. After time-frequency decomposition, we downsampled the data to 100 Hz.

EEG data preprocessing

We preprocessed the EEG data using custom MAT-LAB scripts (MathWorks, USA) that integrated functions from EEGLAB (Delorme and Makeig, 2004 [↗](#)) and ER-PLAB (Lopez-Calderon and Luck, 2014 [↗](#)). High-pass filtering at 0.1 Hz was performed using a finite impulse response (FIR) filter (slope: 12 dB/octave). A CleanLine filter was then used to reduce power line noise at 50 Hz (Mullen, 2012 [↗](#)). Next, we segmented the data into epochs ranging from -1.5 to 2.8 seconds, time-locked to stimulus presentations, followed by baseline correction to remove residual signal drift. Epochs and channels were manually inspected to identify and exclude segments with excessive artefacts. Independent component analysis decomposition was performed using the AMICA algorithm (Palmer et al., 2011 [↗](#)), applied to the data bandpass filtered between 1 and 60 Hz; the solution was later applied to the non-bandpass filtered data. We manually inspected the independent components to identify and remove those representing blinks, horizontal eye movements, muscle noise, line noise, and other artefacts. After the ICA, channel time series reconstruction and channel interpolation were performed, and the data were referenced to the average reference.

Subsequent EEG analyses were conducted only on subjects and conditions that met the required standards, defined as having a minimum of 30 good epochs after cleaning and an accuracy rate of at least 0.6 within a given condition. Exclusion criteria were applied at the level of individual conditions within subjects, rather than excluding entire participants from the analyses. Consequently, only 21% and 10% of participants remained for the visuospatial and verbal lure conditions, respectively. As a result, the lure condition was excluded from all subsequent analyses.

Estimation of periodic and aperiodic component

First, we excluded trials with incorrect or missing responses from all subsequent analyses. ERPs related to early sensory potentials can affect estimates of aperiodic activity (Gyurkovics et al., 2022), therefore, average ERPs were removed prior to time-frequency decomposition, as has been previously done in similar studies (Gyurkovics et al., 2022; Kałamała et al., 2024). This involved computing ERPs for each subject/condition and then subtracting these ERPs from single-trial data for each subject/condition. For completeness, we also analysed the data without subtracted ERPs.

To estimate both periodic and aperiodic components of EEG activity, we performed time-frequency decomposition using superlets, with a range of frequencies from 3 to 50 Hz, in steps of 1 Hz (width of the base wavelet: 3 cycles, order: linearly spaced from 1 to 20, multiplicative combination of wavelets) as implemented in Fieldtrip (Oostenveld et al., 2011). Superlets, a variant of the Morlet wavelet transformation, involve multiple decompositions with varying cycles, the results of which are combined using a geometric mean to achieve enhanced temporal and spectral resolution, known as ‘super-resolution’ (Moca et al., 2021). The results of the time-frequency decomposition were averaged across trials for each condition and participant, and then decimated to 100 Hz in order to save space and facilitate further analysis.

We then estimated the periodic and aperiodic components from the decomposed data for each time point, condition, and participant using the FOOOF algorithm from the Python specparam package (Donoghue et al., 2020b). To avoid edge artefacts, FOOOF parameters were estimated for a period between -0.5 and 2 seconds around the stimulus. The FOOOF algorithm parameterises the frequency spectrum in a semi-logarithmic space, where only power values are log-transformed, using a set of parameters separately for periodic and aperiodic activity. Specifically, neural power spectra (NPS) over a set of frequencies (F) are described as the sum of the aperiodic and periodic components:

$$NPS(F) = L(F) + G(F)_n \quad (1)$$

The aperiodic component is modelled as

$$L(F) = b - \log(k + F^\chi) \quad (2)$$

where b is the offset, k is the knee (bend), and χ is the aperiodic exponent (sometimes referred to as the spectral slope).

The periodic component is modelled as a sum of Gaussians:

$$G(F)_n = a * \exp\left(\frac{-(F - c)^2}{2 * w^2}\right) \quad (3)$$

Each Gaussian is described by three parameters: c , a , and w , representing center frequency, power, and bandwidth, respectively.

Fitting was performed using default specparam parameters: peak width limits = 0.5–12.0 Hz, maximum number of peaks to fit = Inf, minimum peak height = 0, peak threshold = 2.0, aperiodic mode = ‘fixed’ (i.e. the knee parameter was set to 0). Note that the FOOOF fitting was performed on trial-averaged data to increase the stability of the parameter estimates. In contrast, Gyurkovics et al. (2022), who also investigated task-related changes in aperiodic activity, stabilised their fits by estimating power spectra over 1-second time windows at the single-trial level. Although effective, this strategy involved a trade-off in temporal resolution that our trial-averaging approach seeks to avoid.

To assess the fit of the FOOOF models, we examined R^2 and model error metrics (**Figure S9** [↗](#)). Model error was defined as the absolute difference between predicted and actual values. However, as we have shown in the simulation (see section Control analyses below), model fit metrics can be misleading. By simulating data and estimating parameters, we observed cases where the model fit was high, but the estimated parameters deviated substantially from the ground truth values. This highlights the limitations of relying solely on fit metrics to assess parameter accuracy.

In addition, we examined the number of peaks identified per model. On average, 1.9–2 peaks were identified, showing high consistency and indicating that the models did not overfit by detecting an unrealistic number of peaks (**Figure S10** [↗](#)). We also repeated the analysis with the maximum number of peaks set to three, which gave virtually identical results (results not shown).

The outcomes of the analyses conducted with the offset and exponent parameters were found to be highly comparable. Consequently, the main text only includes analyses using the exponent parameter, whereas the Supplement contains analyses using the offset parameter.

To allow comparisons with full time-frequency power and to facilitate comparisons across participants and conditions, we reconstructed the periodic component in a semi-log space based on the estimated parameters for each participant and condition using the formula in **Equation 3** [↗](#). This reconstruction includes only the parameterised oscillatory peaks and is referred to as ‘periodic power’ or ‘periodic activity’, a measure we used in all subsequent analyses. Importantly, the reconstructed periodic components were strictly non-negative: although Gaussians asymptotically approach zero, they never reach it, resulting in smooth transitions without explicit zeros between peaks.

We visually assessed the time-frequency plots, both in their original form and after baseline correction, before performing formal statistical analyses. To assess the effect of baseline correction on the results, we compared several baseline correction methods, including: (a) decibel conversion ($10 * \log_{10}(\text{data}/\text{baseline})$), (b) relative change $((\text{data} - \text{baseline})/\text{baseline})$, (c) normalised change $((\text{data} - \text{baseline})/(\text{data} + \text{baseline}))$, (d) absolute change $(\text{data} - \text{baseline})$. In addition, we considered three baseline periods (relative to stimulus onset): (a) from -500 to -200 ms, (b) from -300 to 0 ms, (b) from -500 to 0 ms.

Comparison of baseline-corrected data and FOOOF parameters

We compared baseline-corrected time-frequency activity with both periodic and aperiodic components. To better understand their relationship, we first visually inspected grand-average plots and then performed mass univariate comparisons. Specifically, we examined the relationship between baseline-corrected power and FOOOF parameters for each channel, time, and frequency point.

A simple approach to assessing this relationship would be to compute correlations across subjects at each point. However, this approach would violate the independence assumption of simple linear models, as each subject contributed multiple observations across different experimental conditions. A more appropriate approach was to use a linear mixed model that included both subject and condition as random effects.

Although condition is typically treated as a fixed effect, in this case, we were not interested in the specific conditions themselves. Instead, we treated them as a random sample from a larger population of possible conditions. Baseline-corrected activity was treated as the dependent variable, while periodic or aperiodic activity served as a fixed effect predictor variable.

For each test, we estimated the marginal R^2 (Nakagawa and Schielzeth, 2013), a measure of variance explained for linear mixed models, analogous to R^2 in general linear models. The marginal R^2 is defined as variance explained by the fixed factors using the equation:

$$R^2_{\text{LMM}(m)} = \frac{\sigma_f^2}{\sigma_f^2 + \sigma_\gamma^2 + \sigma_\alpha^2 + \sigma_\varepsilon^2} \quad (4)$$

where σ_f^2 represents the variance explained by the fixed effects, σ_γ^2 and σ_α^2 correspond to variance of random effects (subject- and condition-related variance, respectively), and σ_ε^2 represents the residual variance. The fixed-effects variance σ_f^2 can be computed as the variance of the predicted values.

Although R^2 is strictly non-negative, we assigned the sign based on the beta coefficient from the fixed effect in the mixed model.

Additionally, to assess variability across participants, we visualised time-averaged periodic activity for each participant (Figure S6).

Validation of FOOOF results

As reviewers of the original manuscript raised concerns that FOOOF might miss peaks near the edges of the spectrum (Gerster et al., 2022), we performed two control analyses to test the validity of the FOOOF results.

First, we assessed whether the FOOOF decomposition can accurately resolve low-frequency periodic activity by simulating ground truth data with systematically varied parameter values. Details of this simulation are provided in the Supplement. Briefly, we found that low-frequency periodic components were often conflated with aperiodic components, resulting in undetected periodic activity at very low frequencies. This issue was more pronounced when fitting was restricted to frequencies ≥ 3 Hz (as in our analyses of experimental data). In addition, although model fit metrics were generally good, they did not always indicate accurate parameter estimation. These results suggest that even when true low-frequency periodic activity is present, FOOOF may fail to detect it. Conversely, changes in the aperiodic parameters estimates may actually be a consequence of low-frequency periodic activity.

Second, we used an alternative method to assess whether signals at different frequencies were oscillatory. Specifically, we used the recently developed phase autocorrelation function (pACF) (Myrov et al., 2024). Unlike amplitude-based methods such as FOOOF, pACF is sensitive to the ‘rhythmicity’ or temporal stability of the phase of a signal. pACF achieves this by calculating the cross-spectrum between a complex signal and its time-delayed copy, then normalising to unit magnitude to remove amplitude information. Because pACF isolates phase relationships while being inherently independent of amplitude, it serves as a complementary measure to FOOOF. Consistent results between FOOOF and pACF would strengthen our conclusion that low-frequency (theta) oscillations were truly absent from our recordings, rather than simply undetectable by FOOOF.

Sample pACF is estimated using formula:

$$\overline{\text{pACF}(l)} = \frac{1}{N} \sum_{i=1}^N \text{CS}_{x,x}(l) \quad (5)$$

where $\text{CS}_{x,x}(l)$ represents a normalised crossspectrum between a complex signal X and its delayed copy at lag l , and N is the total number of samples.

To allow comparison with the time-resolved FOOOF decomposition, the time-resolved pACF was estimated as follows:

$$x_t = \sum_{l=0}^{L_{\max}} \frac{\sum_{i=t-W/2}^{t+W/2} z_i^l}{W} / N \quad (6)$$

where t is a timestep, l is a lag, W is a window size (in samples), N is the number of lags, and z is the phase difference between a signal and its delayed version. The size of the moving window was set to 2.5 cycles, with lags ranging from 1 to 3 cycles in 0.1 increments. These parameters were chosen based on Myrov et al. (2024) [\[14\]](#). The moving window size was chosen to ensure stability of the results without excessive smoothing. The selected lag range was based on the consideration that too small lags are predominantly influenced by filtering effects, whereas large lags may capture activity from non-local sources (V. Myrov, personal communication, January 21, 2025).

To compute pACF, the signals were convolved with complex Morlet wavelets (width: 3 cycles). We did not use superlets here, as they were primarily designed for power estimation rather than phase analysis. In fact, when we tested superlets, the pACF values increased linearly with frequency – probably due to the way superlets combine wavelets of different widths, which can affect the phase autocorrelation of the convolved signal. To ensure an accurate comparison between FOOOF and pACF, we repeated the FOOOF decomposition using the same Morlet wavelets, which gave results comparable to those obtained with superlets, particularly at lower frequencies.

To directly compare FOOOF and pACF results, we used linear mixed models, following the same approach used for the comparison between baseline-corrected activity and FOOOF. In this analysis, pACF served as the dependent variable, while FOOOF parameters were included as fixed effect predictors, with condition modelled as a random effect. We fitted a separate model for each FOOOF parameter and estimated the marginal R^2 .

Comparisons between experimental conditions

We performed statistical comparisons between conditions using linear mixed models with the `fitlme` function in MATLAB. An advantage of using linear mixed models is their ability to accommodate missing data, which improves parameter estimates through partial pooling. We fitted the models as follows:

$$y \sim 1 + \text{modality} * \text{nback} * \text{stimulus} + \text{rt} + (1 | \text{subject}) \quad (7)$$

Here, y refers to the exponent or offset in the case of aperiodic activity and to the log-power in the case of periodic activity. We included reaction times (rt) in the model to control for differences between conditions and to assess the relationship between motor response and EEG activity.

To estimate representative reaction times for each participant and condition, we removed reaction times above a z -value of 3 and those below 200 ms within each participant and condition. We then fitted an exGaussian distribution to estimate the parameter μ using the maximum likelihood method as implemented in the `retimes` package in R (Massidda, 2013 [\[15\]](#)). The exGaussian distribution, a positively skewed distribution, is useful for modelling reaction time distributions (Balota and Yap, 2011 [\[16\]](#)), and the parameter μ can be interpreted as the mode of the distribution. Other parameters (e.g. σ , τ) were excluded to avoid overfitting and because our aim was to obtain a simple, robust estimate of the central tendency rather than to model the entire distribution.

We fitted models in a mass univariate manner, that is for each channel, frequency (where applicable), and time point separately.

Fixed effect were estimated using the maximum likelihood method, and the degrees of freedom were approximated using the Satterthwaite equation. We corrected the p-values using the Benjamini-Yekutieli false discovery rate (FDR) correction (Benjamini and Yekutieli, 2001 [DOI](#)). Given that under the false discovery rate (FDR) correction, a q proportion of significant p-values are false positives, and that EEG effects occur in clusters (in time, frequency, and/or space) (Van Ede and Maris, 2016 [DOI](#)), we additionally considered as significant only clusters of p-values that spanned either at least three time points or at least three channels. We set the significance threshold at $\alpha = .05$.

For the purposes of visualisation, p-values were averaged across channels (for heatmaps or lines) or across time (for topographies). Consequently, there are some discrepancies between heatmaps/lines and topographies.

Comparison of aperiodic activity with event-related potentials

To rule out the possibility that aperiodic activity merely reflected evoked potentials, we first removed evoked activity before time-frequency decomposition (as described above). We then estimated associations between event-related potentials (ERPs) and aperiodic activity. Specifically, we used linear mixed models to assess this association at each channel-time point, with ERP activity as a predictor of aperiodic activity and condition included as a random effect, similar to the comparison between FOOOF parameters and baselinecorrected data.

In addition, we visually compared the topographies of aperiodic activity with those of stimulus-locked and response-locked ERPs.

The results are shown in the Supplement.

Replication: n-back task

To test the generalisability of the results, we performed two replication analyses using two different EEG datasets: an n-back task and an item-recognition task. For the n-back task, we used an EEG dataset from a previously published study by Nakuci et al. (2023) [DOI](#). This dataset includes data from 21 subjects (17 female, mean age 42 ± 12 years) who performed a verbal n-back task (0-, 1-, and 2-back) while being recorded with a 256-channel HydroCel Geodesic Sensor Net. The task involved 150 trials per condition, 50 of which were targets. Each stimulus was presented for 400 ms, followed by an interstimulus interval of 2000 ms.

The dataset was used in its preprocessed form, as the unprocessed data was not available. The preprocessing included band-pass filtering, artifact subspace reconstruction (ASR) (Mullen et al., 2013 [DOI](#)), independent component analysis, channel interpolation, another round of ASR, re-referencing to the average reference, epoching from -1000 to 2000 ms around stimulus presentation, and baseline correction using the prestimulus period (Nakuci et al., 2023 [DOI](#)). Notably, the data were band-pass filtered between 0.5 and 50 Hz, so we restricted the time-frequency decomposition and FOOOF analysis to frequencies between 3 and 35 Hz.

We carried out all other analytical steps, including time-frequency decomposition, FOOOF analysis, and statistical comparisons, in the same manner as in the primary analysis.

The results of these analyses are presented in the Supplement.

Replication: item-recognition task

To test generalisability across tasks, we analysed EEG recordings from another study, which included 48 participants (10 male, mean age 23.8 ± 5 years). All participants were right-handed, had normal or corrected-to-normal vision, and were free of neurological or psychiatric illness.

Most participants were psychology students at the University of Ljubljana, while the rest responded to invitations sent via email and social media. All participants provided informed consent, approved by the Ethics Committee of the Faculty of Arts, University of Ljubljana.

In this study, we used a Sternberg item-recognition task. In this paradigm, a series of visual target stimuli is presented, followed by a probe stimulus that may or may not be identical to one of the previous targets. Participants' task is to indicate whether the probe (test) stimulus was part of the initial set (Sternberg, 1969 [↗](#)).

The experiment included three task conditions performed under two working memory loads: (1) presentation of two or four stimuli that differed in colour but were displayed in the same location; (2) presentation of two or four stimuli that differed in position but were of the same colour (black); (3) presentation of two or four stimuli that differed in both colour and position (integration). Each condition consisted of 432 trials, with 216 trials per working memory load, divided into 18 blocks of 48 trials each. Participants were randomly assigned to three groups, each performing the task in two of the three conditions, resulting in 32 participants per condition and 16 participants per comparison of conditions. A schematic diagram and a detailed description of the task can be found in **Figure S31** [↗](#).

Stimuli were presented on a 27-inch LCD screen with a refresh rate of 120 Hz. The stimuli were presented on a grey background in the centre of the screen, within a frame 250 pixels wide and 250 pixels high. The stimuli appeared as circles 20 pixels in diameter. When different positions were required, the stimuli were spaced at a distance equal to six times the stimulus radius. The frame spanned approximately 4.05 degrees of vertical and horizontal visual angle, and the diameter of each stimulus spanned approximately 0.32 degrees of visual angle. The task was programmed and ran using PsychoPy 2 (Peirce et al., 2022 [↗](#)).

For the EEG measurements we used a BrainAmp EEG amplifier (Brain Products GmbH, Germany) and a 32-channel ActiCAP EEG cap, configured according to the international 10/20 system. Participants provided responses using a Cedrus response pad, RB series, model 530. We monitored horizontal eye movements using a camera and a dedicated HEOG channel ([F7—F8]/2). We manually removed trials in which participants tracked the target stimuli with their eyes.

We preprocessed the EEG data using the same pipeline as for the n-back task, except that the data were segmented into epochs ranging from -1700 to 5400 milliseconds, time-locked to the cue presentations. Subsequent EEG analyses were performed in a manner identical to the n-back task, including only epochs with correct responses.

Data and code availability

The preprocessed data are available at <https://dx.doi.org/10.17605/OSF.IO/RQ8ZM> [↗](#). The n-back dataset from Nakuci et al. (2023) [↗](#) is available at <https://zenodo.org/records/6897260> [↗](#).

Preprocessing was performed using EEGLAB (<https://github.com/scn/eeqlab> [↗](#)) and EEGLAB plugins. Time-frequency decomposition was performed using the Fieldtrip function `ft_freqanalysis` (https://github.com/fieldtrip/fieldtrip/blob/master/ft_freqanalysis.m [↗](#)). FOOOF decomposition was performed using the Python package `specparam` (<https://github.com/fooof-tools/fooof> [↗](#)). Code for phase-autocorrelation function is available on GitHub: https://github.com/palvalab/discovering_rhythmicity [↗](#). Linear mixed models were estimated using the MATLAB function `fitlme` (<https://www.mathworks.com/help/stats/fitlme.html> [↗](#)). Visualisation was done using the packages `ggplot2` (<https://ggplot2.tidyverse.org/> [↗](#)), `matplotlib` (<https://matplotlib.org/> [↗](#)) and `MNE-Python` (<https://mne.tools/> [↗](#)).

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Author contributions

Tisa Frelih: Conceptualization, Project administration, Data curation, Investigation, Methodology, Visualization, Writing – original draft, Writing – review and editing; Andraž Matkovič: Conceptualization, Investigation, Formal analysis, Methodology, Software, Data curation, Visualization, Writing – original draft, Writing – review and editing; Tjaša Mlinarič: Investigation, Project administration, Writing – review and editing; Jurij Bon: Resources, Supervision, Writing – review and editing; Grega Repovš: Conceptualization, Methodology, Software, Resources, Supervision, Writing – review and editing.

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Supplementary Information

N-back task

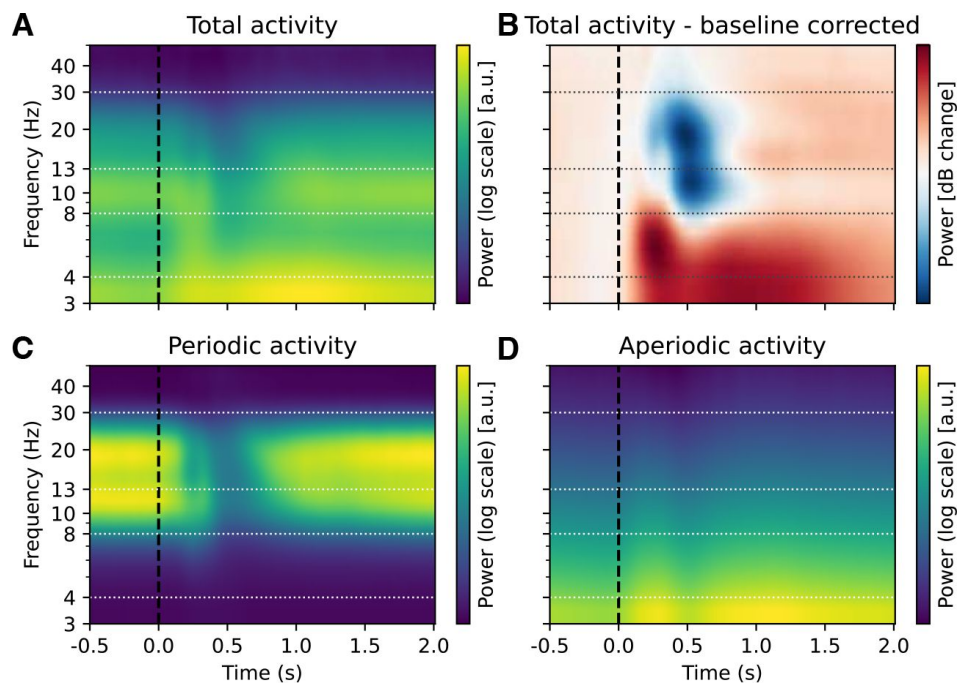


Figure S1.

Effect of baseline correction and FOOOF decomposition on time-frequency decomposition in the n-back task.

Same as [Figure 3](#) but with a logarithmic y-axis.

Figure S2.

The comparison of different baseline corrections on time-frequency decomposition results in the n-back task.

We compared four types of baseline correction (in columns): (a) decibel conversion ($10 * \log_{10}(data/baseline)$), (b) relative change $((data - baseline)/baseline)$, (c) normalised change $((data - baseline)/(data + baseline))$, (d) absolute change $(data - baseline)$. We also considered three baseline periods (in rows): (a) from -500 to -200 ms, (b) from -300 to 0 ms, (b) from -500 to 0 ms. The selection of baseline correction type and period had minimal impact on the outcomes. Among the methods, absolute change correction exhibited a slight reduction in deviations from baseline. Nevertheless, the overall results were qualitatively similar for all baseline corrections. The data presented represent the average across participants, conditions and electrodes.

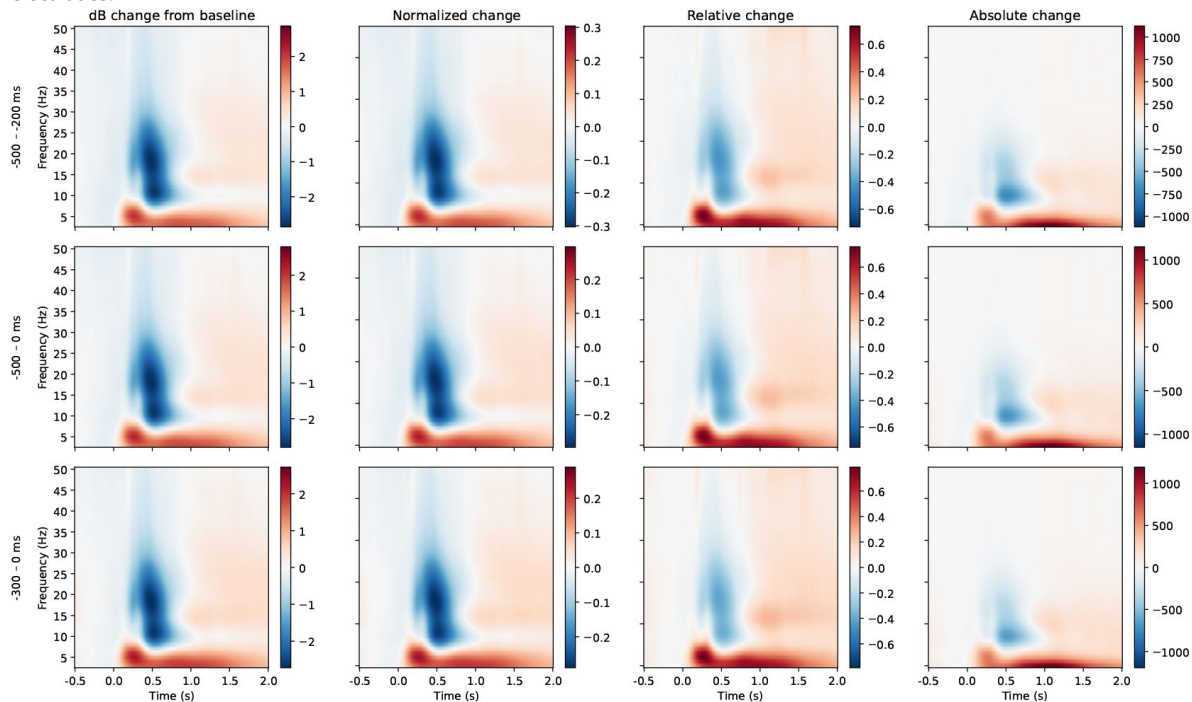


Figure S3.

Topographies of correlations between baseline-corrected time-frequency and FOOOF-decomposed EEG activity.

See Figure 4 for heatmap of average across all channels.

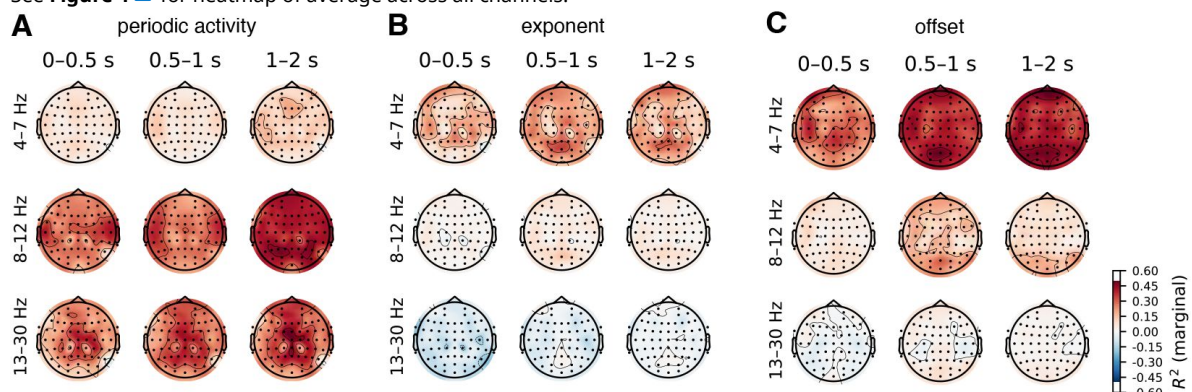


Figure S4.

Changes in phase autocorrelation function (index of rhythmicity) as a function of time.

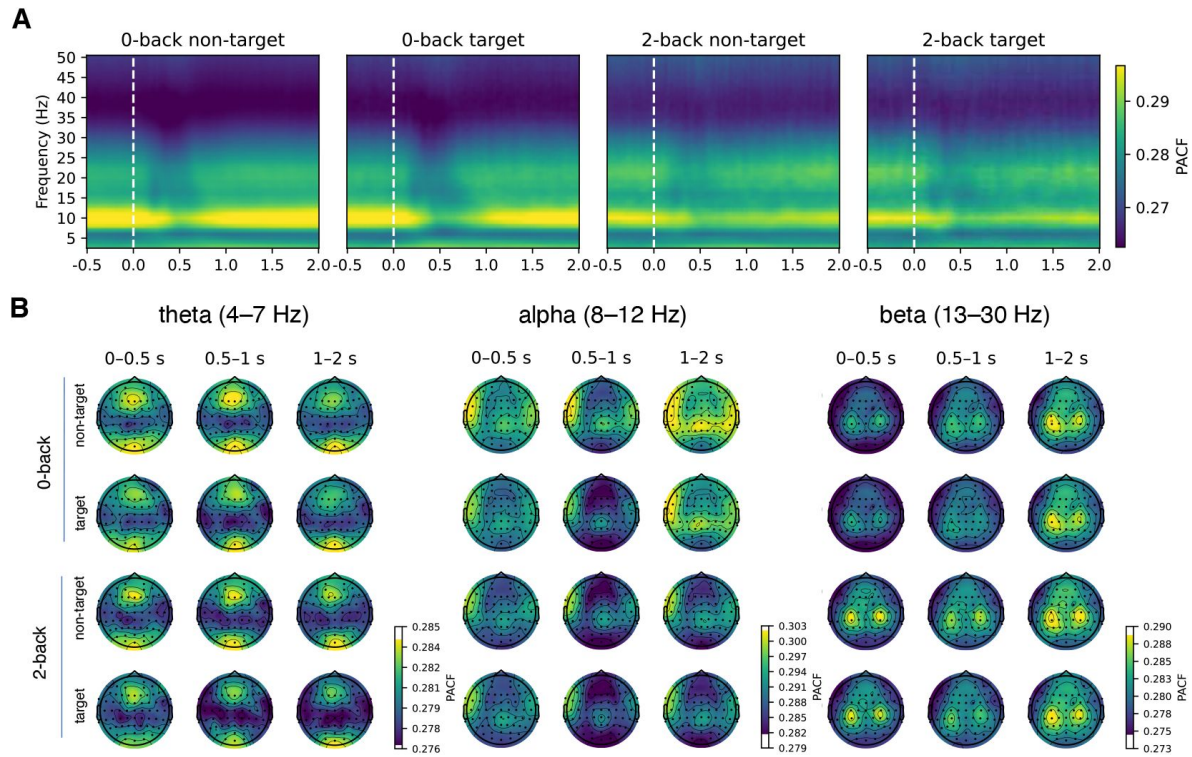
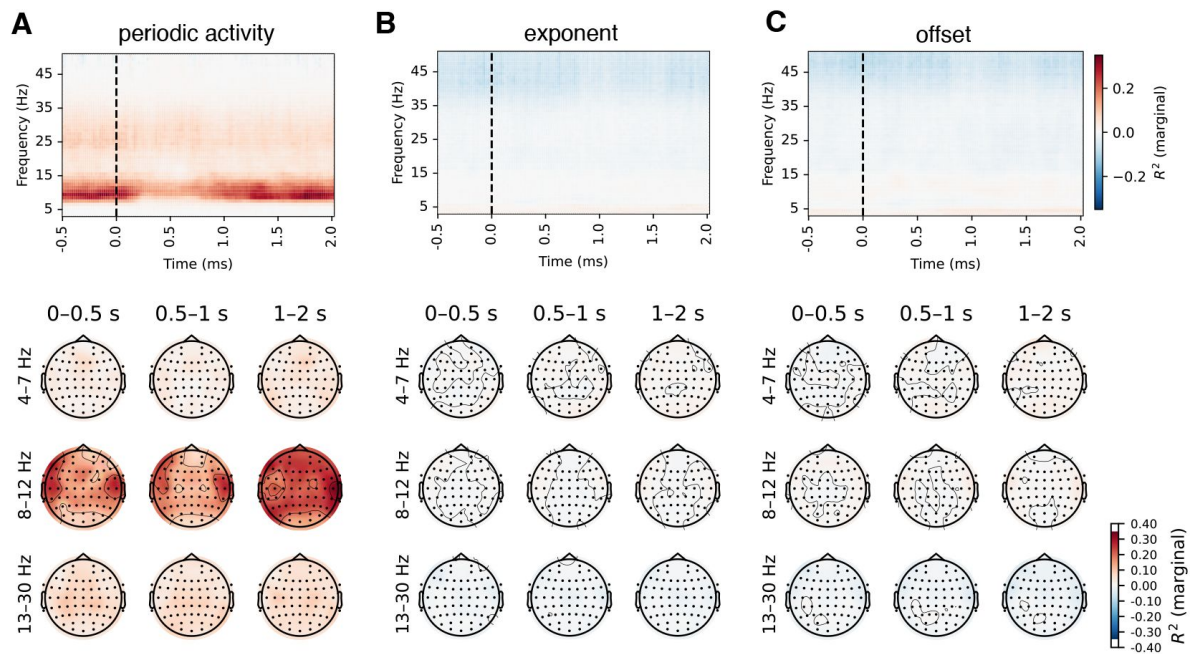


Figure S5.

Correlations between phase-autocorrelation function and FOOOF parameters.



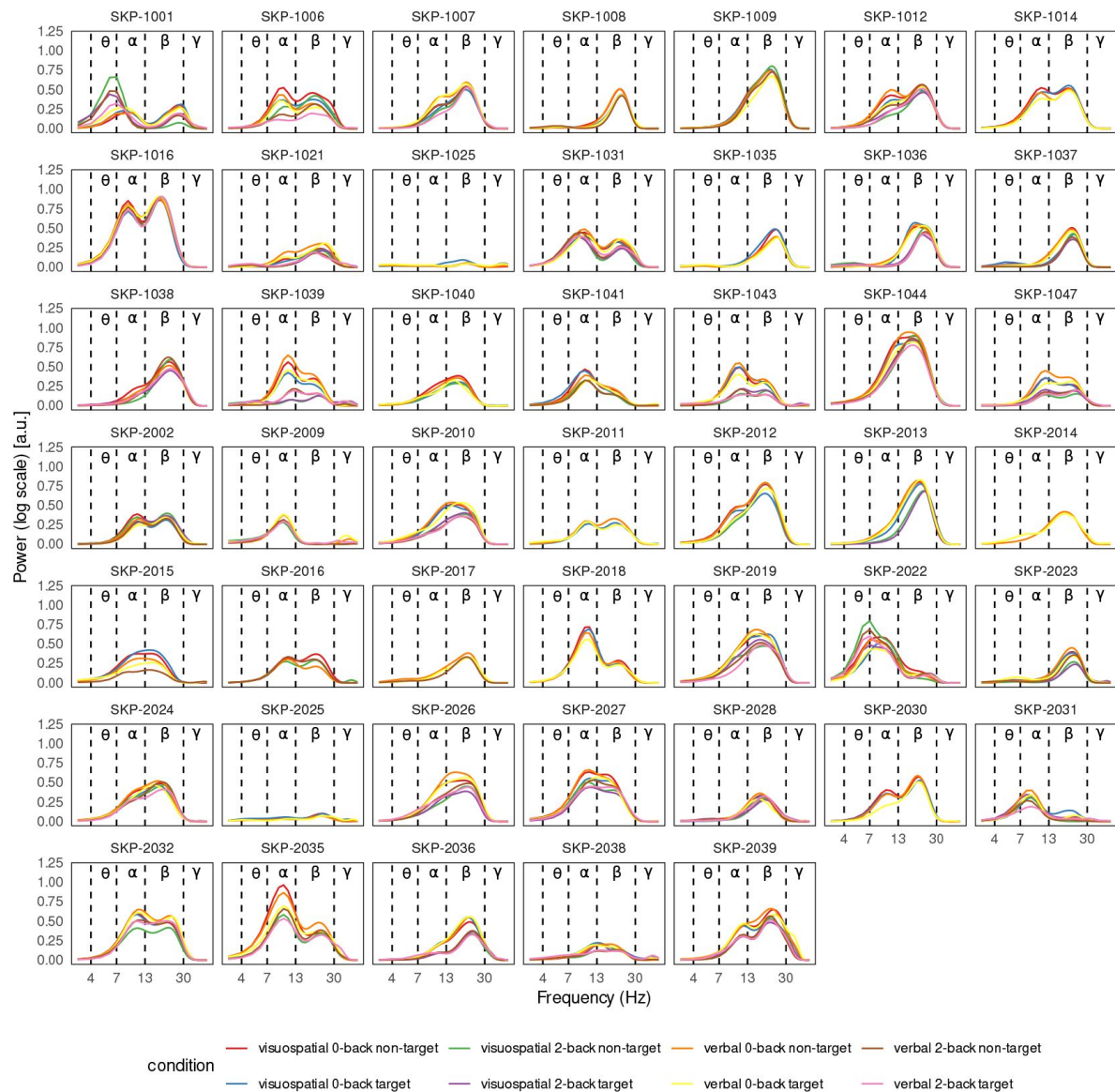


Figure S6.

Power spectra of periodic activity on the Fz channel for each participant, averaged over time (between 0.5 and 2 seconds).

Inspection of the individual periodic power spectra showed that only two participants, SKP-1001 and SKP-2022, exhibited periodic activity in the theta band. Both axes of the spectra are displayed on a logarithmic scale.

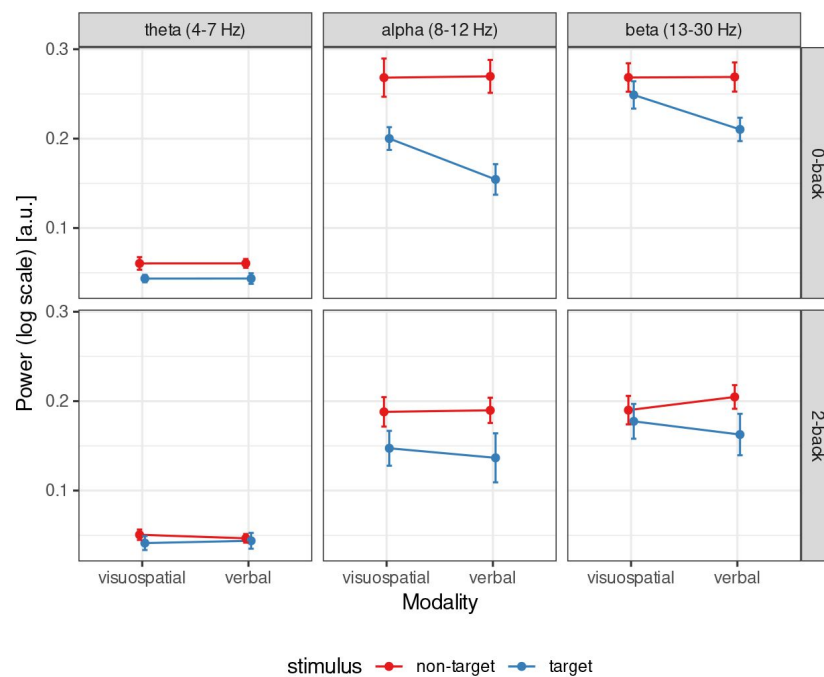


Figure S7.

Power of periodic activity 0.5 to 1 second post-stimulus, averaged across all channels and participants.

This visualisation facilitates the interpretation of the interaction between stimulus type and modality. It illustrates that the difference in power between target and non-target stimuli is more pronounced in the verbal task than in the visuospatial task. Error bars represent 95% Cousineau-Morey (within-subject) confidence intervals (Cousineau, 2017).

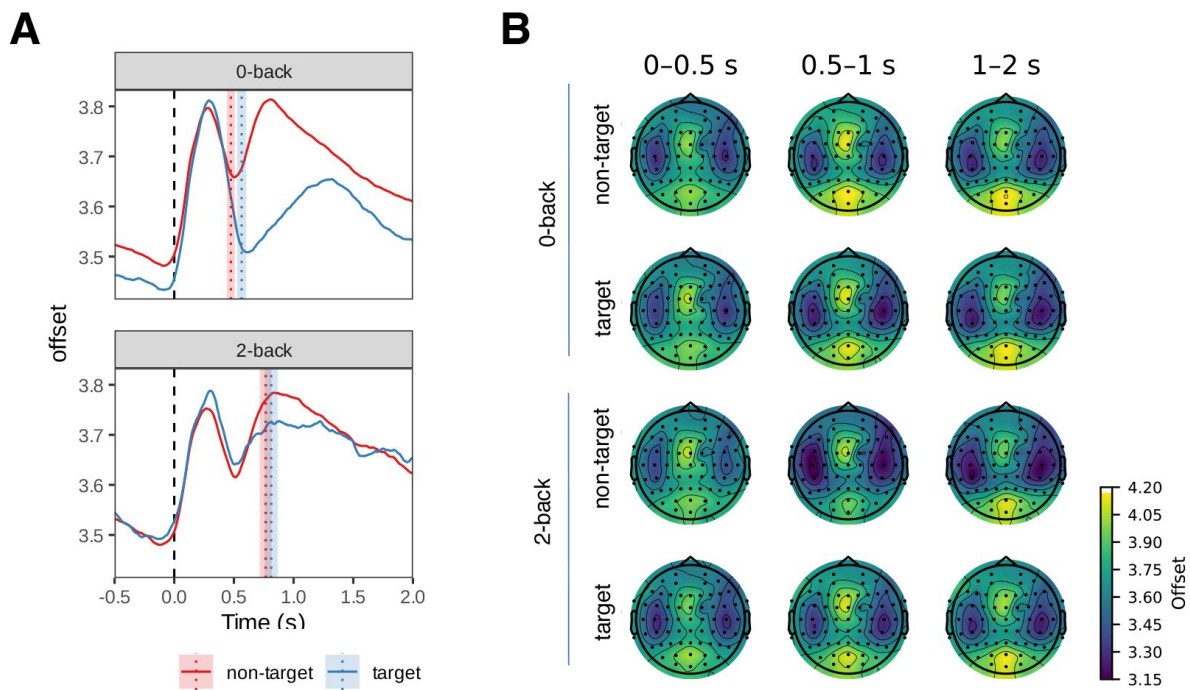


Figure S8.

Temporal changes in aperiodic activity (offset).

Similar as [Figure 8](#), but for the offset parameter. The vertical lines in panel A represent mean reaction time modes. The shaded areas represent 95% Cousineau-Morey within-subjects confidence intervals.

Figure S9.

FOOOF goodness-of-fit measures.

Averaged across channels, subjects and conditions. Goodness-of-fit was lowest around 0.5 s after stimulus onset, corresponding to the decrease in aperiodic and periodic activity.

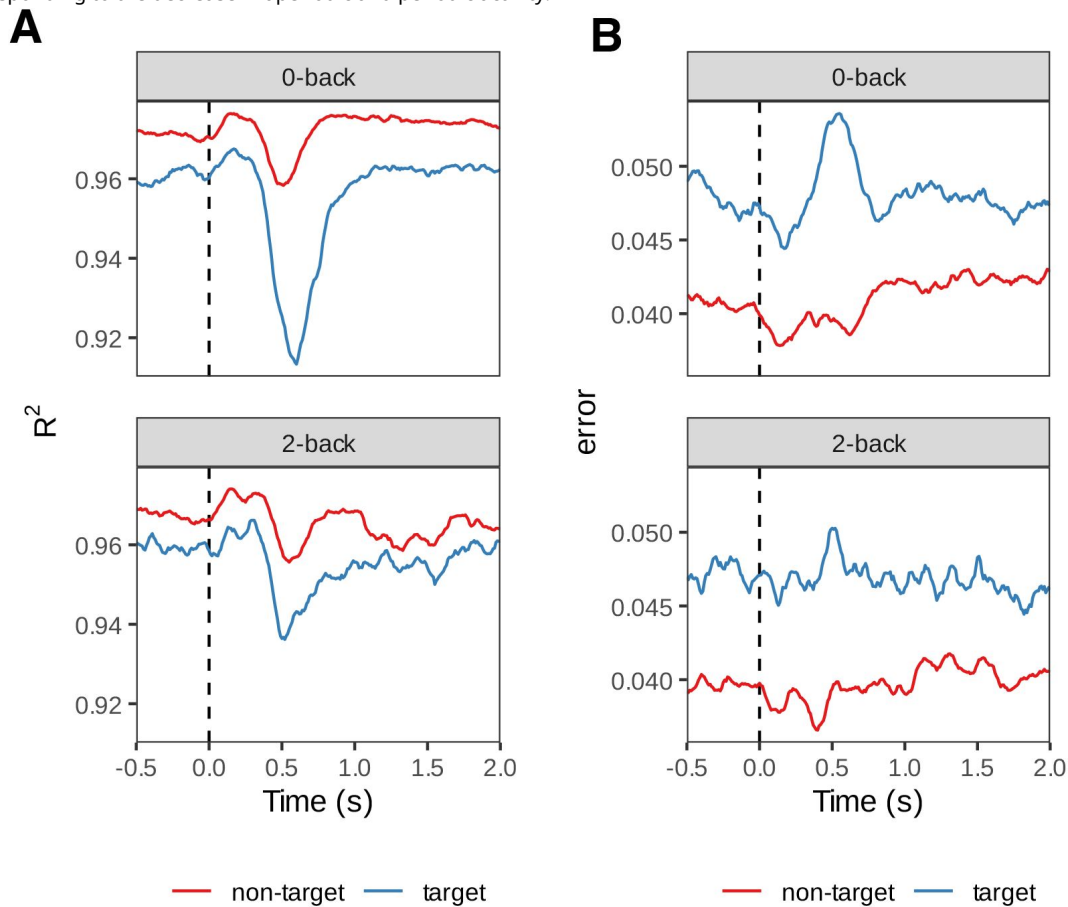
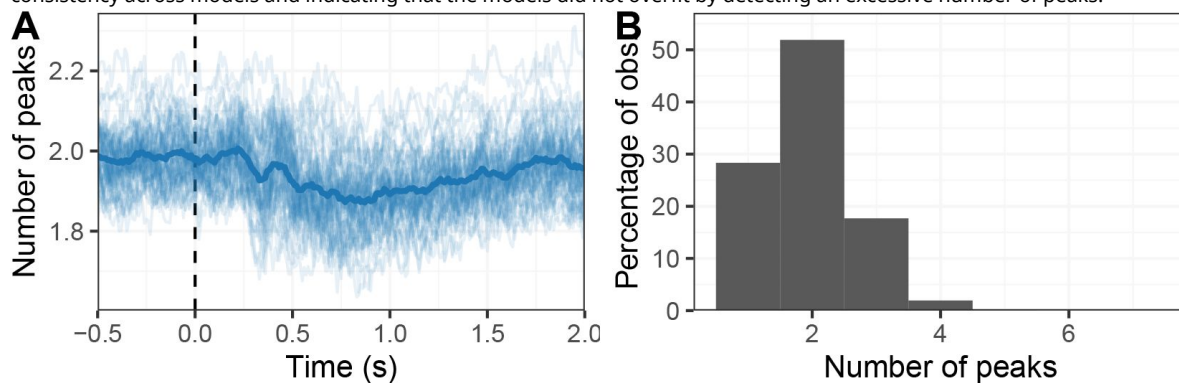


Figure S10.

Number of identified peaks in FOOOF models.

(A) Average number of peaks across all subjects, channels, and conditions. Light lines represent individual channels. (B) Histogram of the number of peaks in all models. On average, 1.9–2 peaks were identified per model, showing high consistency across models and indicating that the models did not overfit by detecting an excessive number of peaks.



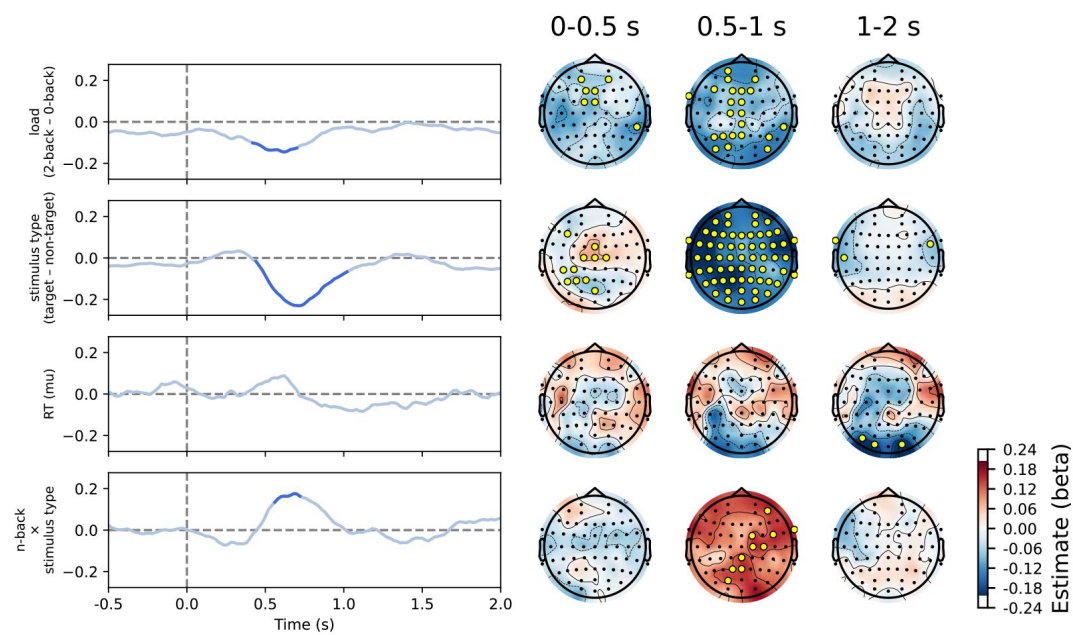


Figure S11.

Results of the linear mixed model on aperiodic activity (offset parameter) of the comparison between conditions.

Significant values are shown in blue on line plots and marked with yellow circles on topographies. Results were similar to the exponent parameter (Figure 9), but there was also an effect of load at about 0.6 seconds post-stimulus.

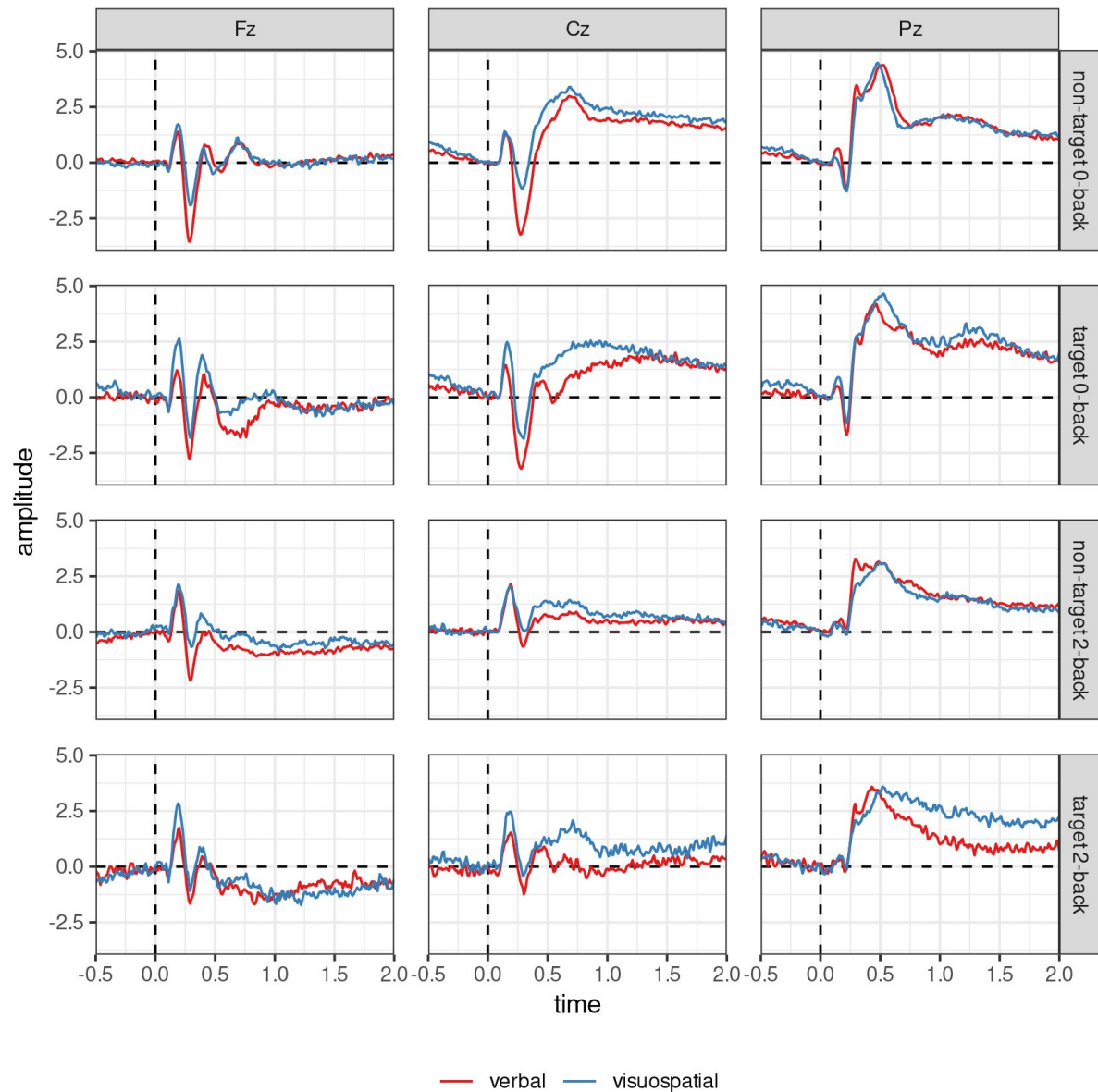


Figure S12.

Grand average event-related potentials (ERPs) on midline electrodes.

ERPs have been low-pass filtered at a 40 Hz cut-off for visualisation.

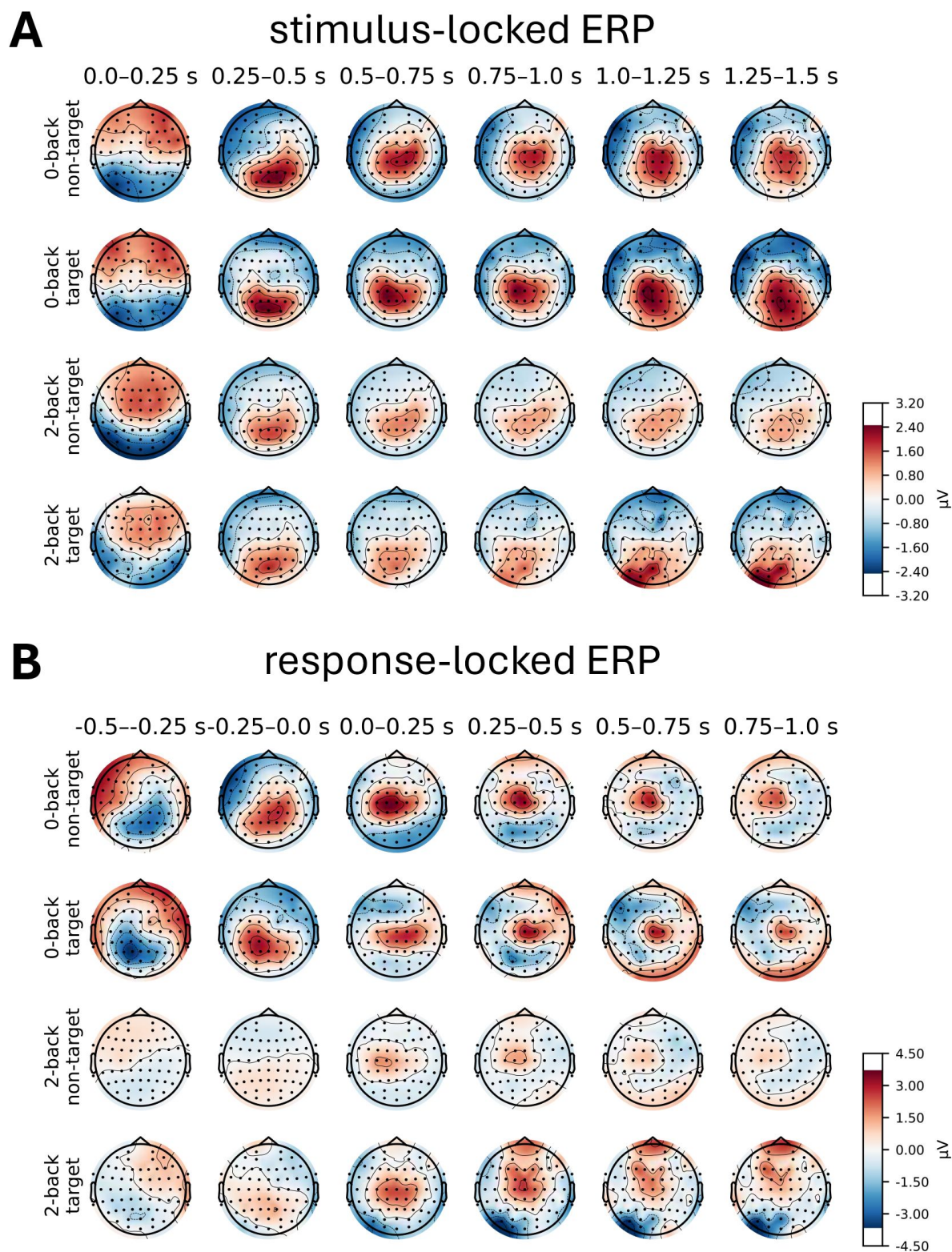


Figure S13.

Topographies of stimulus- and response-locked event-related potentials (ERPs).

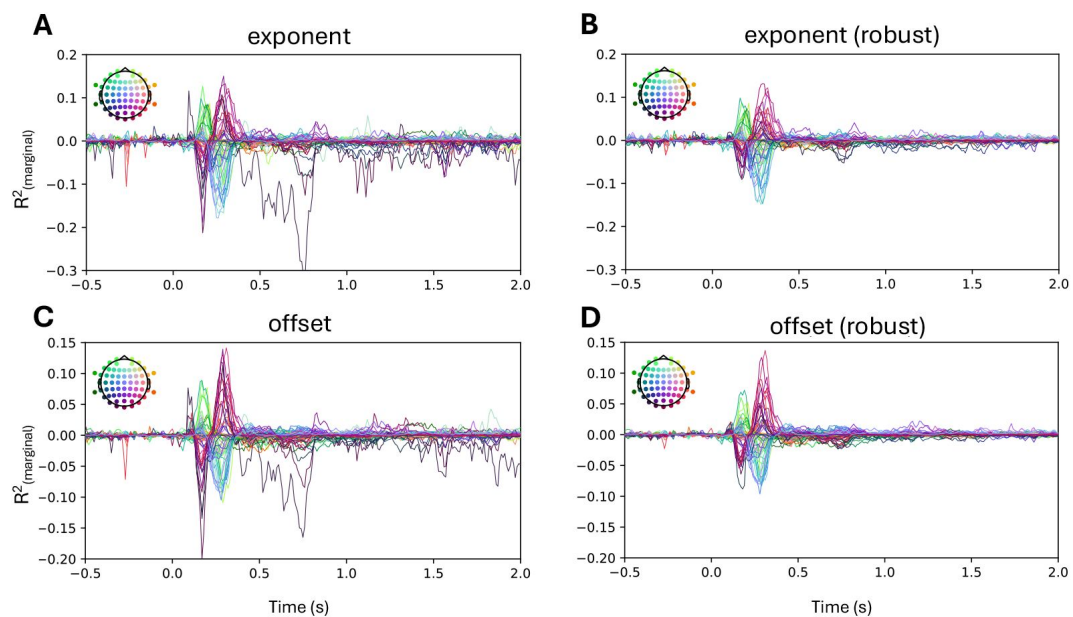


Figure S14.

Correlations between ERPs and aperiodic activity.

Associations between stimulus-locked ERPs and aperiodic activity were estimated using a linear mixed model for each channel-time point. Correlations between ERPs and aperiodic activity peak right after stimulus onset but remain generally low. Notably, moderate correlations are observed between 0.5 and 1 second after stimulus onset on one channel, which we suspected were due to noise. To address this, we recalculated the model by capping values below the 0.1th percentile and above the 99.9th percentile (panels B, D). The results confirm that the associations between stimulus-locked ERPs and aperiodic parameters are very weak beyond 0.5 seconds.

Time-frequency decomposition without subtracted ERPs

In the primary analysis, we subtracted ERPs prior to time-frequency decomposition to avoid effects of phase-locked activity on periodic and aperiodic activity (see Methods for details). For completeness, we also analysed the data without subtracted ERPs, as shown in the figures below.

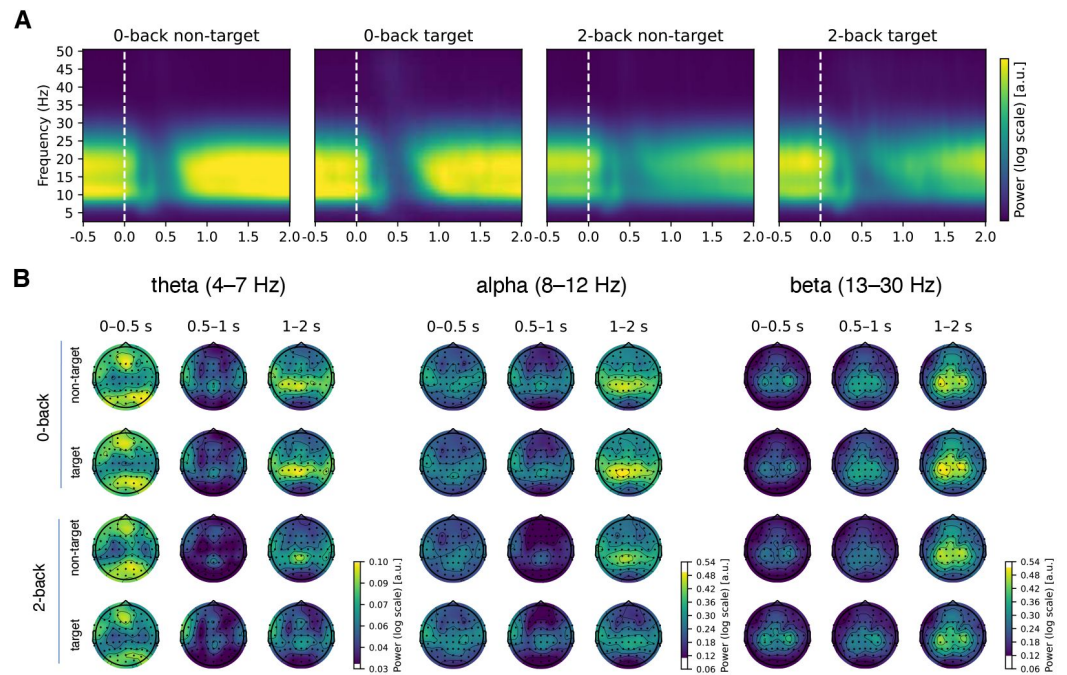


Figure S15.

Changes in periodic (oscillatory) activity as a function of time.

Similar to [Figure 6](#), but without subtracted ERPs. The periodic activity remains very similar to that observed with the subtracted ERPs, with a notable increase in power at occipital and frontal electrodes from 0 to 0.5 seconds post-stimulus.

Figure S16.

Changes in aperiodic activity (slope or exponent) as a function of time.

Similar to **Figure 8**, but without the subtraction of ERPs. Aperiodic activity remains broadly similar to that observed with subtracted ERPs, with an increase in power at occipital electrodes and greater differentiation between frontal and parietal/occipital components. The vertical lines in panel A represent mean reaction time modes. The shaded areas represent 95% Cousineau-Morey within-subjects confidence intervals.

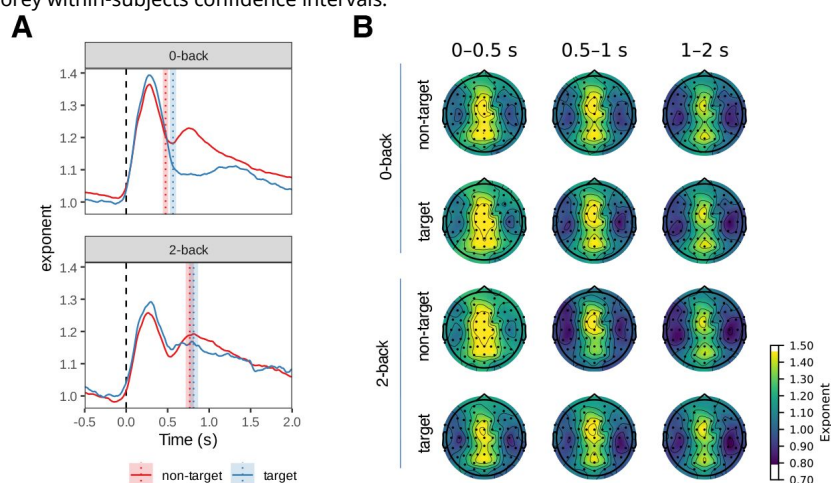
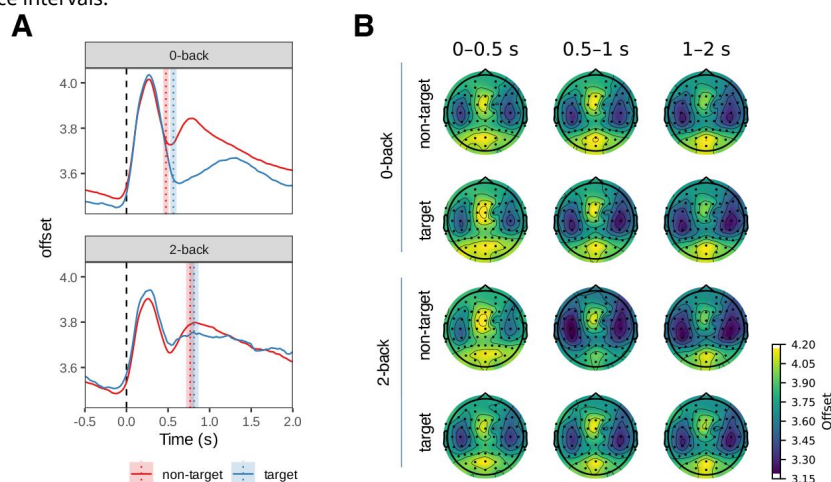


Figure S17.

Changes in aperiodic activity (offset) as a function of time.

Similar to **Figure S8**, a frontal and parietal aperiodic component can also be observed when the ERPs are not subtracted. The vertical lines in panel A represent mean reaction time modes. The shaded areas represent 95% Cousineau-Morey within-subjects confidence intervals.



Replication: n-back

All analyses were repeated on an alternative n-back EEG dataset (Nakuci et al., 2023) and are presented here.

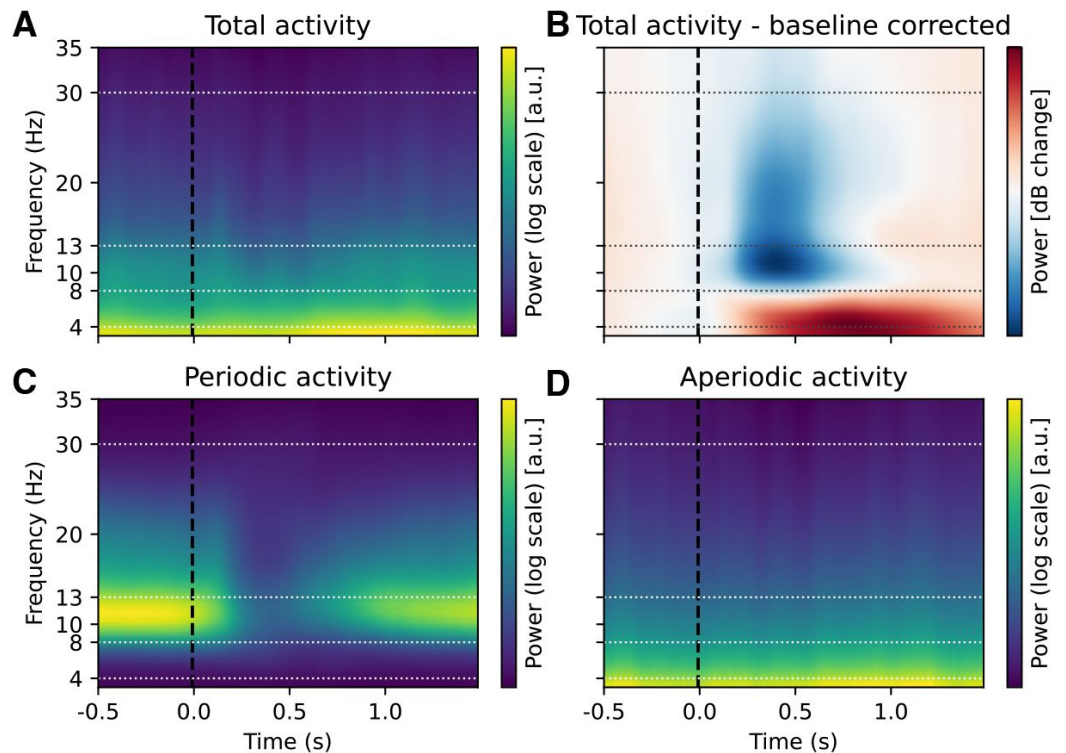


Figure S18.

Effect of baseline correction and FOOOF decomposition on time-frequency decomposition (control dataset).

As in the main analysis (Figure 3), baseline correction (B) revealed changes in alpha, beta and theta power after stimulus presentation. FOOOF decomposition showed that changes in alpha and beta were periodic (C), whereas changes in theta power reflected task-related changes in aperiodic activity (D) (see also Figure S27, Figure S29).

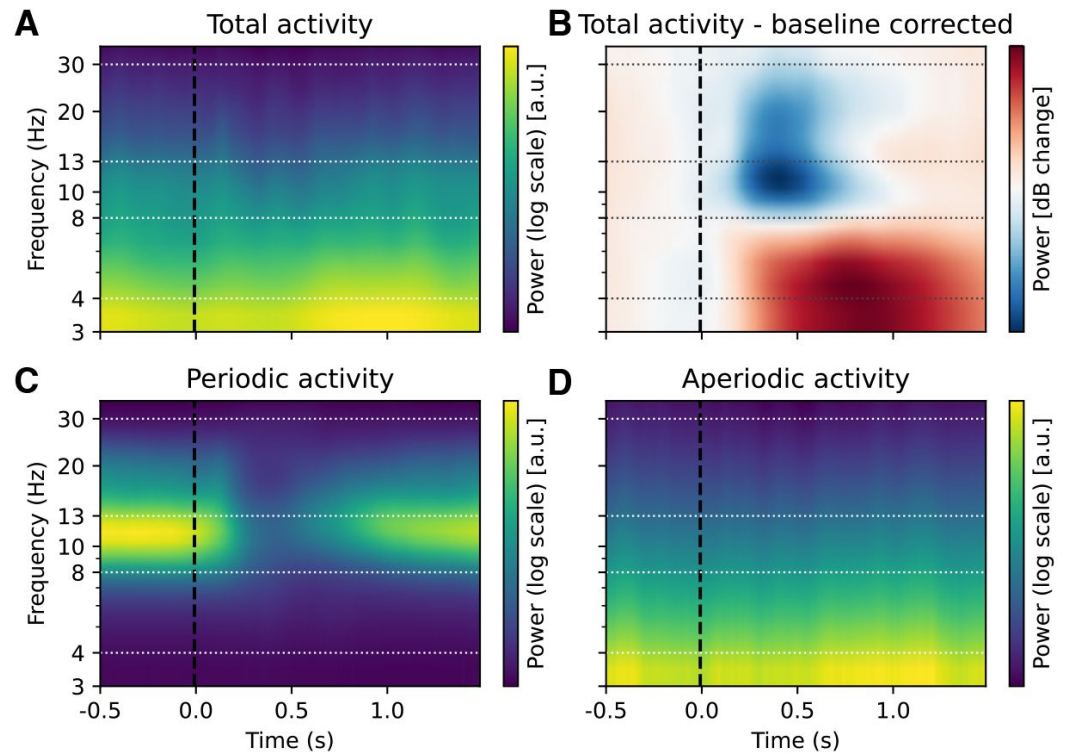


Figure S19.

Effect of baseline correction and FOOOF decomposition on time-frequency decomposition (control dataset).

Same as [Figure S18](#) but with a logarithmic y-axis.

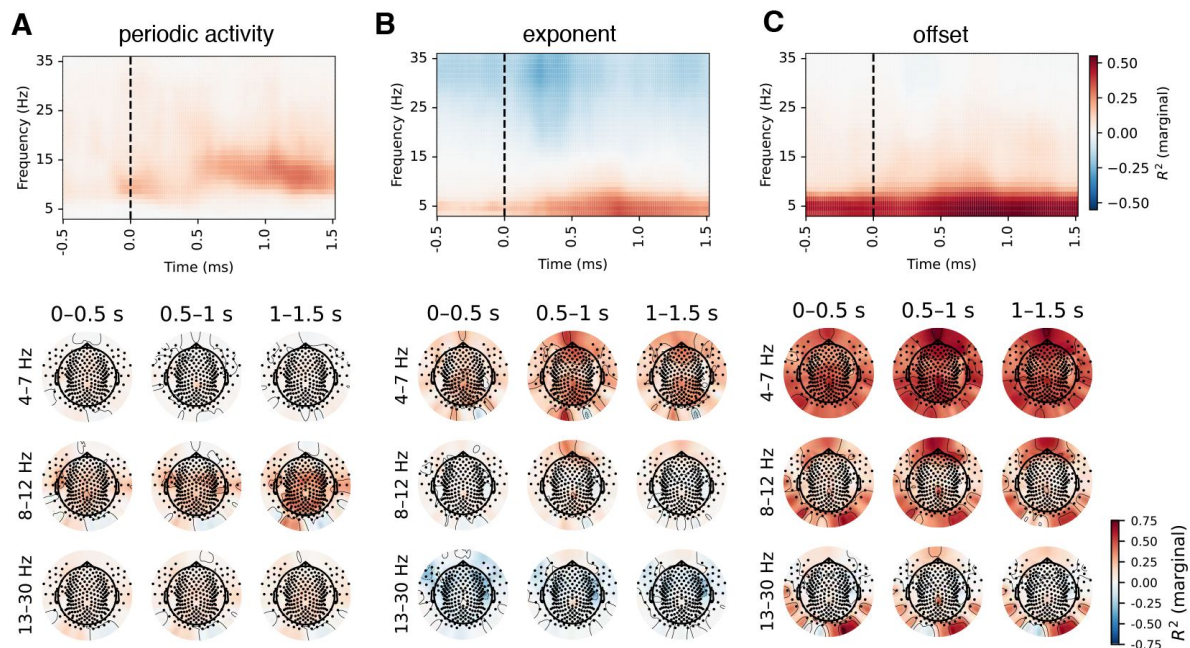


Figure S20.

Correlations between baseline-corrected time-frequency and FOOOF-decomposed EEG activity (control dataset).

The results were very similar to those obtained with the other two datasets (Figure 4, Figure S3, Figure S33).

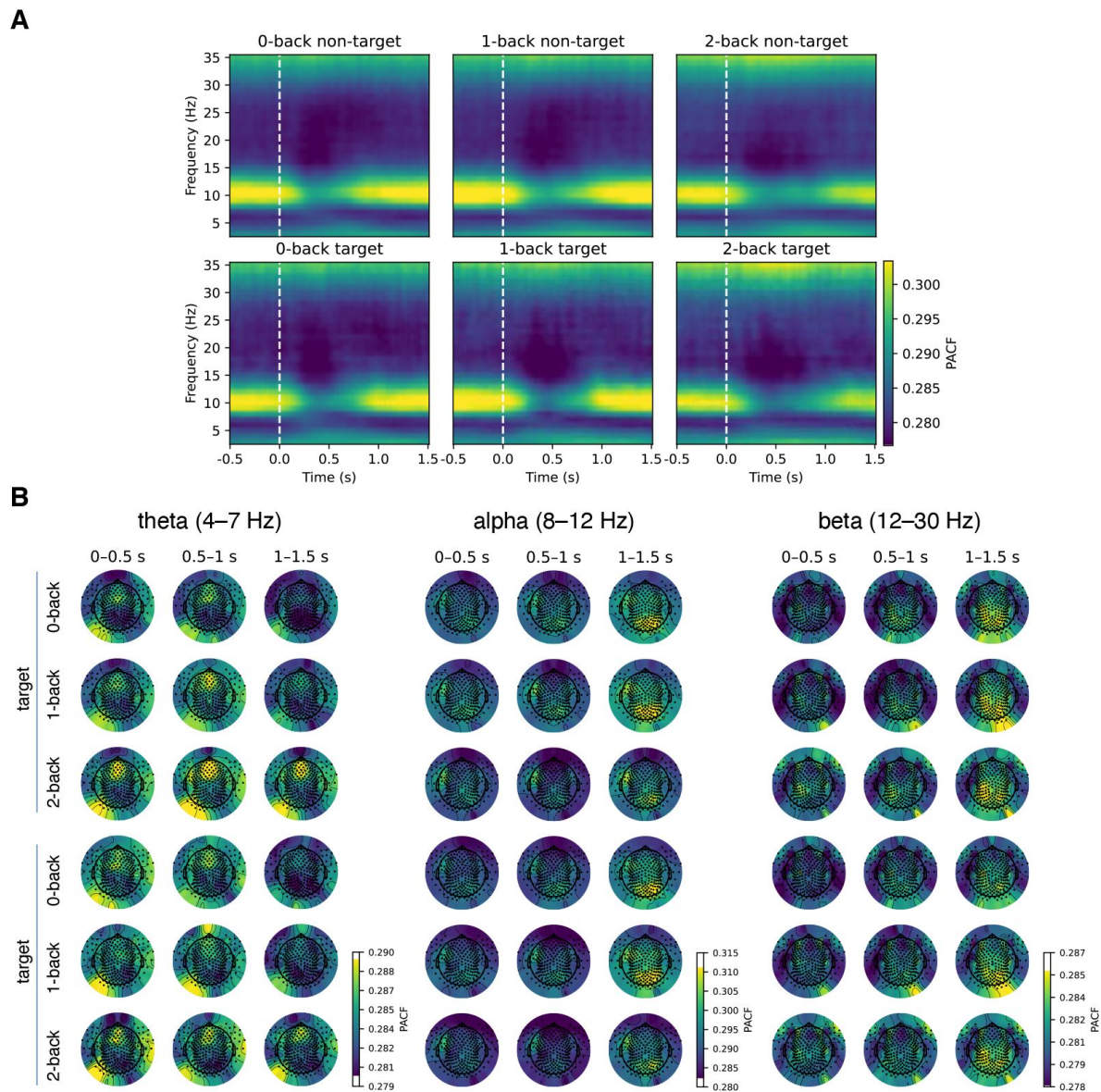


Figure S21.

Changes in phase autocorrelation function (index of rhythmicity) as a function of time (control dataset).

As with the other two datasets, the patterns of pACF were broadly consistent with the periodic activity estimated by FOOOF (see also [Figure S22](#)).

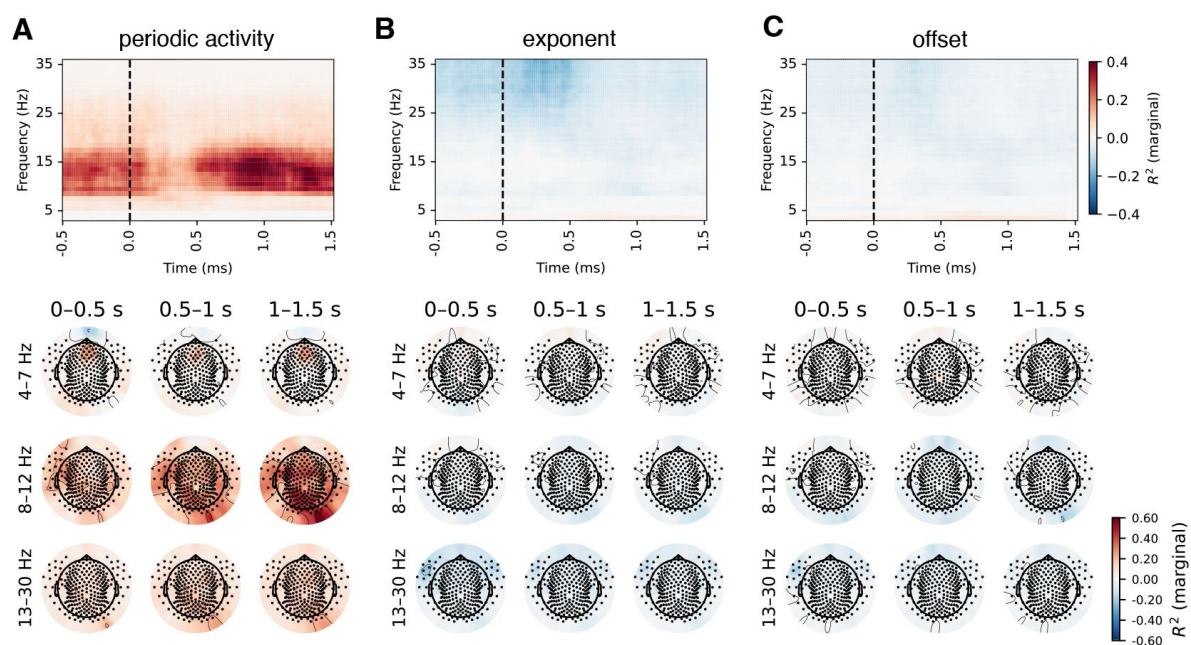


Figure S22.

Correlations between phase-autocorrelation function and FOOOF parameters (control dataset).

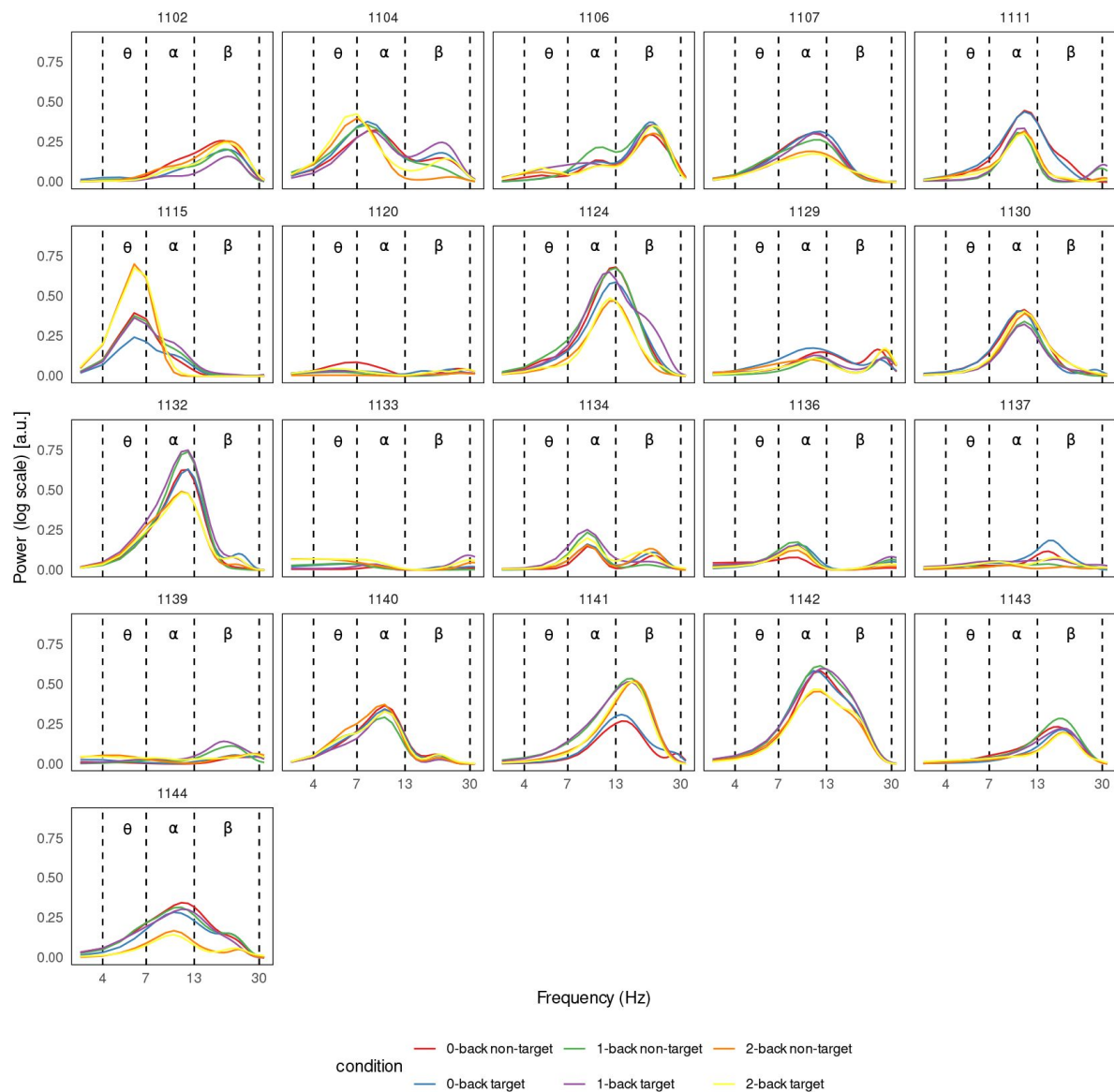
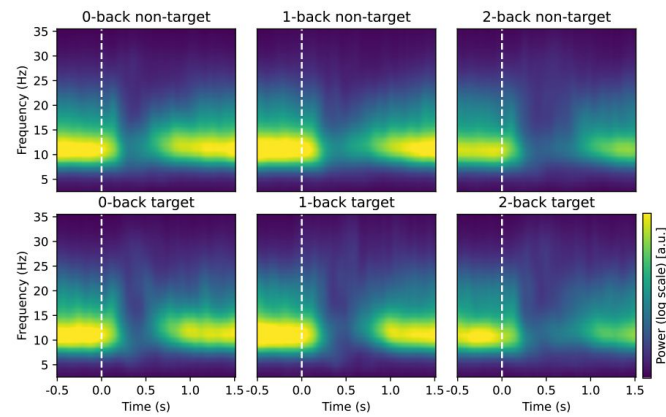


Figure S23.

Power spectra of periodic activity on the E15 channel for each participant, averaged over time (control dataset).

Similar to our main analysis ([Figure S6](#)), theta periodic activity was reliably present in only one participant (1115), with another participant (1104) showing a peak at 7-8 Hz. Both axes are displayed on a logarithmic scale.

A



B

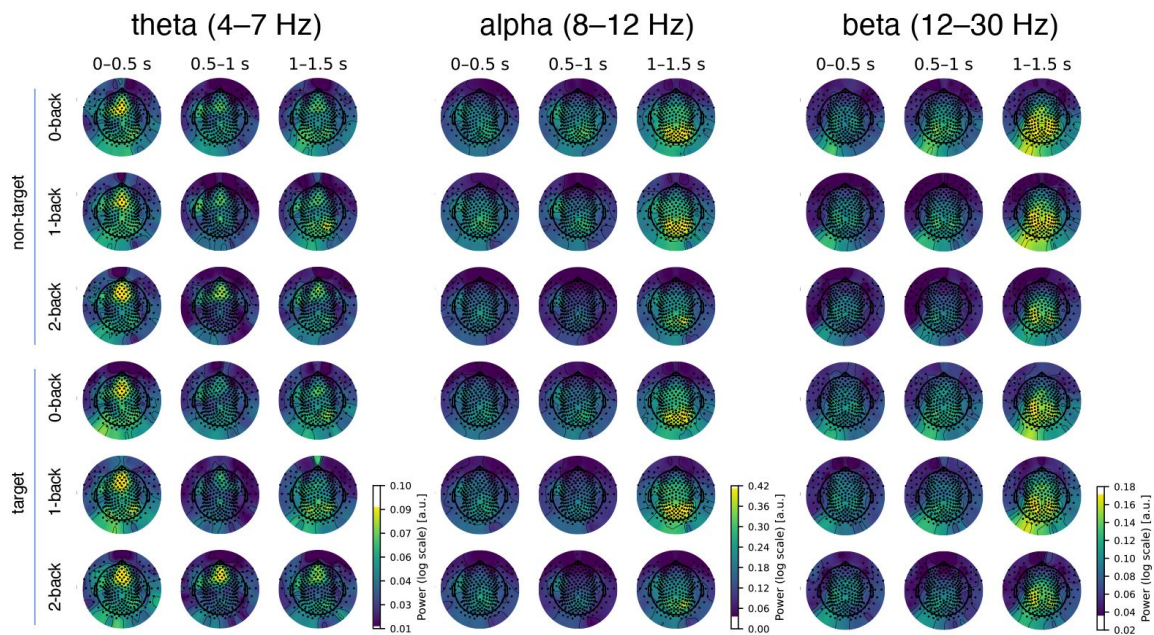


Figure S24.

Changes in periodic (oscillatory) activity as a function of time (control dataset).

(A) Time course of periodic activity averaged across all channels. The periodic activity was most pronounced in the alpha band, with a strong decrease after stimulus onset. (B) Alpha and beta activity was most prominent in frontal channels.

Figure S25.

FOOOF goodness-of-fit measures (control dataset).

Averaged across channels, subjects and conditions. Similar to our primary dataset ([Figure S9](#)), the decrease in goodness-of-fit around 0.5 s after stimulus onset, coincided with the decrease in periodic activity.

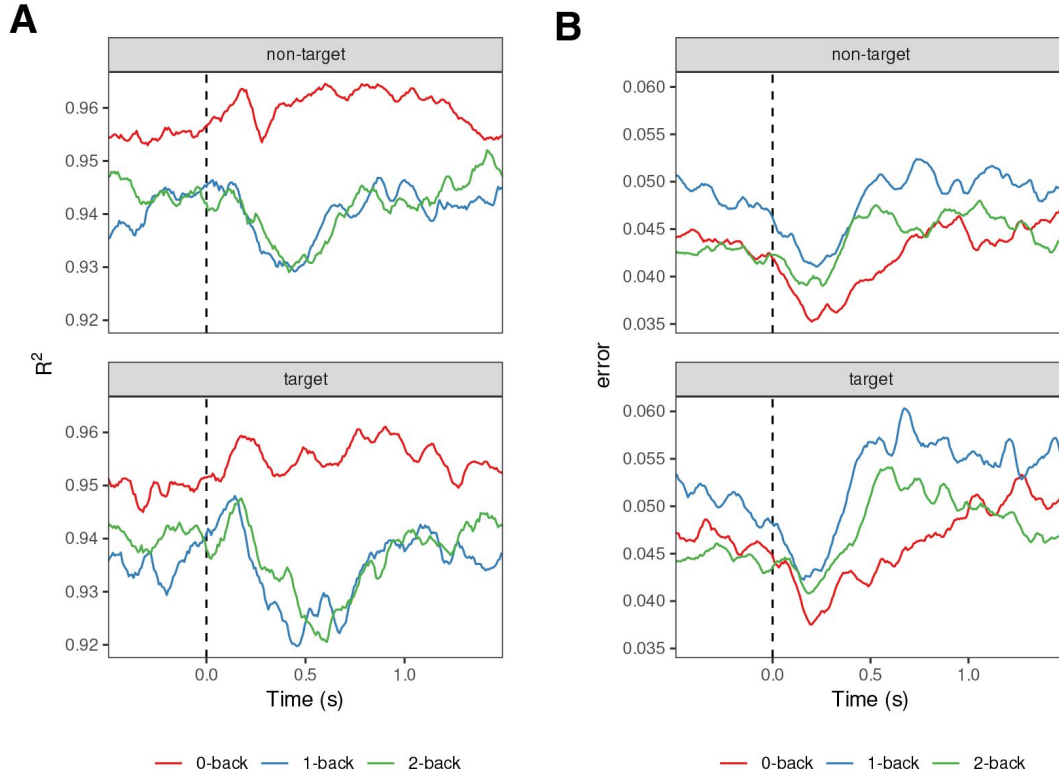


Figure S26.

Results of linear mixed model on periodic activity for comparison between conditions (control data).

There was a significant effect of load with reduced alpha and beta power at higher loads.

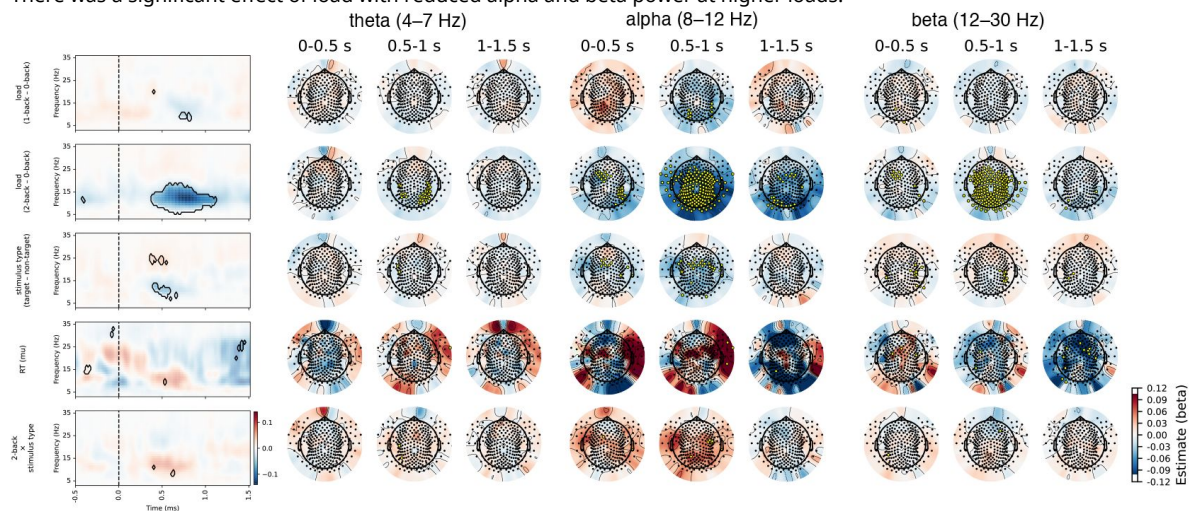


Figure S27.

Changes in aperiodic activity (exponent) as a function of time (control data).

Similar to the main analysis (Figure 8), we observed two peaks, early frontal and late parietal components. The vertical lines in panel A represent mean reaction time modes. The shaded areas represent 95% Cousineau-Morey within-subjects confidence intervals.

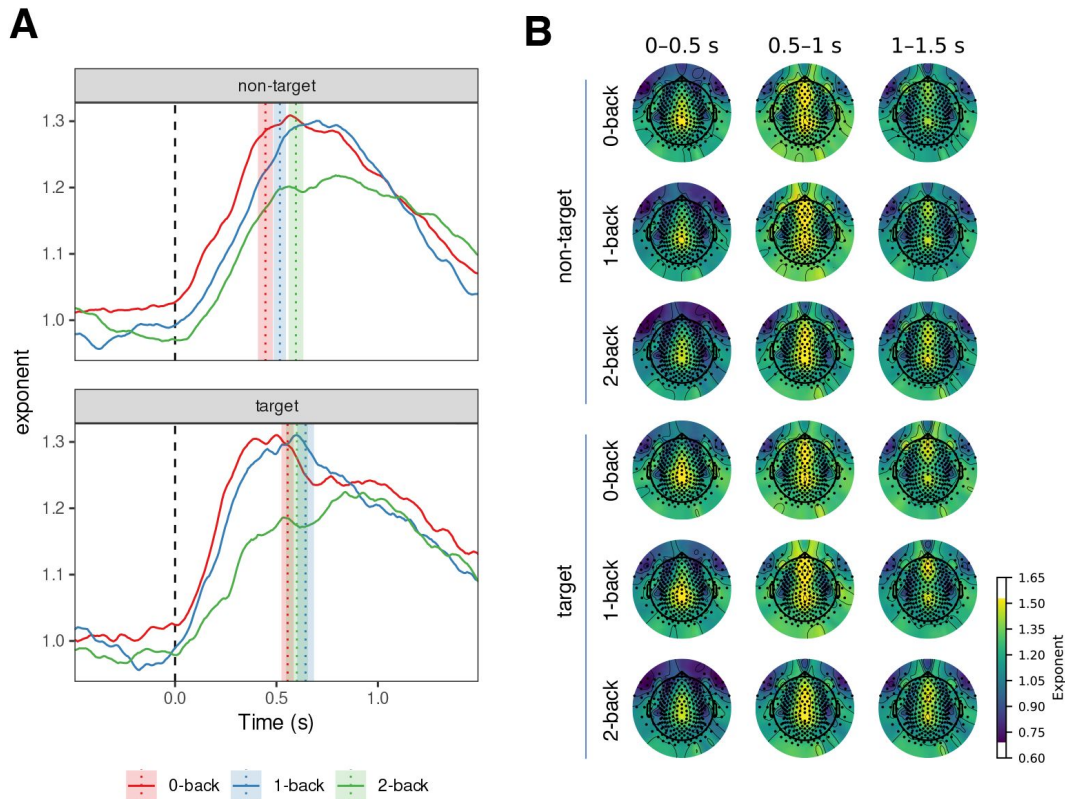


Figure S28.

Results of linear mixed model on aperiodic activity (exponent) for comparison between conditions (control data).

There was a significant effect of load, most evident immediately after stimulus presentation. There was also an association with reaction times, particularly after 1 second post-stimulus.

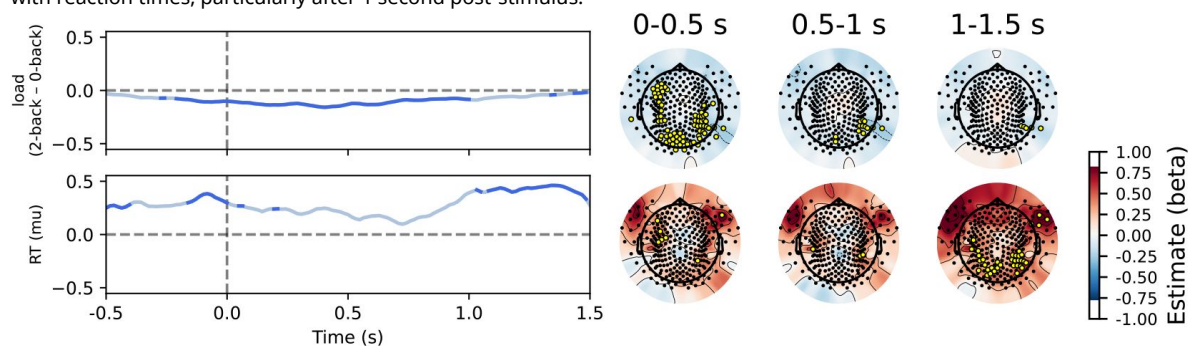


Figure S29.

Changes in aperiodic activity (offset) as a function of time (control data).

The vertical lines in panel A represent mean reaction time modes. The shaded areas represent 95% Cousineau-Morey within-subjects confidence intervals.

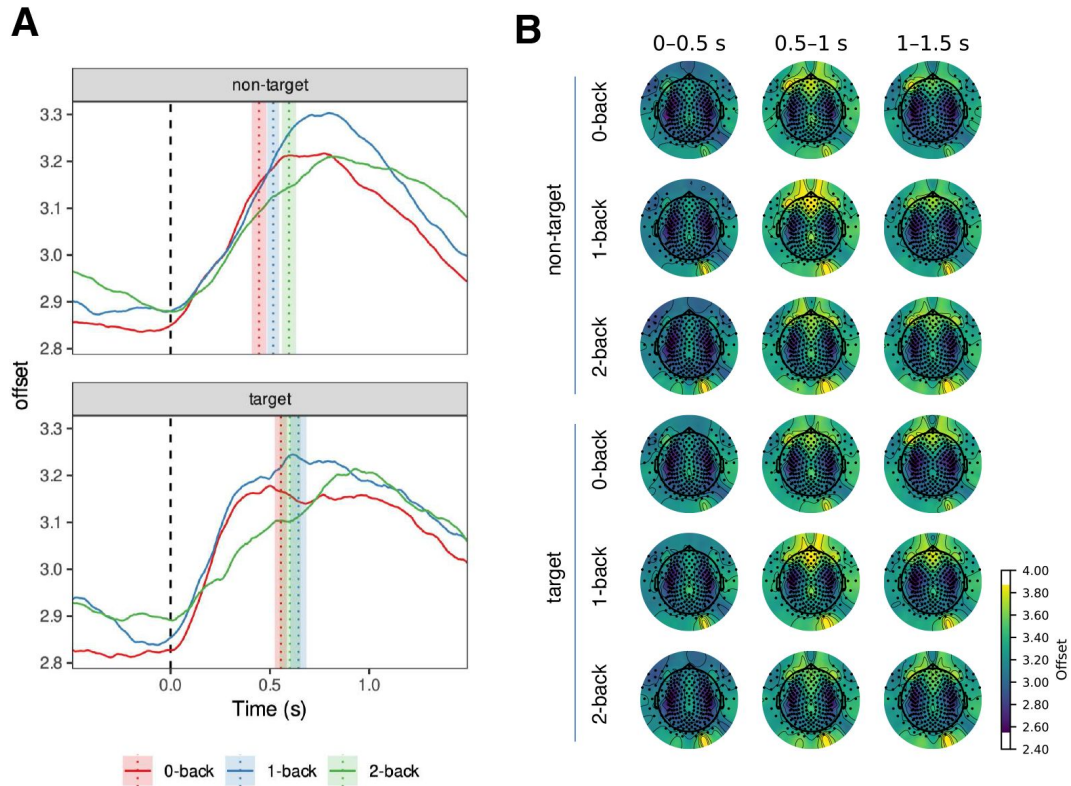
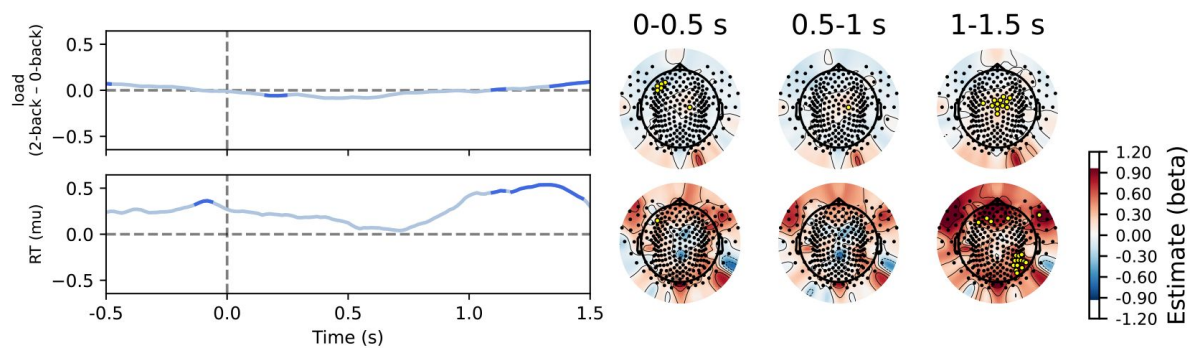


Figure S30.

Results of linear mixed model on aperiodic activity (slope) for comparison between conditions (control data).



Replication: item-recognition task

To test the generalisability of our findings across tasks, we repeated the analysis on an item-recognition task. Although we conducted statistical comparisons between conditions, the differences were negligible, so we have chosen not to report these results here.

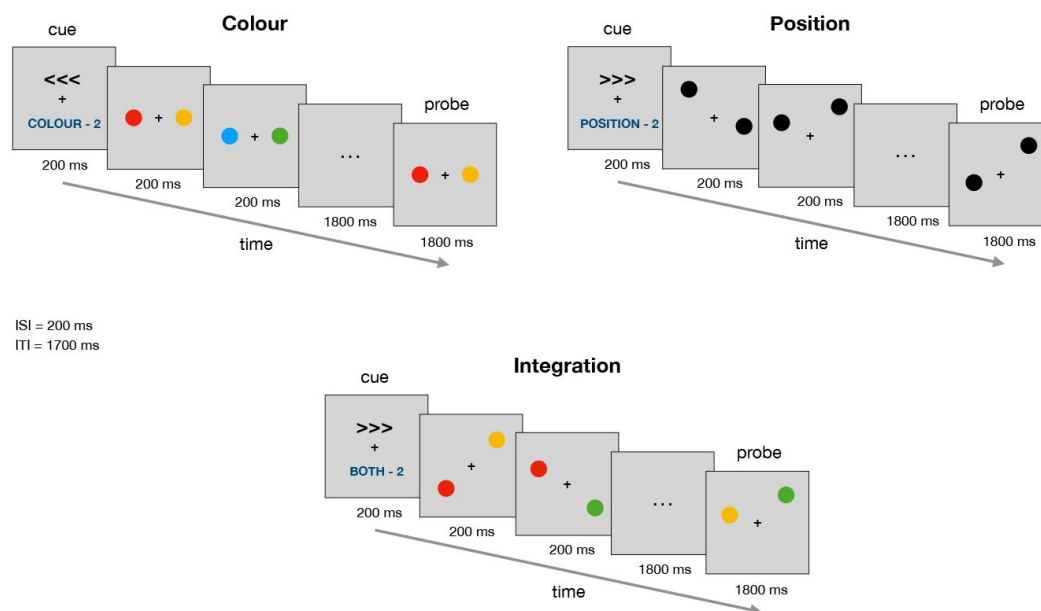


Figure S31.

Schematic representation of the item-recognition task.

In each trial, two or four target stimuli of a given condition were presented sequentially in each half of the visual field while the participant focused on one half. Participants held the relevant visual properties of the stimuli in working memory. This was followed by a probe in which two stimuli were presented, one in each half of the visual field, and participants responded whether the stimulus in the attended half matched one of the previous targets. Each trial began with a fixation cross in the centre of the screen on which participants maintained their gaze. Each trial began with instructions indicating the area of focus and the number of stimuli that would follow. After a blank interval, participants viewed the target stimuli, followed by a maintenance interval and a probe. During this interval, participants indicated whether the probe corresponded to one of the previous targets. Trials were counterbalanced to ensure that an equal number of presentations occurred in both visual fields for each of the conditions and working memory loads. The number of stimuli presented sequentially (i.e., working memory load) was randomised to prevent participants from making predictions.

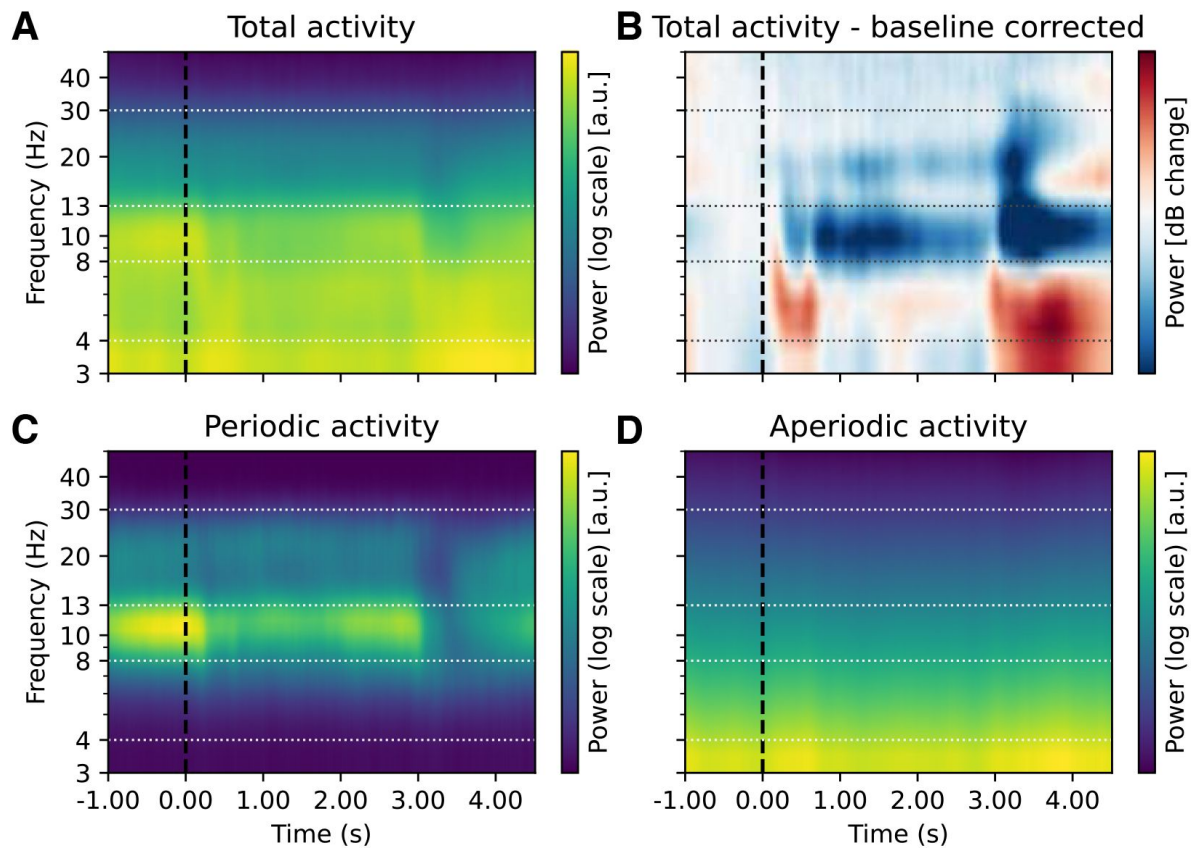


Figure S32.

The effect of baseline correction and FOOOF decomposition on time-frequency decomposition in the item-recognition task.

Same as [Figure 10](#) but with a logarithmic y-axis.

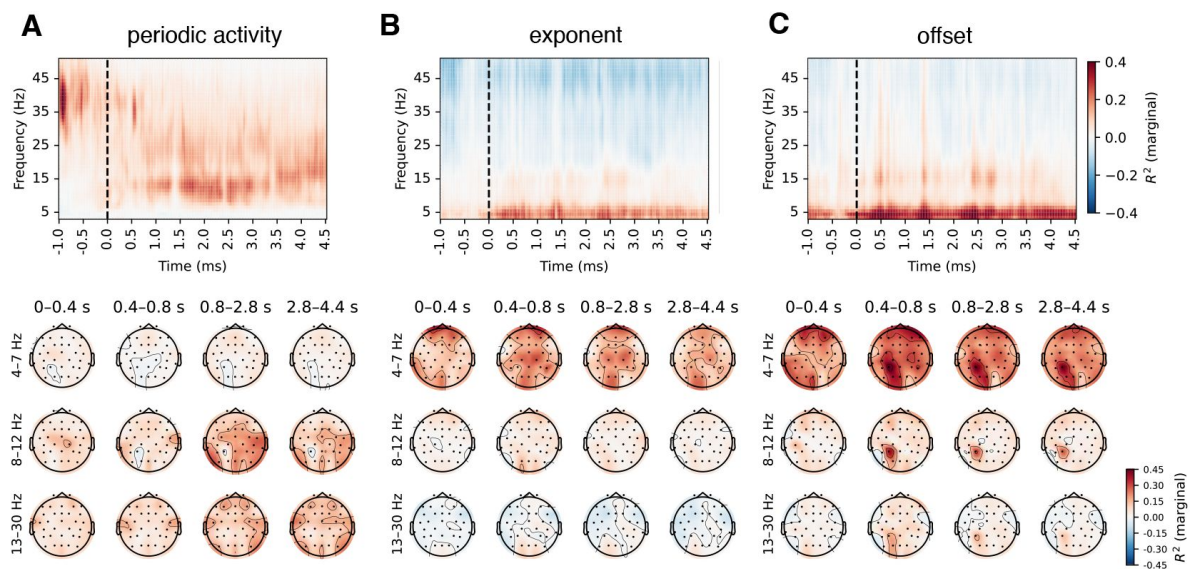


Figure S33.

Correlations between baseline-corrected time-frequency and FOOOF-decomposed EEG activity (item recognition task).

The results were very similar to those obtained with the other two datasets ([Figure 4](#), [Figure S3](#), [Figure S20](#)), but in the item recognition task we also observed smaller correlations with aperiodic parameters in the alpha range.

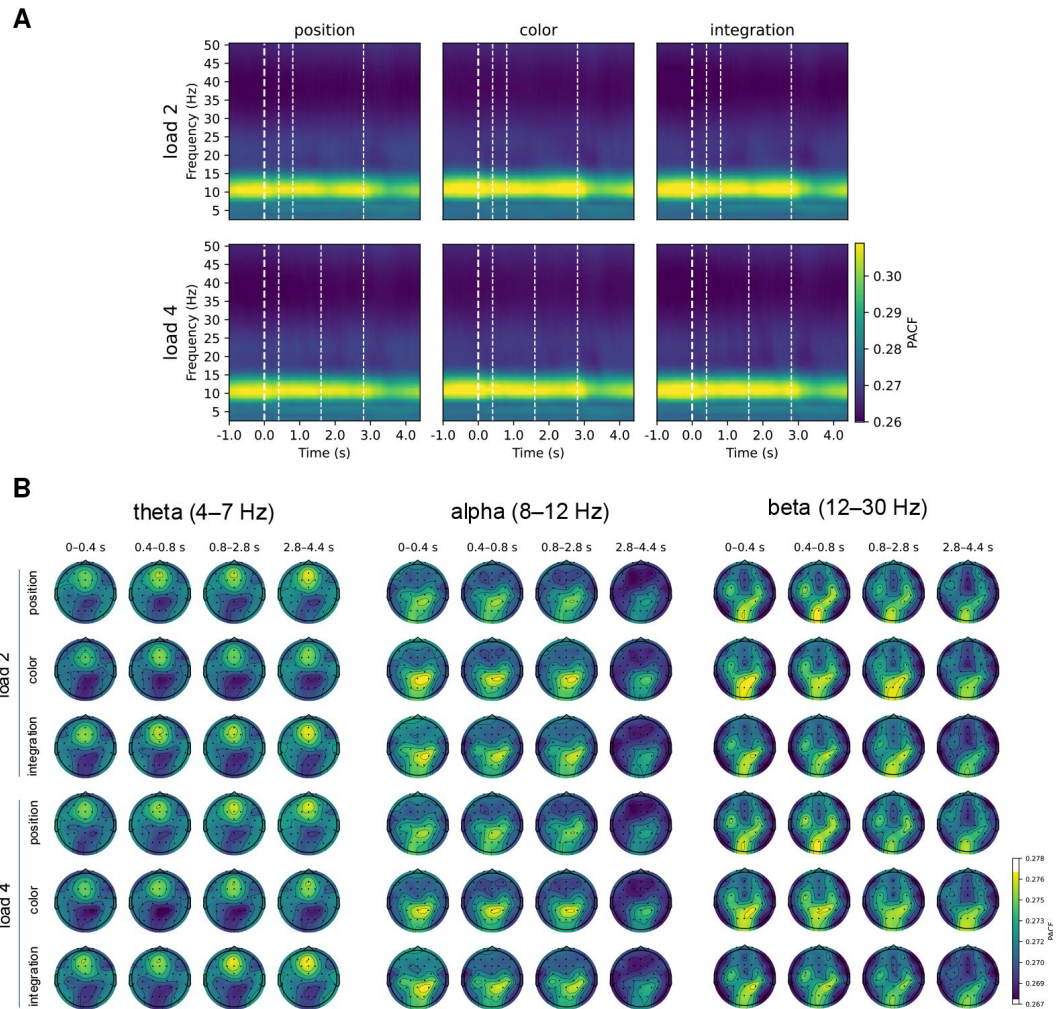


Figure S34.

Changes in phase autocorrelation function (index of rhythmicity) as a function of time (item recognition task).

As with the other two datasets, the patterns of pACF were broadly consistent with the periodic activity estimated by FOOOF (see also [Figure S35](#)).

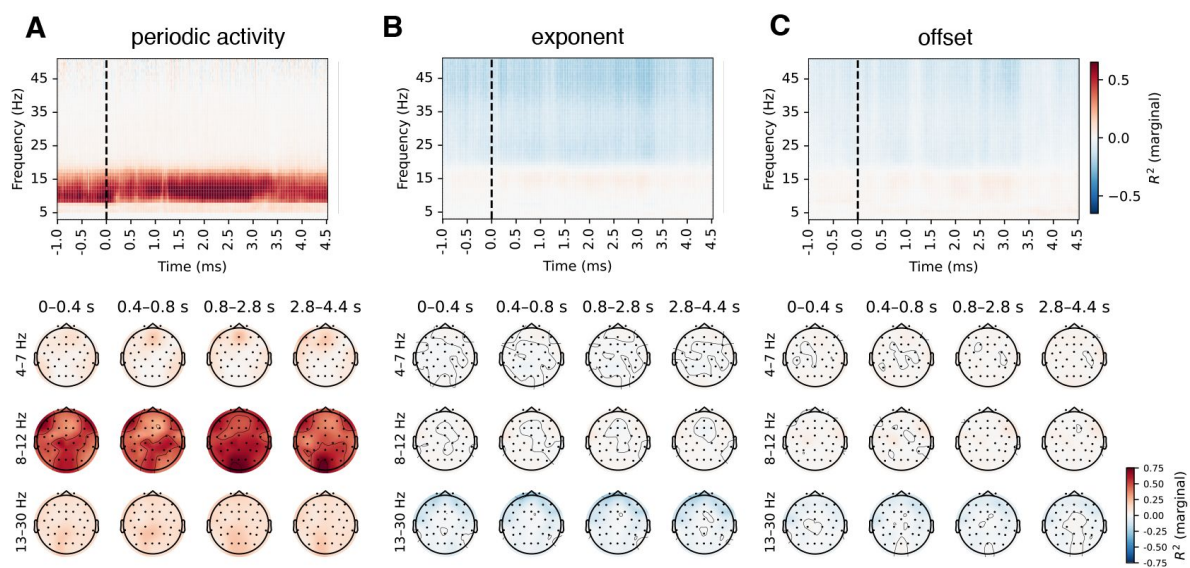


Figure S35.

Correlations between phase-autocorrelation function and FOOOF parameters (item recognition task).

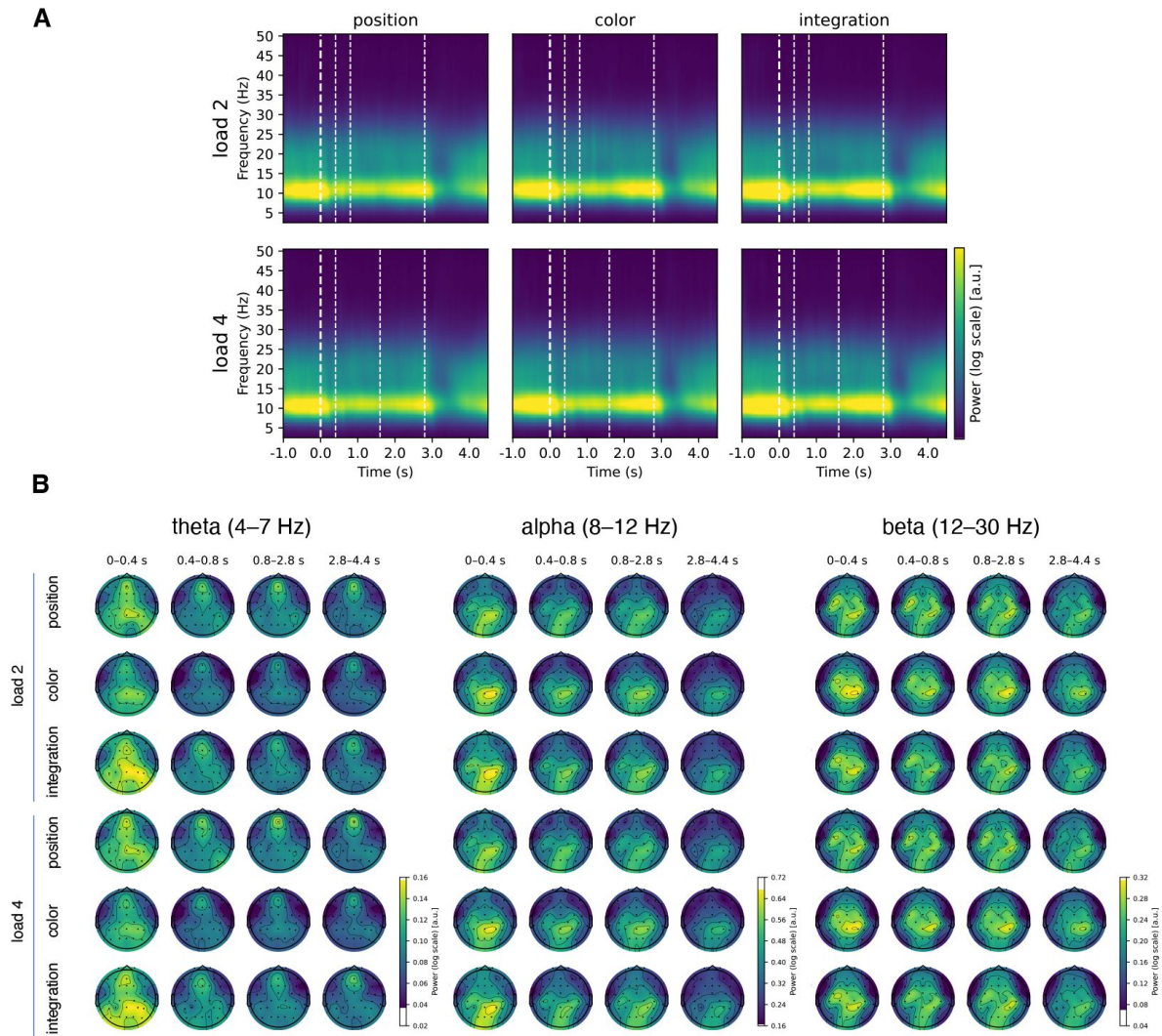


Figure S36.

Periodic activity on item recognition task.

Vertical lines represent: instruction presentation (0 s), stimuli presentation (0.4 s), end of encoding (0.8 s in load 2; 1.6 s in load 4), end of retention period (2.8 s). Prominent task-related modulations were observed in alpha and beta range, with a decrease during stimulus presentation, a relative increase during the retention phase and a decrease during probe presentation.

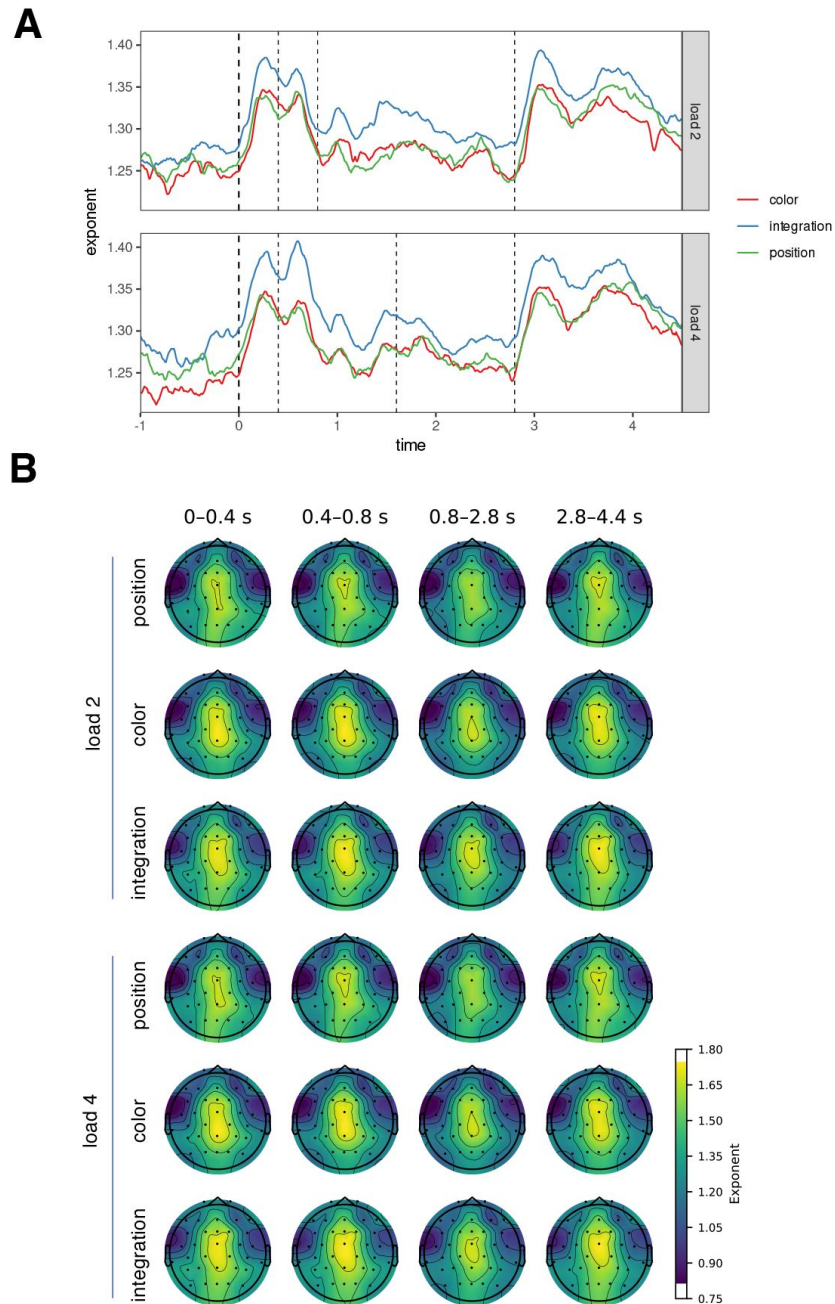


Figure S37.

Aperiodic activity (exponent) in the item-recognition task.

As with periodic activity, we also observed task-related changes in aperiodic activity. Specifically, the aperiodic slope increased during instruction, stimulus presentation and probe presentation compared to the retention period. The topographies were similar to those of the n-back task, with activity spanning frontal and parietal channels. However, the two components could not be distinguished (see [Figure S38](#)).

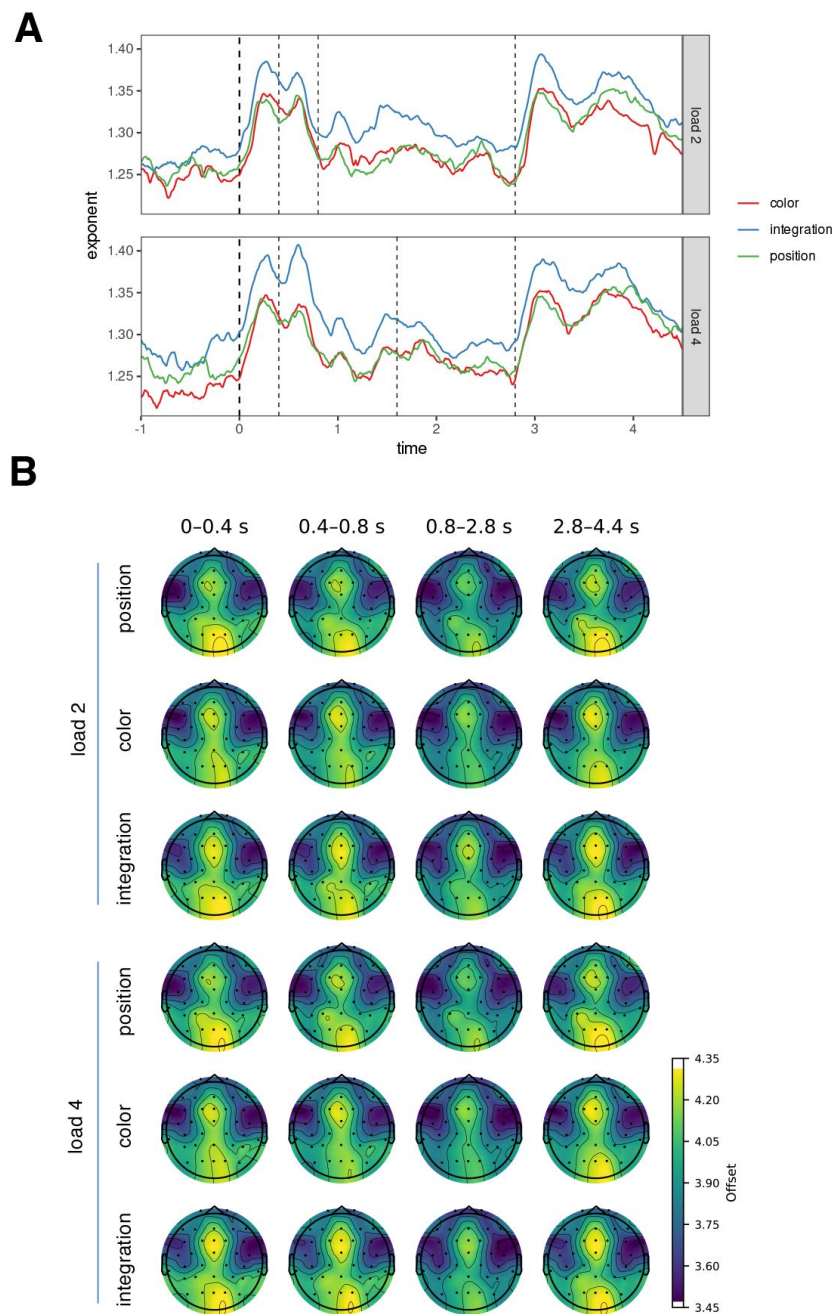


Figure S38.

Aperiodic activity (offset) in the item-recognition task.

The outcomes were comparable to those of exponent ([Figure S37](#)), however, the frontal and parietal/occipital components were discernible.

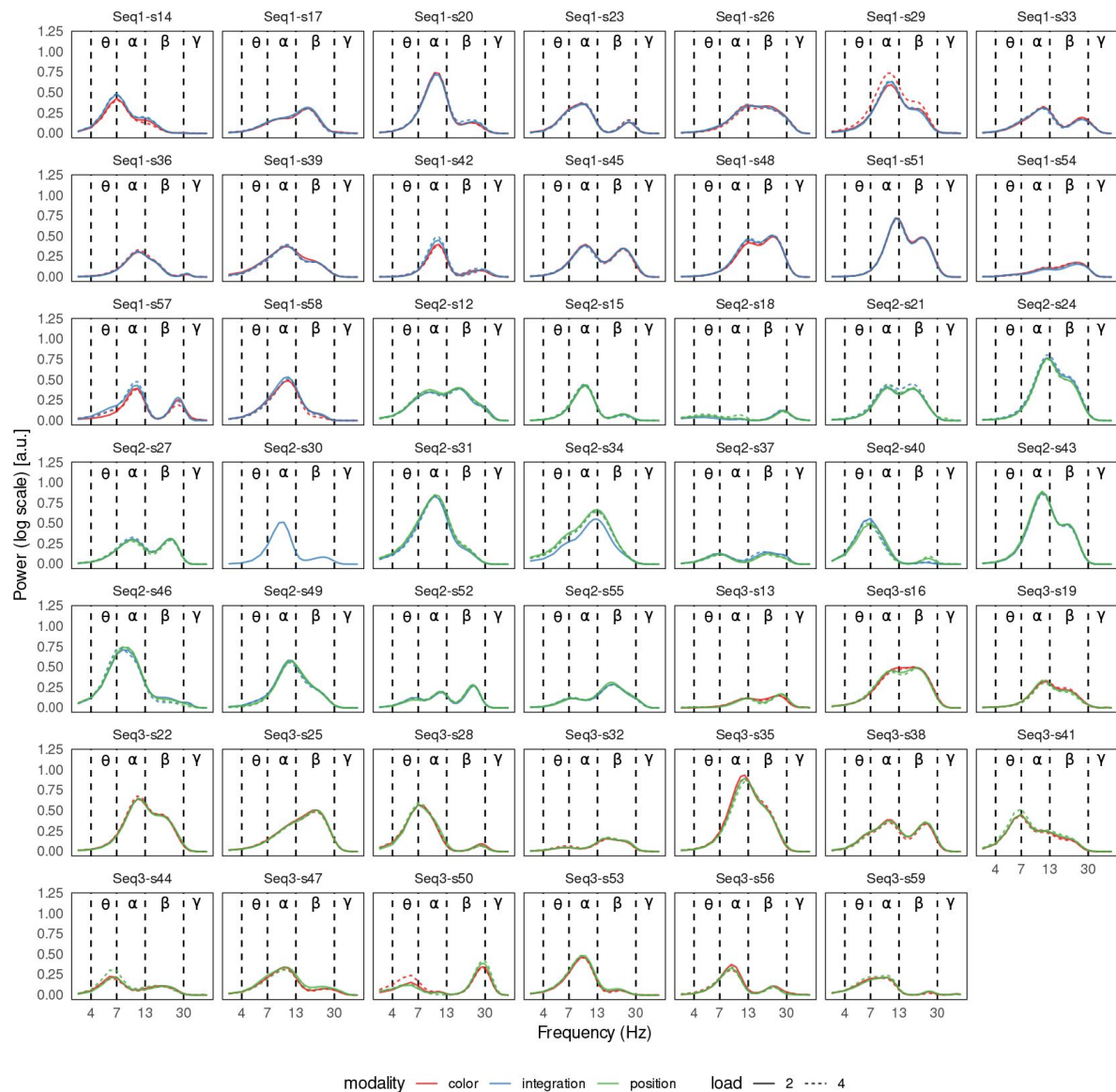


Figure S39.

The power spectra of periodic activity on the Fz channel for each participant in the item-recognition task, averaged over time.

While the group average did not show pronounced theta activity (Figure S36, Figure S40), several participants showed periodic activity in the theta range (Seq2-s18, Seq2-s34, Seq2-s52, Seq3-s22, Seq3-s38, Seq3-s41, Seq3-s44, Seq3-s50). Furthermore, several peaks were observed between 7 and 8 Hz (Seq1-s14, Seq1-s23, Seq2-s37, Seq2-s40, Seq2-s55, Seq3-s28, Seq3-s59). Both axes are displayed on a logarithmic scale.

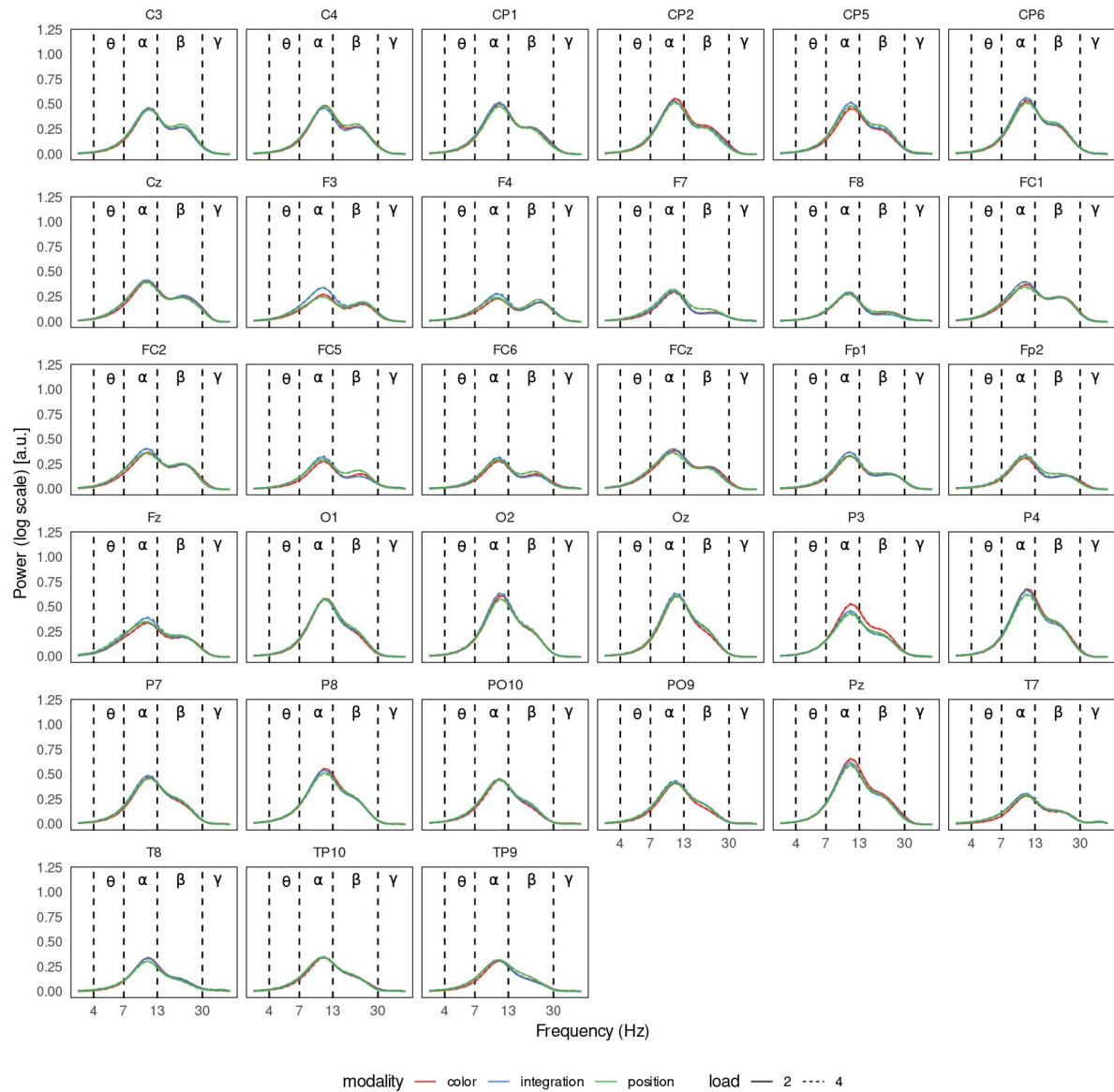


Figure S40.

Group average power spectra of periodic activity in the item-recognition task, averaged over time.

As in [Figure S36](#), only alpha and beta peaks were observed in the group average.

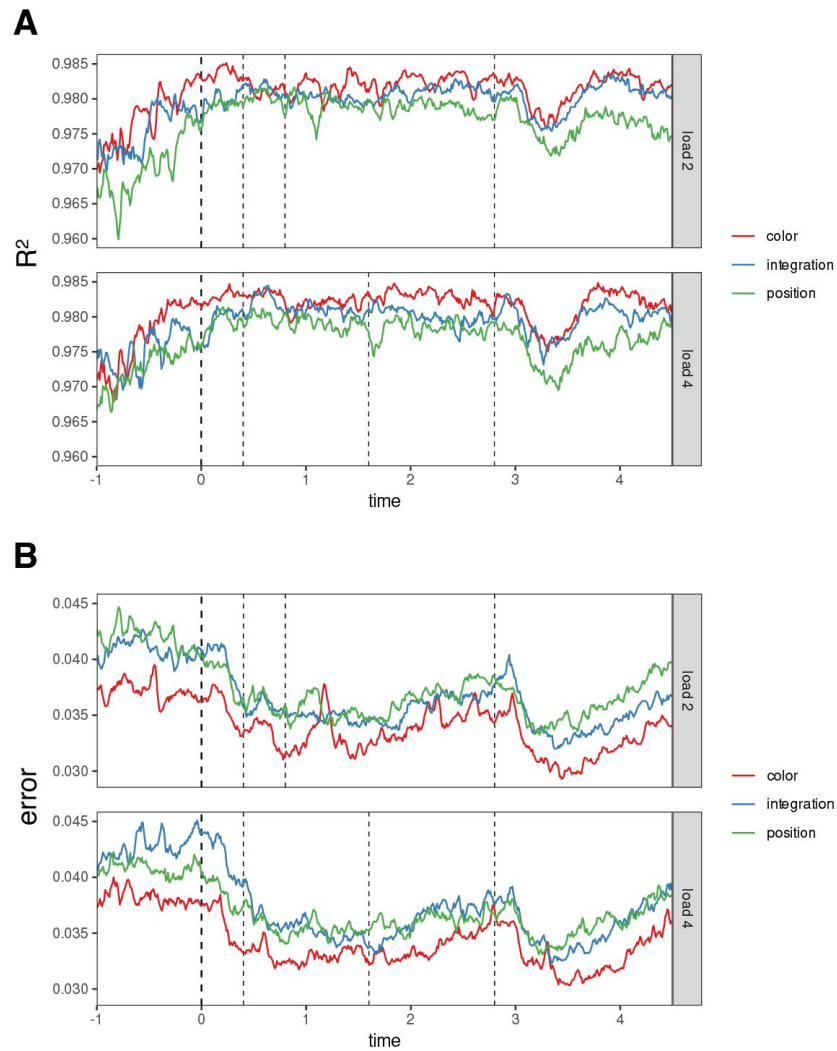


Figure S41.

FOOOF goodness-of-fit measures for the item recognition task, averaged across channels.

R^2 was above 0.97 and model error was below 0.045 throughout the task. There was a small decrease in goodness-of-fit around 3 seconds after stimulus onset, coinciding with a decrease in periodic activity, similar to the other two datasets (Figure S9, Figure S25).

Simulation

To assess whether the FOOOF decomposition can reliably separate low-frequency periodic activity from the aperiodic background, we ran a simulation in which the periodic parameters were systematically varied. Specifically, we used the `sim_spectrum` function from the Python package `specparam` (<https://github.com/fooof-tools/fooof>) to generate 100 power spectrum instances in the range 1–30 Hz, with a frequency resolution of 1 Hz to match our experimental analyses. The offset was set to 1 and the exponent to 1.5. A single periodic component was added to each instance using the following parameter variations:

- central frequency: 3, 4, 5, 6, 7 Hz,
- periodic power: 0.2, 0.4, 0.8, 1.6,
- bandwidth: 0.5, 1, 2, 4,

Gaussian noise with a mean of 0 and a standard deviation of 0.1 was added to each power spectrum. Fitting was performed using the ‘fixed’ method, estimating only the aperiodic offset and exponent parameters (without the knee parameter). We performed two fitting procedures: first, across the entire frequency range (1–30 Hz) (panel **A** in the figures below), and second, within a limited range of 3–30 Hz (panel **B**), consistent with our experimental EEG data analysis, where FOOOF parameters were estimated only for frequencies ≥ 3 Hz. Simulated data are shown in **Figure S42**.

Estimates of the exponent and offset parameters were generally accurate at low bandwidths and low periodic power levels (**Figure S43**, **Figure S44**). However, fitting within the restricted range (≥ 3 Hz, panel **B** in all figures) resulted in reduced accuracy, especially when the periodic component had a lower central frequency. In cases where the periodic component had large bandwidths and/or low power, it was often conflated with the aperiodic component, inflating the aperiodic parameter estimates. This effect was particularly pronounced when only frequencies above 3 Hz were fitted, as indicated by a high number of unestimable periodic components (**Figure S45**, **Figure S46**, **Figure S47**), especially when the simulated central frequency was low (i.e. 3 Hz). Inflated aperiodic parameter estimates were associated with increased central frequency estimates and decreased periodic power estimates. Central frequency and bandwidth estimates showed high variability at low power levels. Conversely, periodic power estimates were more accurate at low simulated power, but less so when both bandwidth and power were high.

Notably, the model fit indices (R^2 and model error) were generally good, even in cases where periodic components were poorly estimated or not detectable at all (**Figure S48**, **Figure S49**). This suggests that model fit metrics can be misleading – high fit does not necessarily imply correct parameter estimation.

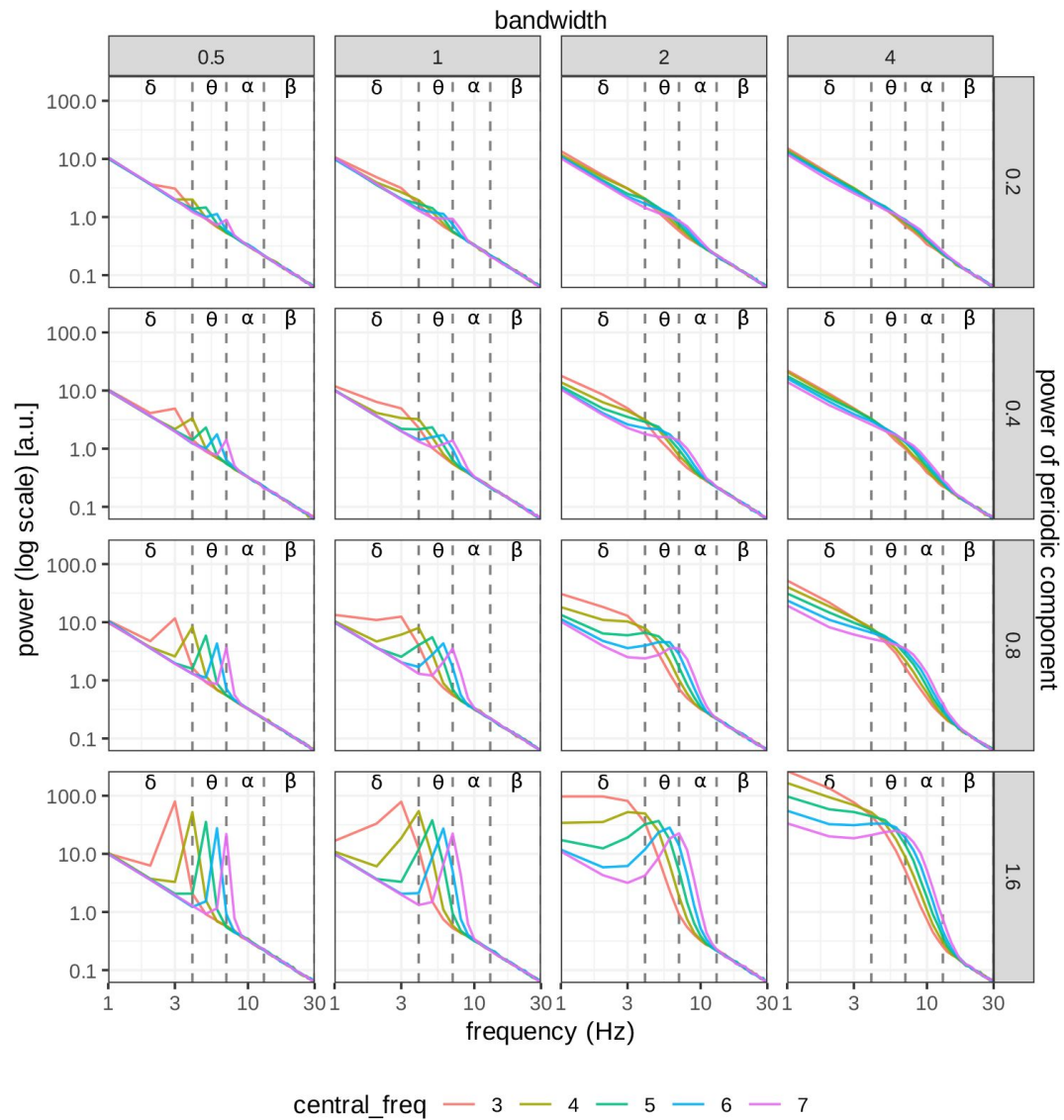


Figure S42.

Simulated power spectra in log-log space.

We added a single periodic component to each power spectrum, with the exponent fixed at 1.5 and the offset at 1.

Figure S43.

Exponent estimates.

Red lines indicate ground truth. Panel A covers 1–30 Hz fits, while panel B covers 3–30 Hz fits. Exponent estimates were inflated at larger bandwidths and higher periodic power, especially when the periodic component's central frequency was low. This suggests a mixing of periodic and aperiodic components.

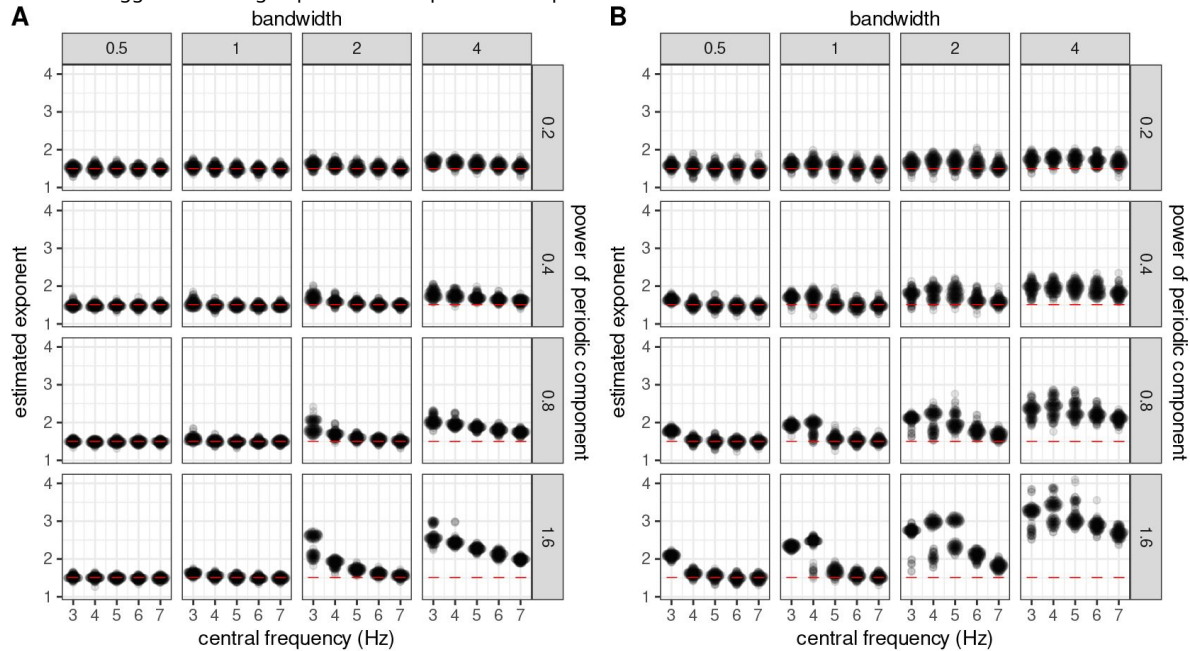


Figure S44.

Offset estimates.

As in [Figure S43](#), estimates tended to be inflated when the periodic component was difficult to detect.

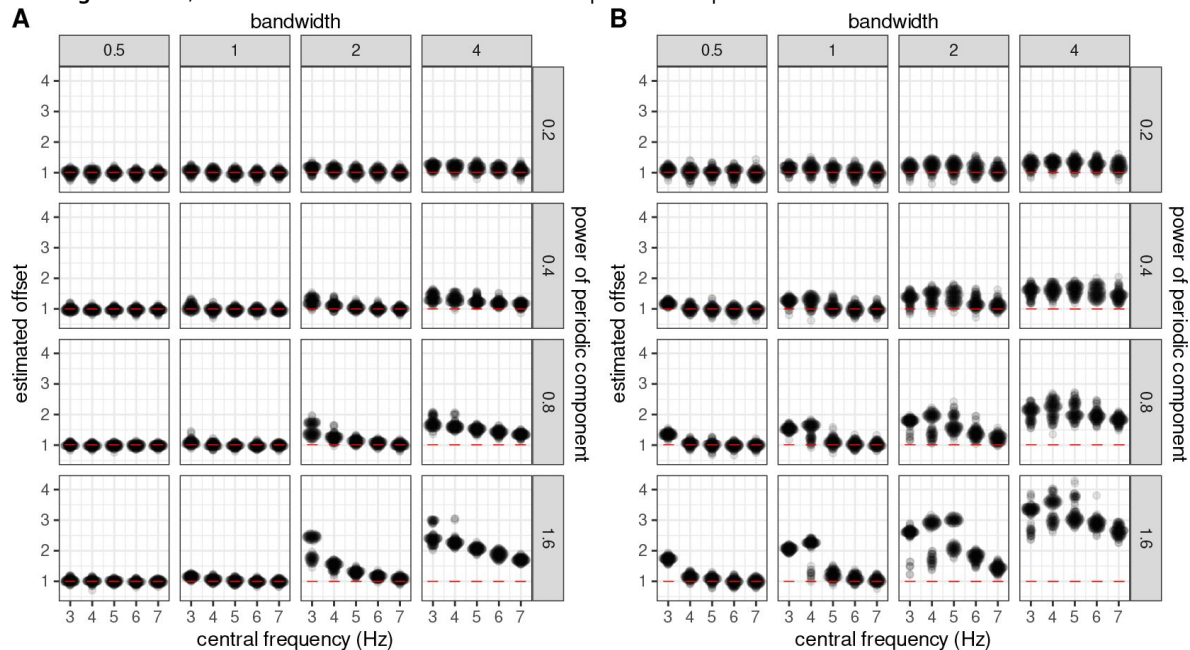


Figure S45.

Bandwidth estimates.

Red lines show ground truth, while red numbers mark how often the periodic component was undetectable (out of 100 cases). Low frequency, low power, and the restricted 3–30 Hz fitting range often made periodic features hard to detect.

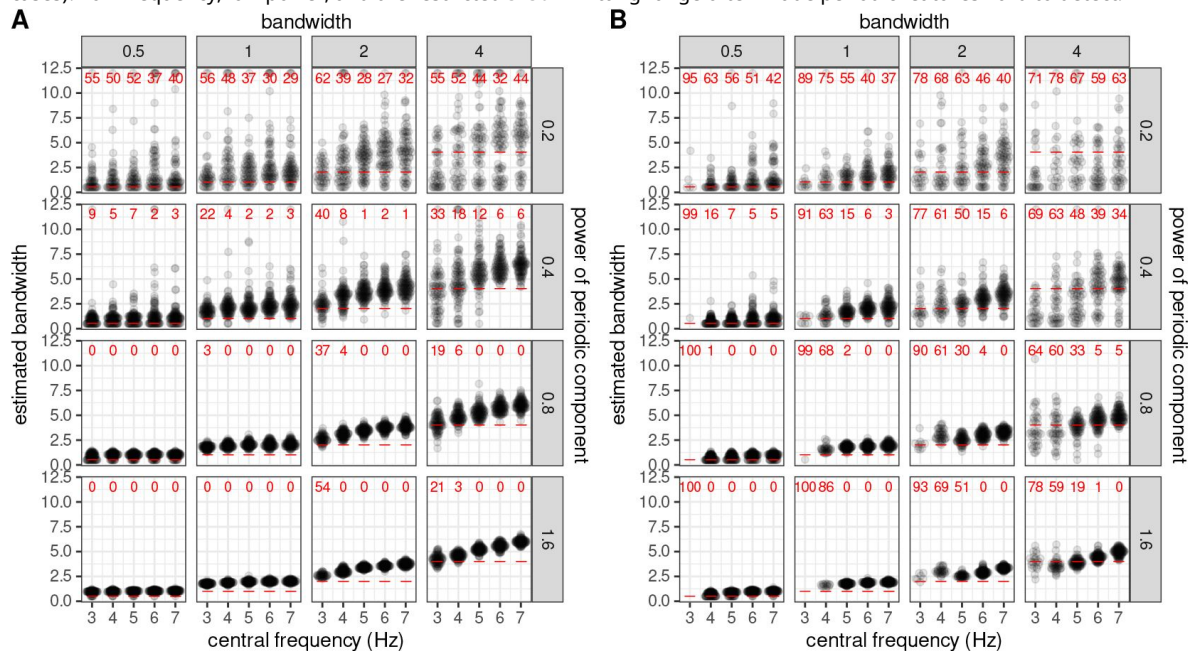


Figure S46.

Central frequency estimates.

Central frequency estimates were inflated for large bandwidths and high power, particularly when fitting was restricted to ≥ 3 Hz (panel B). Variability of estimates also increased at low simulated power. The red numbers indicate how many times out of 100 the periodic component was undetectable.

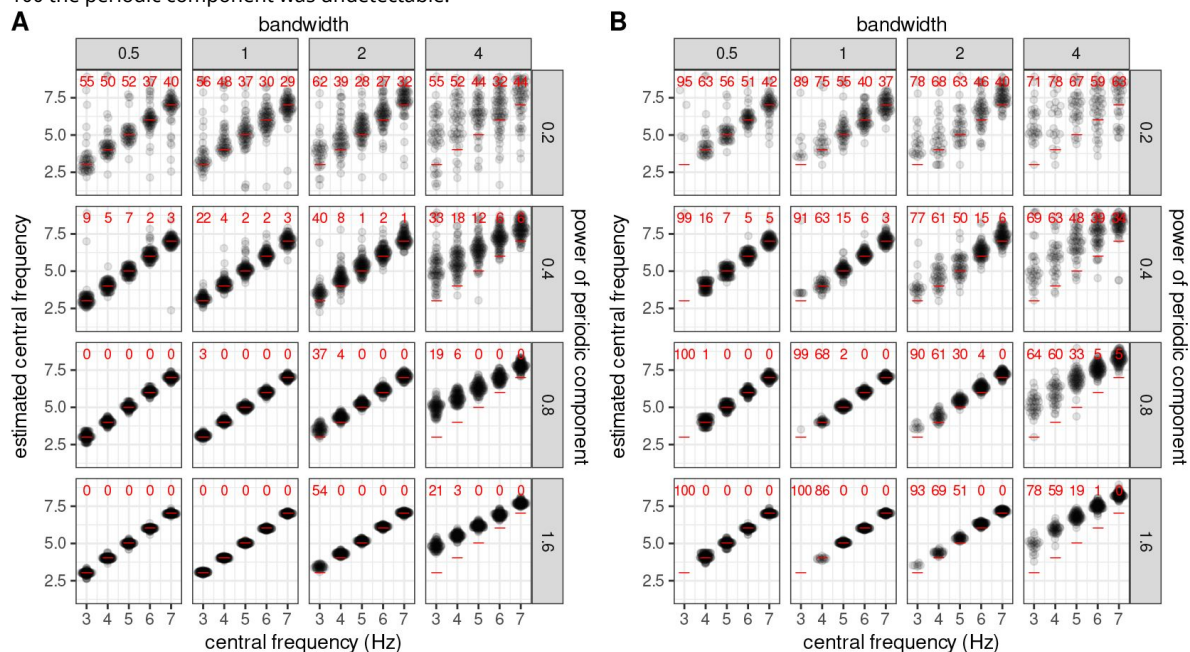


Figure S47.

Power estimates.

In contrast to bandwidth and central frequency estimates, power estimates remained largely accurate at low simulated power, regardless of bandwidth. However, at high simulated power and large bandwidths power estimates were deflated. The red numbers indicate how many times out of 100 the periodic component was undetectable.

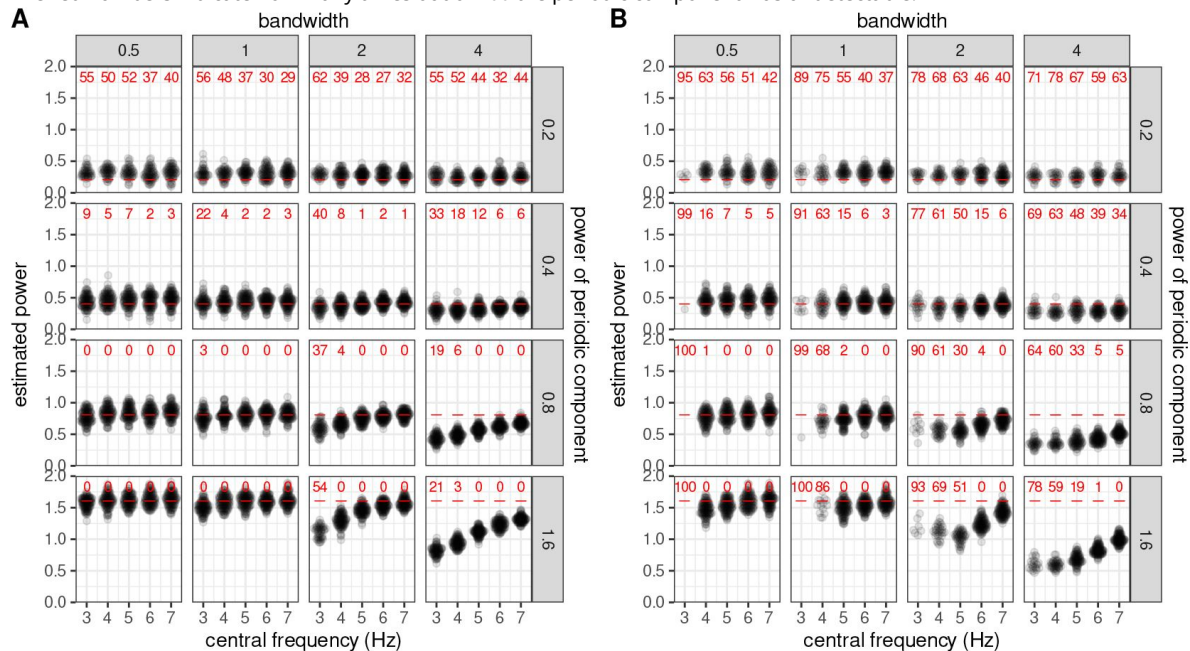
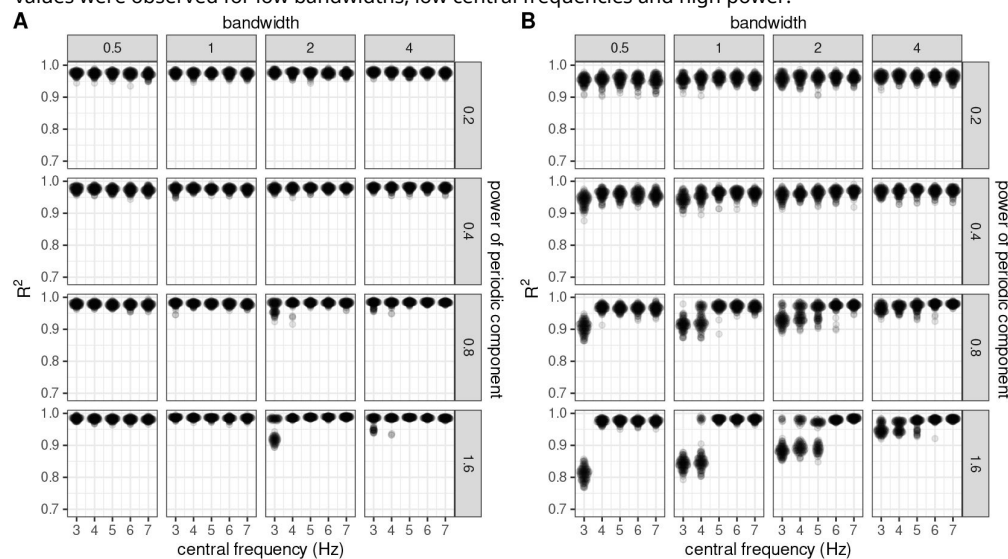


Figure S48.

R^2 , index of model fit.

R^2 values were very high in most cases, even when parameter estimates were largely inaccurate (see previous figures). The lowest R^2 values were observed for low bandwidths, low central frequencies and high power.



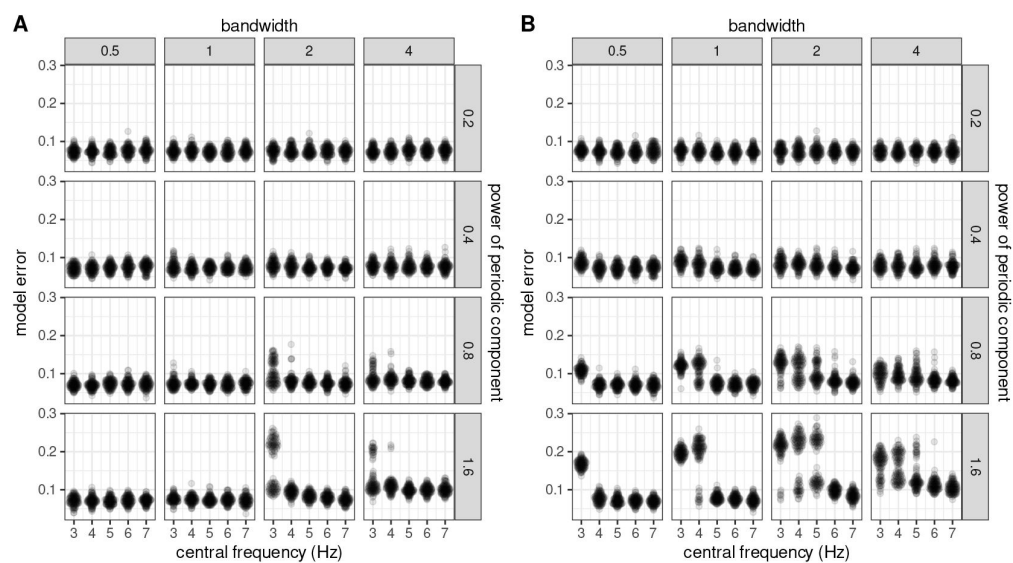


Figure S49.

Model errors.

Similar to R^2 , higher model errors were associated with low central frequencies and large powers, and the effect was more pronounced when only frequencies ≥ 3 Hz were fitted.

References

- Ahmad J., Ellis C., Leech R., Voytek B., Garces P., Jones E., Buitelaar J., Loth E., dos Santos F. P., Amil A. F., Verschure P. F. M. J., Murphy D., McAlonan G. (2022) **From mechanisms to markers: novel noninvasive EEG proxy markers of the neural excitation and inhibition system in humans** *Translational Psychiatry* **12**:1–12 <https://doi.org/10.1038/s41398-022-02218-z> | Google Scholar
- Akbadian F., Rossi C., Costers L., D’hooghe M. B., D’haeseleer M., Nagels G., Van Schependom J. (2023a) **The spectral slope as a marker of excitation/inhibition ratio and cognitive functioning in multiple sclerosis** *Human Brain Mapping* **44**:5784–5794 <https://doi.org/10.1002/hbm.26476> | Google Scholar
- Akbadian F., Rossi C., Costers L., D’hooghe M. B., D’haeseleer M., Nagels G., Van Schependom J. (2023b) **Stimulus-related modulation in the 1/f spectral slope suggests an impaired inhibition of irrelevant information in people with multiple sclerosis** *bioRxiv* <http://biorxiv.org/lookup/doi/10.1101/2023.12.28.573572> | Google Scholar
- Ameen M. S., Jacobs J., Schabus M., Hoedlmoser K., Donoghue T. (2024) **The Temporal Dynamics of Aperiodic Neural Activity Track Changes in Sleep Architecture** <http://biorxiv.org/lookup/doi/10.1101/2024.01.25.577204>
- Baddeley A. (2012) **Working Memory: Theories, Models, and Controversies** *Annual Review of Psychology* **63**:1–29 <https://doi.org/10.1146/annurev-psych-120710-100422> | Google Scholar
- Balota D. A., Yap M. J. (2011) **Moving Beyond the Mean in Studies of Mental Chronometry: The Power of Response Time Distributional Analyses** *Current Directions in Psychological Science* **20**:160–166 <https://doi.org/10.1177/0963721411408885> | Google Scholar
- Barry R. J., De Blasio F. M. (2021) **Characterizing pink and white noise in the human electroencephalogram** *Journal of Neural Engineering* IOP Publishing **18**:034001 <https://doi.org/10.1088/1741-2552/abe399> | Google Scholar
- Benjamini Y., Yekutieli D. (2001) **The control of the false discovery rate in multiple testing under dependency** *The Annals of Statistics* **29** <https://doi.org/10.1214/aos/1013699998> | Google Scholar
- Brooks H., Goodman M. S., Bowie C. R., Zomorodi R., Blumberger D. M., Butters M. A., Daskalakis Z. J., Fischer C. E., Flint A., Herrmann N., Kumar S., Mah L., Mulsant B. H., Pollock B. G., Voineskos A. N., Rajji T. K. (2020) **Theta-gamma coupling and ordering information: a stable brain-behavior relationship across cognitive tasks and clinical conditions** *Neuropsychopharmacology* **45**:2038–2047 <https://doi.org/10.1038/s41386-020-0759-z> | Google Scholar
- Bullock T. H., McClune M. C., Enright J. T. (2003) **Are the electroencephalograms mainly rhythmic? Assessment of periodicity in wide-band time series** *Neuroscience* **121**:233–252 [https://doi.org/10.1016/S0306-4522\(03\)00208-2](https://doi.org/10.1016/S0306-4522(03)00208-2) | Google Scholar
- Cavanagh J. F., Frank M. J. (2014) **Frontal theta as a mechanism for cognitive control** *Trends in Cognitive Sciences* **18**:414–421 <https://doi.org/10.1016/j.tics.2014.04.012> | Google Scholar

- Chen Y., Huang X. (2016) **Modulation of Alpha and Beta Oscillations during an n-back Task with Varying Temporal Memory Load** *Frontiers in Psychology* **6** <https://doi.org/10.3389/fpsyg.2015.02031> | [Google Scholar](#)
- Cohen M. X. (2014) **Analyzing Neural Time Series Data: Theory and Practice** The MIT Press <https://doi.org/10.7551/mitpress/9609.001.0001> | [Google Scholar](#)
- Coleman S. C., Seedat Z. A., Whittaker A. C., Lenartowicz A., Mullinger K. J. (2023) **Beyond the Beta Rebound: Post-Task Responses in Oscillatory Activity follow Cessation of Working Memory Processes** *NeuroImage* **265**:119801 <https://doi.org/10.1016/j.neuroimage.2022.119801> | [Google Scholar](#)
- Cousineau D. (2017) **Varieties of Confidence Intervals** *Advances in Cognitive Psychology* **13**:140–155 <https://doi.org/10.5709/acp-0214-z> | [Google Scholar](#)
- Cunningham E., Zimnicki C., Beck D. M. (2023) **The Influence of Prestimulus 1/f-Like versus Alpha-Band Activity on Subjective Awareness of Auditory and Visual Stimuli** *Journal of Neuroscience Society for Neuroscience Section: Research Articles* **43**:6447–6459 <https://doi.org/10.1523/JNEUROSCI.0238-23.2023> | [Google Scholar](#)
- Delorme A., Makeig S. (2004) **EEGLAB: an open source toolbox for analysis of single-trial EEG dynamics including independent component analysis** *Journal of Neuroscience Methods* **134**:9–21 <https://doi.org/10.1016/j.jneumeth.2003.10.009> | [Google Scholar](#)
- Donoghue T., Dominguez J., Voytek B. (2020a) **Electrophysiological Frequency Band Ratio Measures Conflate Periodic and Aperiodic Neural Activity** *eNeuro Society for Neuroscience* **7** <https://doi.org/10.1523/ENEURO.0192-20.2020> | [Google Scholar](#)
- Donoghue T., Haller M., Peterson E. J., Varma P., Sebastian P., Gao R., Noto T., Lara A. H., Wallis J. D., Knight R. T., Shestyuk A., Voytek B. (2020b) **Parameterizing neural power spectra into periodic and aperiodic components** *Nature Neuroscience* **23**:1655–1665 <https://doi.org/10.1038/s41593-020-00744-x> | [Google Scholar](#)
- Donoghue T., Schaworonkow N., Voytek B. (2022) **Methodological considerations for studying neural oscillations** *European Journal of Neuroscience* **55**:3502–3527 <https://doi.org/10.1111/ejn.15361> | [Google Scholar](#)
- Eggermont J. J. (2021) **Chapter 2 - The gamma, beta, and theta rhythms and nested oscillations** In: Eggermont J. J., editors. *Brain Oscillations, Synchrony, and Plasticity* Academic Press pp. 23–42 <https://doi.org/10.1016/B978-0-12-819818-6.00014-5> | [Google Scholar](#)
- ElShafei H. A., Zhou Y. J., Haegens S. (2022) **Shaping Information Processing: The Role of Oscillatory Dynamics in a Working Memory Task** *eNeuro* **9** <https://doi.org/10.1523/ENEURO.0489-21.2022> | [Google Scholar](#)
- Gao R., Peterson E. J., Voytek B. (2017) **Inferring synaptic excitation/inhibition balance from field potentials** *NeuroImage* **158**:70–78 <https://doi.org/10.1016/j.neuroimage.2017.06.078> | [Google Scholar](#)
- Gazzaley A., Nobre A. C. (2012) **Top-down modulation: bridging selective attention and working memory** *Trends in Cognitive Sciences Elsevier* **16**:129–135 <https://doi.org/10.1016/j.tics.2011.11.014> | [Google Scholar](#)

- Gerster M., Waterstraat G., Litvak V., Lehnertz K., Schnitzler A., Florin E., Curio G., Nikulin V. (2022) **Separating Neural Oscillations from Aperiodic 1/f Activity: Challenges and Recommendations** *Neuroinformatics* **20**:991–1012 <https://doi.org/10.1007/s12021-022-09581-8> | [Google Scholar](#)
- Gevins A., Smith M. E., McEvoy L., Yu D. (1997) **High-resolution EEG mapping of cortical activation related to working memory: effects of task difficulty, type of processing, and practice** *Cerebral Cortex* **7**:374–385 <https://doi.org/10.1093/cercor/7.4.374> | [Google Scholar](#)
- Goodman M. S., Kumar S., Zomorodi R., Ghazala Z., Cheam A. S. M., Barr M. S., Daskalakis Z. J., Blumberger D. M., Fischer C., Flint A., Mah L., Herrmann N., Bowie C. R., Mulsant B. H., Rajji T. K. (2018) **Theta-Gamma Coupling and Working Memory in Alzheimer’s Dementia and Mild Cognitive Impairment** *Frontiers in Aging Neuroscience* **10** <https://doi.org/10.3389/fnagi.2018.00101> | [Google Scholar](#)
- Gyurkovics M., Clements G. M., Low K. A., Fabiani M., Gratton G. (2021) **The impact of 1/f activity and baseline correction on the results and interpretation of timefrequency analyses of EEG/MEG data: A cautionary tale** *NeuroImage* **237**:118192 <https://doi.org/10.1016/j.neuroimage.2021.118192> | [Google Scholar](#)
- Gyurkovics M., Clements G. M., Low K. A., Fabiani M., Gratton G. (2022) **Stimulus-Induced Changes in 1/ f -like Background Activity in EEG** *The Journal of Neuroscience* **42**:7144–7151 <https://doi.org/10.1523/JNEUROSCI.0414-22.2022> | [Google Scholar](#)
- Haegens S., Cousijn H., Wallis G., Harrison P. J., Nobre A. C. (2014) **Inter- and intra-individual variability in alpha peak frequency** *NeuroImage* **92**:46–55 <https://doi.org/10.1016/j.neuroimage.2014.01.049> | [Google Scholar](#)
- HajiHosseini A., Hutcherson C. A., Holroyd C. B. (2020) **Beta oscillations following performance feedback predict subsequent recall of task-relevant information** *Scientific Reports* Nature Publishing Group **10**:15114 <https://doi.org/10.1038/s41598-020-72128-x> | [Google Scholar](#)
- Hanslmayr S., Staresina B. P., Bowman H. (2016) **Oscillations and Episodic Memory: Addressing the Synchronization/Desynchronization Conundrum** *Trends in Neurosciences* Elsevier **39**:16–25 <https://doi.org/10.1016/j.tins.2015.11.004> | [Google Scholar](#)
- Kaiser J., Birbaumer N., Lutzenberger W. (2001) **Event-related beta desynchronization indicates timing of response selection in a delayed-response paradigm in humans** *Neuroscience Letters* **312**:149–152 [https://doi.org/10.1016/S0304-3940\(01\)02217-0](https://doi.org/10.1016/S0304-3940(01)02217-0) | [Google Scholar](#)
- Kałamała P., Gyurkovics M., Bowie D. C., Clements G. M., Low K. A., Dolcos F., Fabiani M., Gratton G. (2024) **Event-induced modulation of aperiodic background EEG: Attention-dependent and age-related shifts in E:I balance, and their consequences for behavior** *Imaging Neuroscience* **2**:1–18 https://doi.org/10.1162/imag_a_00054 | [Google Scholar](#)
- Krause C. M., Pesonen M., Hämäläinen H. (2010) **Brain oscillatory 4–30 Hz electroencephalogram responses in adolescents during a visual memory task** *NeuroReport* **21**:767 <https://doi.org/10.1097/WNR.0b013e32833bfcbb> | [Google Scholar](#)
- Köster M., Fries U., Schöne B., Trujillo-Barreto N., Gruber T. (2014) **Theta-gamma coupling during episodic retrieval in the human EEG** *Brain Research* **1577**:57–68 <https://doi.org/10.1016/j.brainres.2014.06.028> | [Google Scholar](#)

- Lim S., Goldman M. S. (2013) **Balanced cortical microcircuitry for maintaining information in working memory** *Nature neuroscience* **16**:1306–1314 <https://doi.org/10.1038/nn.3492> | [Google Scholar](#)
- Lopez-Calderon J., Luck S. J. (2014) **ERPLAB: an open-source toolbox for the analysis of event-related potentials** *Frontiers in Human Neuroscience* **8** <https://doi.org/10.3389/fnhum> | [Google Scholar](#)
- (2013) **Massidda, D., (2013). URL <https://CRAN.R-project.org/package=retimes>. <https://CRAN.R-project.org/package=retimes>**
- McGill M. B., Kieffaber P. D. (2024) **Event-Related Theta and Gamma Band Oscillatory Dynamics During Visuo-Spatial Sequence Memory in Younger and Older Adults** *PLOS One* **19**:e0297995 <https://doi.org/10.1371/journal.pone.0297995> | [Google Scholar](#)
- Merkin A., Sghirripa S., Graetz L., Smith A. E., Hordacre B., Harris R., Pitcher J., Semmler J., Rogasch N. C., Goldsworthy M. (2023) **Do age-related differences in aperiodic neural activity explain differences in resting EEG alpha?** *Neurobiology of Aging* **121**:78–87 <https://doi.org/10.1016/j.neurobiolaging.2022.09.003> | [Google Scholar](#)
- Moca V. V., Bârzan H., Nagy-Dabâcan A., Muresan RC. (2021) **Time-frequency super-resolution with superlets** *Nature Communications* **12**:337 <https://doi.org/10.1038/s41467-020-20539-9> | [Google Scholar](#)
- Monchy N., Modolo J., Houvenaghel J.-F., Voytek B., Duprez J. (2023) **Changes in electro-physiological aperiodic activity during cognitive control in Parkinson’s disease. preprint Neuroscience** <http://biorxiv.org/lookup/doi/10.1101/2023.10.09.561166>
- (2012) **Mullen, T., (2012). URL <https://www.nitrc.org/projects/cleanline>. <https://www.nitrc.org/projects/cleanline>**
- Mullen T., Kothe C., Chi Y. M., Ojeda A., Kerth T., Makeig S., Cauwenberghs G., Jung Tzyy-Ping (2013) **Real-time modeling and 3D visualization of source dynamics and connectivity using wearable EEG** In: *2013 35th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)* IEEE pp. 2184–2187 <https://doi.org/10.1109/EMBC.2013.6609968> | [Google Scholar](#)
- Myrov V., Siebenhühner F., Juvonen J. J., Arnulfo G., Palva S., Palva J. M. (2024) **Rhythmicity of neuronal oscillations delineates their cortical and spectral architecture** *Communications Biology* **7**:405 <https://doi.org/10.1038/s42003-024-06083-y> | [Google Scholar](#)
- Nakagawa S., Schielzeth H. (2013) **A general and simple method for obtaining R2 from generalized linear mixed-effects models** *Methods in Ecology and Evolution* **4**:133–142 <https://doi.org/10.1111/j.2041-210x.2012.00261.x> | [Google Scholar](#)
- Nakuci J., Covey T. J., Shucard J. L., Shucard D. W., Muldoon S. F. (2023) **Single trial variability in neural activity during a working memory task reveals multiple distinct information processing sequences** *NeuroImage* **269**:119895 <https://doi.org/10.1016/j.neuroimage.2023.119895> | [Google Scholar](#)

- Olesen P. J., Westerberg H., Klingberg T. (2004) **Increased prefrontal and parietal activity after training of working memory** *Nature Neuroscience* Nature Publishing Group **7**:75–79 <https://doi.org/10.1038/nn1165> | [Google Scholar](#)
- Oostenveld R., Fries P., Maris E., Schoffelen J.-M. (2011) **FieldTrip: Open Source Software for Advanced Analysis of MEG, EEG, and Invasive Electrophysiological Data** *Computational Intelligence and Neuroscience* **2011**:1–9 <https://doi.org/10.1155/2011/156869> | [Google Scholar](#)
- Owen A. M., McMillan K. M., Laird A. R., Bullmore E. (2005) **N-back working memory paradigm: A meta-analysis of normative functional neuroimaging studies** *Human Brain Mapping* **25**:46–59 <https://doi.org/10.1002/hbm.20131> | [Google Scholar](#)
- Palmer J. A., Kreutz-Delgado K., Makeig S. (2011) **AMICA: An Adaptive Mixture of Independent Component Analyzers with Shared Components** San Diego: Swartz Center for Computational Neuroscience [Google Scholar](#)
- Palomäki J., Kivikangas M., Alafuzoff A., Hakala T., Krause C. M. (2012) **Brain oscillatory 4–35Hz EEG responses during an n-back task with complex visual stimuli** *Neuroscience Letters* **516**:141–145 <https://doi.org/10.1016/j.neulet.2012.03.076> | [Google Scholar](#)
- Park J. Y., Jhung K., Lee J., An S. K. (2013) **Theta-gamma coupling during a working memory task as compared to a simple vigilance task** *Neuroscience Letters* **532**:39–43 <https://doi.org/10.1016/j.neulet.2012.10.061> | [Google Scholar](#)
- Pavlov Y. G., Kotchoubey B. (2022) **Oscillatory brain activity and maintenance of verbal and visual working memory: A systematic review** *Psychophysiology* **59**:e13735 <https://doi.org/10.1111/psyp.13735> | [Google Scholar](#)
- Peirce J., Hirst R., MacAskill M. (2022) **Building experiments in PsychoPy** Los Angeles: SAGE [Google Scholar](#)
- Peng W., Hu Y., Mao Y., Babiloni C. (2015) **Widespread cortical α -ERD accompanying visual oddball target stimuli is frequency but non-modality specific** *Behavioural Brain Research* **295**:71–77 <https://doi.org/10.1016/j.bbr.2015.04.051> | [Google Scholar](#)
- Pesonen M., Hämäläinen H., Krause C. M. (2007) **Brain oscillatory 4–30 Hz responses during a visual n-back memory task with varying memory load** *Brain Research* **1138**:171–177 <https://doi.org/10.1016/j.brainres.2006.12.076> | [Google Scholar](#)
- Pfurtscheller G., Zalaudek K., Neuper C. (1998) **Event-related beta synchronization after wrist, finger and thumb movement** *Electroencephalography and Clinical Neurophysiology/Electromyography and Motor Control* **109**:154–160 [https://doi.org/10.1016/S0924-980X\(97\)00070-2](https://doi.org/10.1016/S0924-980X(97)00070-2) | [Google Scholar](#)
- Rajji T. K., Zomorodi R., Barr M. S., Blumberger D. M., Mulsant B. H., Daskalakis Z. J. (2016) **Ordering Information in Working Memory and Modulation of Gamma by Theta Oscillations in Humans** *Cerebral Cortex* :bhv326 <https://doi.org/10.1093/cercor/bhv326> | [Google Scholar](#)
- Ratcliffe O., Shapiro K., Staresina B. P. (2022) **Fronto-medial theta coordinates posterior maintenance of working memory content** *Current Biology* **32**:2121–2129 <https://doi.org/10.1016/j.cub.2022.03.045> | [Google Scholar](#)

- Rossi C., Vidaurre D., Costers L., Akbarian F., Woolrich M., Nagels G., Van Schependom J. (2023) **A data-driven network decomposition of the temporal, spatial, and spectral dynamics underpinning visual-verbal working memory processes** *Communications Biology* **6**:1–12 <https://doi.org/10.1038/s42003-023-05448-z> | [Google Scholar](#)
- Roux F., Uhlhaas P. J. (2014) **Working memory and neural oscillations: alpha-gamma versus theta-gamma codes for distinct WM information?** *Trends in Cognitive Sciences* **18**:16–25 <https://doi.org/10.1016/j.tics.2013.10.010> | [Google Scholar](#)
- Scharinger C., Soutschek A., Schubert T., Gerjets P. (2017) **Comparison of the Working Memory Load in N-Back and Working Memory Span Tasks by Means of EEG Frequency Band Power and P300 Amplitude** *Frontiers in Human Neuroscience* **11** <https://doi.org/10.3389/fnhum.2017.00006> | [Google Scholar](#)
- Schmidt R., Herrojo Ruiz M., Kilavik B. E., Lundqvist M., Starr P. A., Aron A. R. (2019) **Beta Oscillations in Working Memory, Executive Control of Movement and Thought, and Sensorimotor Function** *The Journal of Neuroscience* **39**:8231–8238 <https://doi.org/10.1523/JNEUROSCI.1163-19.2019> | [Google Scholar](#)
- Sternberg S. (1969) **Memory-Scanning: Mental Processes Revealed by Reaction-Time Experiments** *American Scientist* Sigma Xi, The Scientific Research Society **57**:421–457 [Google Scholar](#)
- Thuwal K., Banerjee A., Roy D. (2021) **Aperiodic and Periodic Components of Ongoing Oscillatory Brain Dynamics Link Distinct Functional Aspects of Cognition across Adult Lifespan** *eneuro* **8**:ENEURO.0224–21.2021 <https://doi.org/10.1523/ENEURO.0224-21.2021> | [Google Scholar](#)
- Tort A. B. L., Komorowski R., Eichenbaum H., Kopell N. (2010) **Measuring Phase-Amplitude Coupling Between Neuronal Oscillations of Different Frequencies** *Journal of Neurophysiology* **104**:1195–1210 <https://doi.org/10.1152/jn.00106.2010> | [Google Scholar](#)
- Tseng Y.-L., Liu H.-H., Liou M., Tsai A. C., Chien V. S. C., Shyu S.-T., Yang Z.-S. (2019) **Lingering Sound: Event-Related Phase-Amplitude Coupling and Phase-Locking in Fronto-Temporo-Parietal Functional Networks During Memory Retrieval of Music Melodies** *Frontiers in Human Neuroscience* **13**:150 <https://doi.org/10.3389/fnhum.2019.00150> | [Google Scholar](#)
- Van Der Meij R., Kahana M., Maris E. (2012) **Phase-Amplitude Coupling in Human Electroencephalography Is Spatially Distributed and Phase Diverse** *The Journal of Neuroscience* **32**:111–123 <https://doi.org/10.1523/JNEUROSCI.4816-11.2012> | [Google Scholar](#)
- Van Ede F., Maris E. (2016) **Physiological Plausibility Can Increase Reproducibility in Cognitive Neuroscience** *Trends in Cognitive Sciences* **20**:567–569 <https://doi.org/10.1016/j.tics.2016.05.006> | [Google Scholar](#)
- Van Engen Q., Chau G., Smith A., Adam K., Donoghue T., Voytek B. (2024) **Dissociating Contributions of Theta and Alpha Oscillations from Aperiodic Neural Activity in Human Visual Working Memory** <http://biorxiv.org/lookup/doi/10.1101/2024.12.16.628786>
- Virtue-Griffiths S., Fornito A., Thompson S., Biabani M., Tiego J., Thapa T., Rogasch N. C. (2022) **Task-related changes in aperiodic activity are related to visual working memory capacity**

independent of event-related potentials and alpha oscillations. preprint *Neuroscience*
<http://biorxiv.org/lookup/doi/10.1101/2022.01.18.476852>

Vogels T., Abbott L. (2009) **Gating Multiple Signals through Detailed Balance of Excitation and Inhibition in Spiking Networks** *Nature neuroscience* **12**:483–491 <https://doi.org/10.1038/nn.2276> | Google Scholar

Voytek B., Kramer M. A., Case J., Lepage K. Q., Tempesta Z. R., Knight R. T., Gazzaley A. (2015) **Age-Related Changes in 1/f Neural Electrophysiological Noise** *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience* **35**:13257–13265 <https://doi.org/10.1523/JNEUROSCI.2332-14.2015> | Google Scholar

Wager T. D., Smith E. E. (2003) **Neuroimaging studies of working memory** *Cognitive, Affective, & Behavioral Neuroscience* **3**:255–274 <https://doi.org/10.3758/CABN.3.4.255> | Google Scholar

Waschke L., Donoghue T., Fiedler L., Smith S., Garrett D. D., Voytek B., Obleser J. (2021) **Modality-specific tracking of attention and sensory statistics in the human electrophysiological spectral exponent** *eLife* **10**:e70068 <https://doi.org/10.7554/eLife.70068> | Google Scholar

Wilson L. E., Da Silva Castanheira J., Baillet S. (2022) **Time-resolved parameterization of aperiodic and periodic brain activity** *eLife* **11**:e77348 <https://doi.org/10.7554/eLife.77348> | Google Scholar

Cousineau D. (2017) **Varieties of Confidence Intervals** *Advances in Cognitive Psychology* **13**:140–155 <https://doi.org/10.5709/acp-0214-z> | Google Scholar

Nakuci J., Covey T. J., Shucard J. L., Shucard D. W., Muldoon S. F. (2023) **Single trial variability in neural activity during a working memory task reveals multiple distinct information processing sequences** *NeuroImage* **269**:119895 <https://doi.org/10.1016/j.neuroimage.2023.119895> | Google Scholar

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Reviewer #1 (Public review):**Summary:**

Frelih et al. investigated both periodic and aperiodic activity in EEG during working memory tasks. In terms of periodic activity, they found post-stimulus decreases in alpha and beta activity, while in terms of aperiodic activity, they found a bi-phasic post-stimulus steepening of the power spectrum, which was weakly predictive of performance. They conclude that it is crucial to properly distinguish between aperiodic and periodic activity in event-related designs as the former could confound the latter. They also add to the growing body of research highlighting the functional relevance of aperiodic activity in the brain.

Strengths:

This is a well-written, timely paper that could be of interest to the field of cognitive neuroscience, especially to researchers investigating the functional role of aperiodic activity. The authors describe a well-designed study that looked at both the oscillatory and non-oscillatory aspects of brain activity during a working memory task. The analytic approach is appropriate, as a state-of-the-art toolbox is used to separate these two types of activity. The results support the basic claim of the paper that it is crucial to properly distinguish between aperiodic and periodic activity in event-related designs as the former could confound the latter. They also add to the growing body of research highlighting the functional relevance of aperiodic activity in the brain. Commendably, the authors include replications of their key findings on multiple independent data sets.

Weaknesses:

The authors also claim that their results speak to the interplay between oscillatory and non-oscillatory activity, and crucially, that task-related changes in the theta frequency band - often attributed to neural oscillations in the field - are in fact only a by-product of non-oscillatory changes. I believe these claims are too bold and are not supported by compelling evidence in the paper. Some control analyses - e.g., contrasting the scalp topographies of purportedly theta and non-oscillatory effects - could help strengthen the latter argument, but it may be safest to simply soften these two claims.

In terms of the methodology used, I suggest the authors make it clearer to readers that the primary results were obtained on a sample of middle-aged-to-older-adults, some with subjective cognitive complaints, and note that while stimulus-locked event-related potentials (ERPs) were removed from the data prior to analyses, response-locked ERPs were not. This could potentially confound aperiodic findings. Contrasting the scalp topographies of response-related ERPs and the identified aperiodic components, especially the later one, could bring some clarity here too.

I have also found certain parts of the introduction to be somewhat confusing.

Comments on the latest version:

The authors have addressed several of the weaknesses I noted in my original review, specifically, they softened their claims regarding the theta findings, while simultaneously strengthening these findings with additional analyses (using simulations as well as a new measure of rhythmicity, the phase autocorrelation function, pACF). Most of the other suggested control analyses were also implemented. While I believe the fact that the participants in the main sample were not young adults could be made even more explicit, and the potential interaction between age and aperiodic changes could be unpacked a little in the discussion, the age of the sample is definitely addressed upfront.

<https://doi.org/10.7554/eLife.101071.2.sa3>

Reviewer #2 (Public review):

Summary:

In this manuscript, Frelüh et al, investigate the relationship between aperiodic neural activity, as measured by EEG, and working memory performance, and compares this to the more commonly analyzed periodic, and in particular theta, measures that are often associated with such tasks. To do so, they analyze a primary dataset of 57 participants engaging in an n-back task, as well as a replication dataset, and use spectral parameterization to measure periodic and aperiodic features of the data, across time. In the revision, the authors have clarified some key points, and added a series of additional analyses and controls, including the use of an additional method, that helps to complement the original analyses and further corroborates their claims. In doing so, they find both periodic and aperiodic features that relate to the task dynamics, but importantly, the aperiodic component appears to explain away what otherwise looks like theta activity in a more traditional analysis. This study therefore helps to establish that aperiodic activity is a task-relevant dynamic feature in working memory tasks and may be the underlying change in many other studies that reported 'theta' changes, but did not use methods that could differentiate periodic and aperiodic features.

Strengths:

Key strengths of this paper include that it addresses an important question - that of properly adjudicating which features of EEG recordings relate to working memory tasks - and in doing

so provides a compelling answer, with important implications for considering prior work and contributing to understanding the neural underpinnings of working memory. The revision is improved by showing this using an additional analysis method. I do not find any significant faults or error with the design, analysis, and main interpretations as presented by this paper, and as such, find the approach taken to be a valid and well-enacted. The use of multiple variants of the working memory task, as well as a replication dataset significantly strengthens this manuscript, by demonstrating a degree of replicability and generalizability. This manuscript is also an important contribution to motivating best practices for analyzing neuro-electrophysiological data, including in relation to using baselining procedures. I think the updates in the revision have helped to clarify the findings and impact of this study.

Weaknesses:

Overall, I do not find any obvious weaknesses with this manuscript and its analyses that challenge the key results and conclusions. Updates through the revision have addressed my previous points about adding some additional notes on the methods and conclusions.

<https://doi.org/10.7554/eLife.101071.2.sa2>

Reviewer #3 (Public review):

Summary:

Using a specparam (1/f) analysis of task-evoked activity, the authors propose that "substantial changes traditionally attributed to theta oscillations in working memory tasks are, in fact, due to shifts in the spectral slope of aperiodic activity." This is a very bold and ambitious statement, and the field of event-related EEG would benefit from more critical assessments of the role of aperiodic changes during task events. Unfortunately, the data shown here does not support the main conclusion advanced by the authors.

Strengths:

The field of event-related EEG would benefit from more critical assessments of the role of aperiodic changes during task events. The authors perform a number of additional control analyses, including different types of baseline correction, ERP subtraction, as well as replication of the experiment with two additional datasets.

Weaknesses:

The authors did not first show that their first task successfully evoked theta power, nor that specparam is capable of quantifying the background around a short theta burst, nor that theta effects are different between baseline corrected vs. spectral parameterized quantification.

Comments on revisions:

The authors have completed a substantial revision based on the comments from all of the reviewers. Overall, the major claims of the initial report have been profoundly tempered, but more of the conclusions are supported by the data.

<https://doi.org/10.7554/eLife.101071.2.sa1>

Author Response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Expressed concern that FOOOF may not be sensitive to peaks located at the edges of the spectrum and suggested using rhythmicity as an alternative measure of oscillatory activity.

To address this concern, we first conducted a simulation in which we generated power spectra with a single periodic component while varying its parameters. The results confirmed that FOOOF may indeed have reduced sensitivity to low-frequency periodic components. In such cases, periodic activity can be conflated with aperiodic activity, leading to inflated estimates of the aperiodic component. These simulation results are presented in detail at the end of the Supplement.

To further investigate whether the low-frequency activity in our datasets may be oscillatory, we employed the phase-autocorrelation function (pACF), a measure of rhythmicity developed by Myrov et al. (2024). We compared pACF and FOOOF-derived parameters using linear mixed models at each channel–frequency–time point (see Methods for details). Our analyses showed that pACF activity closely resembles periodic activity across all three datasets, and is dissimilar to aperiodic parameters (see Figures 5, S4, S5, S21, S22, S34, S35). This supports the interpretation that, in our data, aperiodic activity is not conflated with periodic activity.

I was concerned that “there were no dedicated analyses in the paper to show that the aperiodic changes account for the theta changes.”

To address this concern, we used linear mixed models to estimate the association between FOOOF parameters and baseline-corrected time-frequency activity. These models were fitted at each channel-frequency-time point. Our results indicate that aperiodic activity is correlated with low-frequency (theta) baseline-corrected activity, while periodic activity is correlated primarily with activity in the alpha/beta range, but not with theta (see Figures 4, S3, S20, S33). Additionally, the exponent parameter exhibited a negative correlation in the gamma frequency range.

These findings support the reviewer's hypothesis: “I would also like to note that if the theta effect is only the aperiodic shift in disguise, we should see a concomitant increase in delta activity too – maybe even a decrease at high frequencies.” Overall, the results are consistent with our interpretation that low-frequency baseline-corrected activity reflects changes in aperiodic, rather than periodic, activity.

“On page 7 it is noted that baseline correction might subtract a significant amount of ongoing periodic activity. I would replace the word “subtract” with “remove” as not all baseline correction procedures are subtractive. Furthermore, while this sentence makes it sound like a problem, this is, to my mind, a feature, not a bug - baseline correction is meant to take away whatever is ongoing, be it oscillatory or not, and emphasise changes compared to that, in response to some event.”

We thank the reviewer for this helpful clarification. We have revised the sentence accordingly to read: “Our results show that classical baseline correction can remove continuous oscillatory activity that is present both during baseline and after stimulus onset, because it treats all baseline signals as ‘background’ to be removed without distinguishing between transient and continuous oscillations. While this is consistent with the intended purpose of baseline correction—to highlight changes relative to ongoing activity—it may also lead to unintended consequences, such as misinterpreting aperiodic activity as an increase in poststimulus theta oscillations.”

In addition, we have made several broader revisions throughout the manuscript to improve clarity and accuracy in response to the reviewer's feedback:

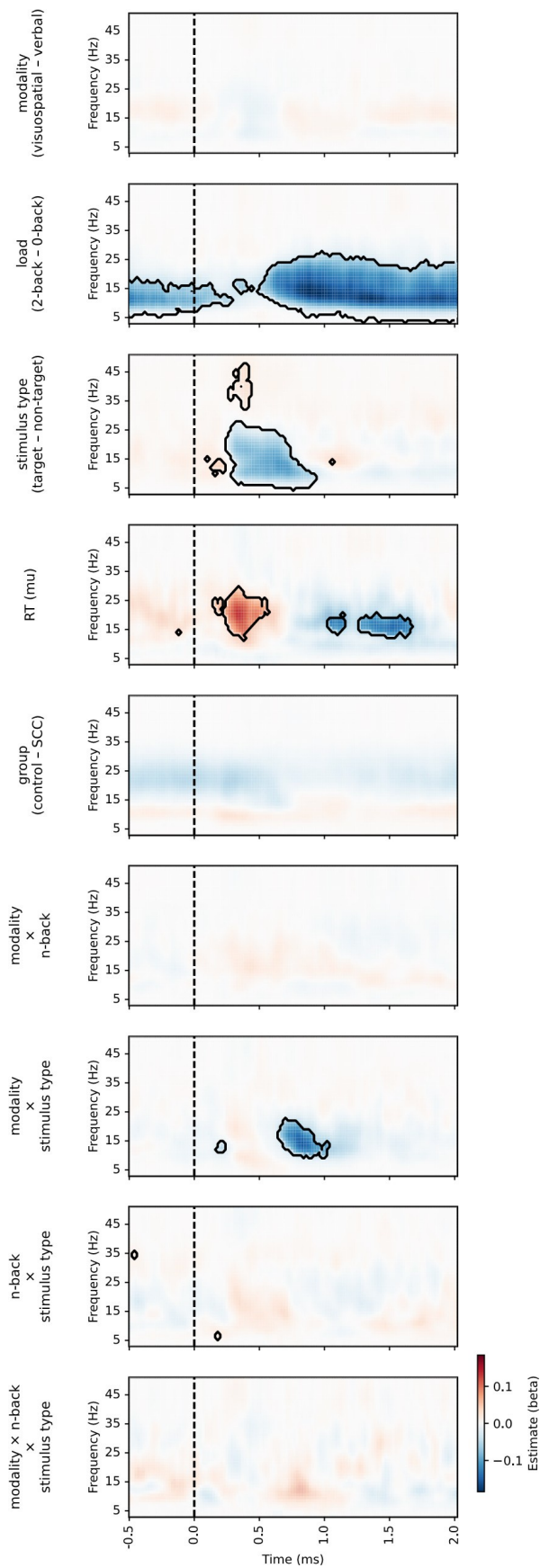
- (1) We have softened our interpretation of changes in the theta range. We no longer claim that these effects are solely due to aperiodic activity; rather, we now state that our findings suggest a *potential contribution* of aperiodic activity to signals typically interpreted as theta oscillations.
- (2) We have revised our language to avoid suggesting a direct “interplay” between periodic and aperiodic components. Instead, we emphasize the concurrent presence of both components, using more precise and cautious formulations.
- (3) We have clarified our discussion of baseline normalization approaches, explicitly noting that our findings hold regardless of whether a subtractive or divisive baseline correction was applied.
- (4) Finally, we have restructured the introduction to improve readability and address points of potential confusion. Specifically, we have clarified the definition and role of 1/f activity, refined the discussion linking baseline correction to aperiodic activity, and improved transitions between key concepts.

Reviewer suggested that “it might be good to show that the findings were not driven by the cognitive-complaint subgroup (although the internal replications suggest they were not).”

We agree that it is important to demonstrate that our findings are not driven solely by the cognitive-complaint subgroup. While we did not include additional figures in the manuscript due to their limited relevance to the primary research question, we have attached figures that explicitly show the comparison between the clinical and control groups here in the response to reviewers. These figures include non-significant effects.

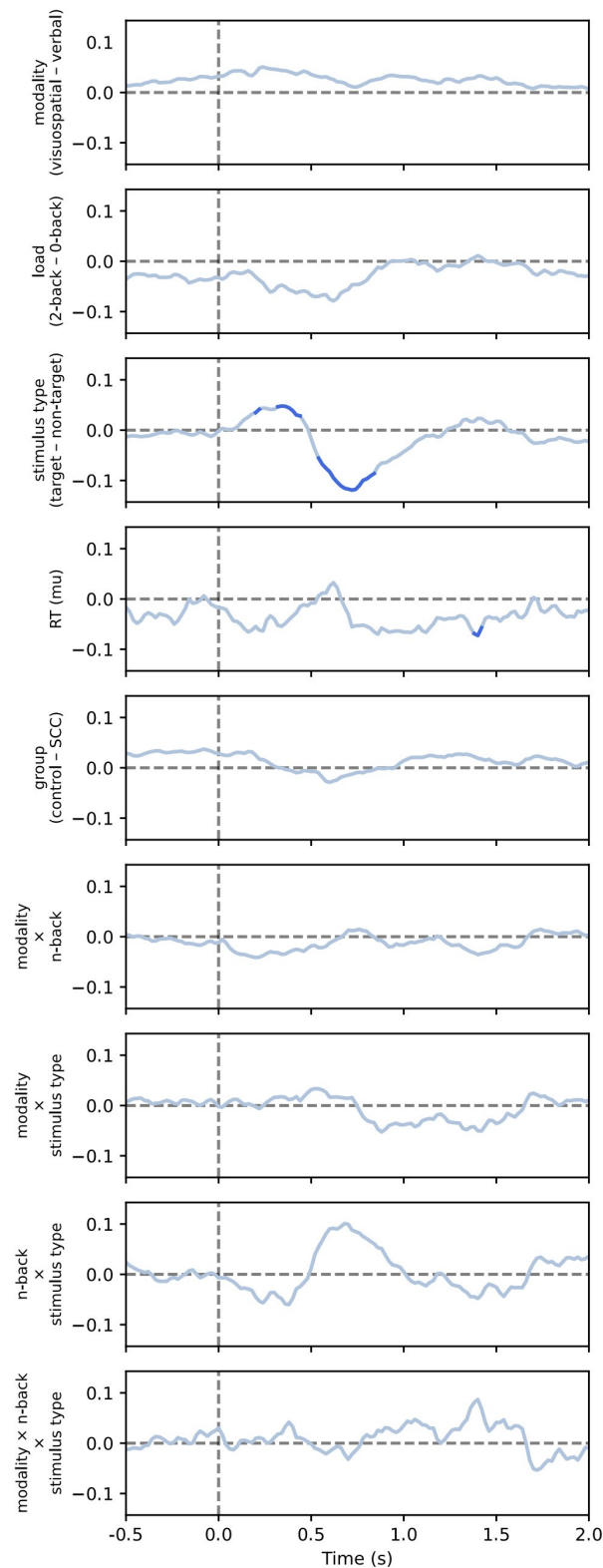
Author response image 1.

Results of the linear mixed model analysis of periodic activity for comparison between conditions, including non-significant effect (see also Figure 7 in the paper)



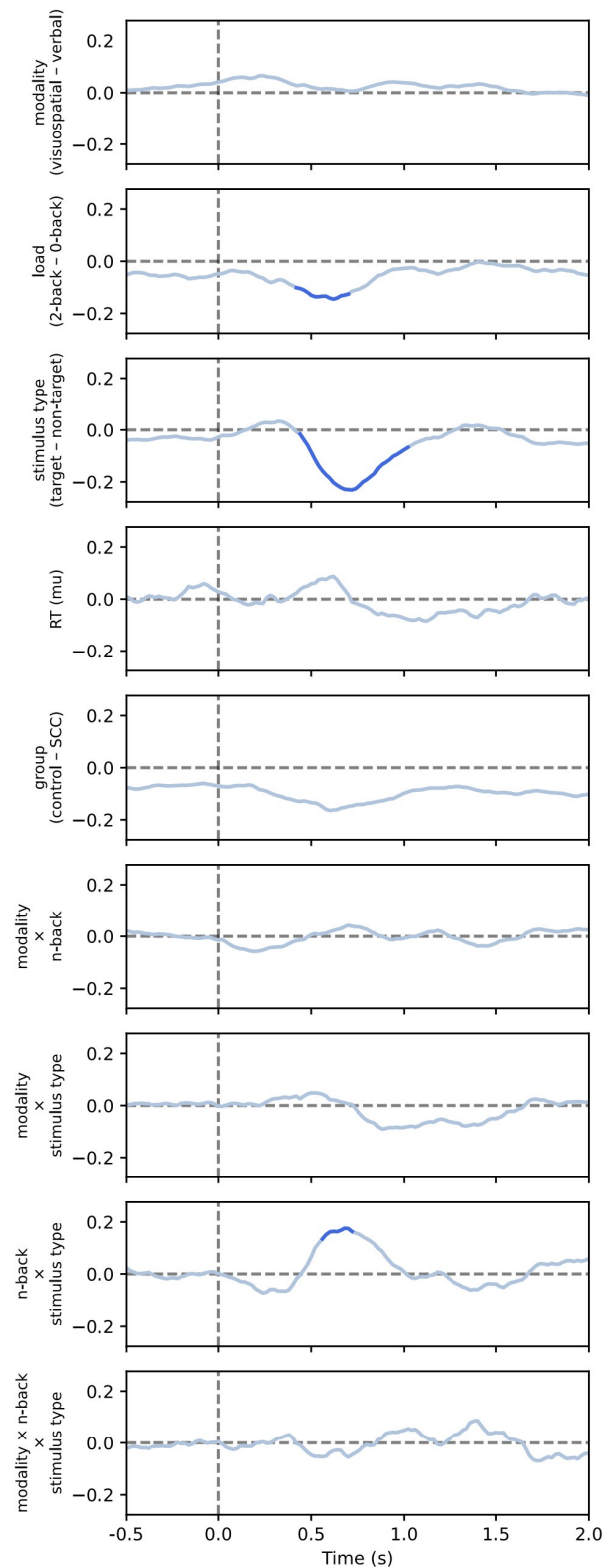
Author response image 2.

Results of the linear mixed model analysis of aperiodic exponent for comparison between conditions, including nonsignificant effects (see also Figure 9 in the paper)



Author response image 3.

Results of the linear mixed model analysis of aperiodic offset for comparison between conditions, including non-significant effects (see also Figure S11 in the paper)



"Were lure trials discarded completely, or were they included in the non-target group?"

Thank you for the question. As described in the Methods section (EEG data preprocessing), lure trials were discarded entirely from further analysis and were not included in the non-target group.

"Also, just as a side note, while this time-resolved approach is definitely new, it is not novel to this paper, at least two other groups have tried similar approaches, e.g., Wilson, da Silva Castanheira, & Baillet, 2022; Ameen, Jacobs, et al., 2024."

Thank you for drawing our attention to these relevant studies. We have now cited both Wilson et al. (2022) and Ameen et al. (2024) in our manuscript. While these papers did indeed use time-resolved approaches, to our knowledge our study is the first to use such an approach within a task-based paradigm.

noted that it was unclear how the periodic component was reconstructed: "I understand that a Gaussian was recreated based on these parameters, but were frequencies between and around the Gaussians just zeroed out? Or rather, given a value of 1, so that it would be 0 after taking its log10."

The periodic component was reconstructed by summing the Gaussians derived from the FOOOF model parameters. Since the Gaussians asymptotically approach, but never reach, zero, there were no explicit zeros between them. We have included this explanation in the manuscript.

"If my understanding is correct, the periodic and aperiodic analyses were not run on the singletrial level, but on trial-averaged TF representations. Is that correct? In that case, there was only a single observation per participant for each within-subject cell at each TF point. This means that model (4) on p. 15 just simplifies to a repeated-measures ANOVA, does it not? As hinted at later in this section, the model was run at each time point for aperiodic analyses, and at each TF point for periodic analyses, resulting in a series of p-values or a map of p-values, respectively, is that correct?"

We thank the reviewer for this careful reading and helpful interpretation. The reviewer is correct that analyses were conducted on trial-averaged time-frequency representations. Model presented in equation 7 (as referred to in the current version of the manuscript) is indeed conceptually similar to a repeated-measures ANOVA in that it tests within-subject effects across conditions. However, due to some missing data (i.e., excluded conditions within subjects), we employed linear mixed-effects models (LMER), which can handle unbalanced data without resorting to listwise deletion. This provides more flexibility and preserves statistical power.

The reviewer is also correct that the models were run at each channel-time point for the aperiodic analyses, and at each channel-time-frequency point for the periodic analyses, resulting in a series or map of p-values, respectively.

suggested marking the mean response time and contrasting scalp topographies of response-related ERPs with those of aperiodic components.

We thank the reviewer for this helpful suggestion. In response, we have now marked the mean response time and associated confidence intervals on the relevant figures (Figures 8 and S8). Additionally, we have included a new figure (Figure S13) presenting both stimulus- and response-locked ERP scalp topographies for comparison with aperiodic activity.

In the previous version of the manuscript, we assessed the relationship between ERPs and aperiodic parameters by computing correlations between their topographies at each time point. However, to maintain consistency with our other analyses and to provide a more fine-grained view, we revised this approach and now compute correlations at each channel–time point. This updated analysis is presented in Figure S14. The results confirm that the correlation between ERPs and aperiodic activity remains low, and we discuss these findings in the manuscript.

Regardless of the low correlation, we have added the following statement to the manuscript to clarify our conceptual stance: “While contrasting response-related ERPs with aperiodic components can help address potential confounds, we believe that ERPs are not inherently separate from aperiodic or periodic activity. Instead, ERPs may reflect underlying changes in aperiodic and periodic activity. Therefore, different approaches to studying EEG activity should be seen as providing complementary rather than competing perspectives.”

“On page 3, it is noted that distinct theta peaks were only observed in 2 participants. Was this through visual inspection?”

Yes, this observation was based on visual inspection of the individual power spectra. We have included this explanation in the text.

suggested improving the plots by reducing the number of conditions (e.g., averaging across conditions), increasing the size of the colorbars, and using different color scales for different frequency bands, given their differing value ranges. Additionally, the reviewer noted that the theta and alpha results appeared surprising and lacked their expected topographical patterns, possibly due to the color scale.

We appreciate these thoughtful suggestions and have implemented all of them to improve the clarity and interpretability of the figures. Specifically, we reduced the number of conditions by averaging across them where appropriate, enlarged the colorbars for better readability, and applied separate color scales for different frequency bands to account for variability in dynamic range.

In the process, we also identified and corrected an error in the code that had affected the topographies of periodic activity in the previous version of the manuscript. With this correction, the resulting topographical patterns are now more consistent with canonical findings and are easier to interpret. For example, activity in the beta range now shows a clear central distribution (see Figure 6B and Figure S5B), and frontal activity in the theta range is more apparent.

This correction also directly addresses the reviewer’s concern that the “theta and alpha results (where visible) look surprising – the characteristic mid-frontal and posterior topographies, respectively, are not really present.” These unexpected patterns were primarily due to the aforementioned error.

“Relatedly, why is the mu parameter used here for correlations? Why not simply the RT mean/median, or one of the other ex-Gaussian parameters? Was this an a priori decision?”

We appreciate the reviewer’s thoughtful question. While mean and median RTs are indeed commonly used as summary measures, we chose the mu parameter because it provides a more principled estimate of central tendency that explicitly accounts for the positive skew typically observed in RT distributions. Although we did not directly compare mu, mean and median in this dataset, our experience with similar datasets suggests that differences

between them are typically small. We chose not to include other ex-Gaussian parameters (e.g., sigma, tau) to avoid unnecessary model complexity and potential overfitting, especially since our primary interest was not in modelling the full distribution of response variability. This decision was made a priori, although we note that the study was not pre-registered. We have now added a clarification in the manuscript to reflect this rationale.

“Relatedly, were (some) analyses of the study preregistered?”

The analyses were not preregistered. Our initial aim was to investigate differences in phaseamplitude coupling (PAC) between the clinical and control groups. However, we did not observe clear PAC in either group—an outcome consistent with recent concerns about the validity of PAC measures in scalp EEG data (see: <https://doi.org/10.3390/a16120540>). This unexpected finding prompted us to shift our focus toward examining the presence of theta activity and assessing its periodicity.

The reviewer suggested examining whether there might be differences between trials preceded by a target versus trials preceded by a non-target, potentially reflecting a CNV-like mechanism.

We appreciate the reviewer’s insightful suggestion. The idea of investigating differences between trials preceded by a target versus a non-target, possibly reflecting a CNV-like mechanism, is indeed compelling. However, this question falls outside the scope of the current study and was not addressed in our analyses. We agree that this represents an interesting direction for future research.

Reviewer #2 (Public review):

“For the spectral parameterization, it is recommended to report goodness-of-fit measures, to demonstrate that the models are well fit and the resulting parameters can be interpreted.”

We thank the reviewer for this suggestion. We have added reports of goodness-of-fit measures in the supplementary material (Fig. S9, S25, S41). However, we would like to note that our simulation results suggest that high goodness-of-fit values are not always indicative of accurate parameter estimation. For example, in our simulations, the R^2 values remained high even when the periodic component was not detectable or when it was conflated with the aperiodic component (e.g., compare Fig. S48 with Fig. S47). We now mention this limitation in the revised manuscript to clarify the interpretation of the goodness-of-fit metrics.

“Relatedly, it is typically recommended to set a maximum number of peaks for spectral parameterization (based on the expected number in the analyzed frequency range). Without doing so, the algorithm can potentially overfit an excessive number of peaks. What is the average number of peaks fit in the parameterized spectra? Does anything change significantly in setting a maximum number of peaks? This is worth evaluating and reporting.”

We report the average number of peaks, which was 1.9—2 (Figure S10). The results were virtually identical when setting number of peaks to 3.

“In the main text, I think the analyses of ‘periodic power’ (e.g. section ‘Periodic activity...’ and Figures 4 & 5 could be a little clearer / more explicit on the measure being analyzed. ‘Periodic’ power could in theory refer to the total power across different frequency bands, the parameterized peaks in the spectral models, the aperiodic-removed power across frequencies, etc. Based on the methods, I believe it is either the aperiodic power or an

estimate of the total power in the periodic-only model fit. The methods should be clearer on this point, and the results should specify the measure being used."

We thank the reviewer for highlighting this point. In our analyses, "periodic power" (or "periodic activity") refers specifically to the periodic-only model fit. We have added clarifications under Figure 3 and in the Methods section to make this explicit in the revised manuscript.

"The aperiodic component was further separated into the slope (exponent) and offset components". These two parameters describe the aperiodic component but are not a further decomposition per se - could be rephrased."

We thank the reviewer for alerting us to this potential misunderstanding. We have now rephrased the sentence to read: "The aperiodic component was characterised by the aperiodic slope (the negative counterpart of the exponent parameter) and the offset, which together describe the underlying broadband spectral shape."

"In the figures (e.g. Figure 5), the channel positions do not appear to be aligned with the head layout (for example - there are channels that extend out in front of the eyes)."

Corrected.

"Page 2: aperiodic activity 'can be described by a linear slope when plotted in semi-logarithmic space'. This is incorrect. A 1/f distributed power spectrum has a linear slope in log-log space, not semi-log."

Corrected.

Page 7: "Our results clearly indicate that the classical baseline correction can subtract a significant amount of continuous periodic activity". I am unclear on what this means - it could be rephrased.

We thank the reviewer to pointing out that the statement is not clear. We have now rephrased is to read: "Our results show that classical baseline correction can remove continuous oscillatory activity that is present both during baseline and after stimulus onset, because it treats all baseline signals as 'background' to be removed without distinguishing between transient and continuous oscillations."

"Page 14: 'the FOOOF algorithm estimates the frequency spectrum in a semi-log space'. This is not quite correct - the algorithm parameterizes the spectrum in semi-log but does not itself estimate the spectrum."

Again, we thank the reviewer for alerting us to imprecise description. We have now changed the sentence to: "The FOOOF algorithm parameterises the frequency spectrum in a semi-logarithmic space".

We have made refinements to improve clarity, consistency, and flow of the main text. First, we streamlined the introduction by removing redundancies and ensuring a more concise presentation of key concepts. We also clarified our use of terminology, consistently referring to the 'aperiodic slope' throughout the manuscript, except where methodological descriptions necessitate the term 'exponent.' Additionally, we revised the final section of the introduction to better integrate the discussion of generalisability, ensuring that the inclusion of additional datasets feels more seamlessly connected to the study's main objectives rather than appearing as an addendum. Finally, we carefully reviewed the entire manuscript to enhance

coherence, particularly ensuring that discussions of periodic and aperiodic activity remain precise and do not imply an assumed interplay between the two components. We believe these revisions align with the reviewer's suggestions and improve the overall readability and logical structure of the manuscript.

Reviewer #3 (Public review):

Raised concerns regarding the task's effectiveness in evoking theta power and the ability of our spectral parameterization method (specparam) to adequately quantify background activity around theta bursts.

We thank Reviewer #3 for their constructive feedback. To address the concerns regarding the task's effectiveness in evoking theta power and the adequacy of our spectral parameterization method, we have added additional visualizations using a log-y axis **** (Figures S1, S19, S32). These figures demonstrate that, in baseline-corrected data, low-frequency activity during working memory tasks appears as both theta and delta activity. Additionally, we have marked the borders between frequency ranges with dotted lines to facilitate clearer visual differentiation between these bands. We believe these additions help clarify the results and address the reviewer's concerns.

The reviewer noted that "aperiodic activity seems specifically ~1–2 Hz."

In our data baseline-corrected low-frequency post-stimulus increase in EEG activity spans from approximately 3 to 7 Hz, with no prominent peak observed in the canonical theta band (4–7 Hz). While we did not analyze frequencies below 3 Hz, we agree with the reviewer that some of this activity could potentially fall within the delta range.

Nonetheless, we would like to emphasize that similar patterns of activity have often been interpreted as theta in the literature, even in the absence of a distinct spectral peak (see: <https://doi.org/10.1016/j.neulet.2012.03.076>; <https://doi.org/10.1016/j.brainres.2006.12.076>; <https://doi.org/10.1111/psyp.12500>; <https://doi.org/10.1038/s42003-023-05448-z> — particularly, see the interpretation of State 1 as a "theta prefrontal state").

To accommodate both interpretations, we have opted to use the more neutral term "low-frequency activity" where appropriate. However, we also clarify that such activity is frequently referred to as "theta" in prior studies, even in the absence of a clear oscillatory peak.

"Figure 4 [now Figure 6]: there is no representation of periodic theta."

Yes, this is one of the main findings of our study - periodic theta is absent in the vast majority of participants. A similar finding was found in a recent preprint on a working memory task (<https://doi.org/10.1101/2024.12.16.628786>), which further supports our results.

"Figure 5 [now Figure 7]: there is some theta here, but it isn't clear that this is different from baseline corrected status-quo activity."

This figure shows comparisons of periodic activity between conditions. Although there are differences between conditions in the theta band, this does not indicate the presence of theta oscillations. Instead, the differences between the conditions in the theta band are most likely due to alpha components extending into the theta band (see Figure S6). This is further supported by the large overlap of significant channels between theta and alpha in Figure 7.

"Figure 8: On the item-recognition task, there appears to be a short-lived burst in the high delta / low theta band, for about 500 ms. This is a short phenomenon, and there is

no evidence that specparam techniques can resolve such time-limited activity."

We thank the reviewer for their comment. As we noted in our preliminary response, specparam, in the form we used, does not incorporate temporal information; it can be applied to any power spectral density (PSD), regardless of how the PSD is derived. Therefore, the ability of specparam to resolve temporal activity depends on the time-frequency decomposition method used. In particular, the performance of specparam is limited by the underlying time-frequency decomposition method and the data available for it. In fact, Wilson et al. (2022, <https://doi.org/10.7554/eLife.77348>), who have developed an approach for timeresolved estimation of aperiodic parameters, actually compare two approaches that differ only in their underlying time-frequency estimation method, while the specparam algorithm is the same in both cases. For the time-frequency decomposition we used superlets (<https://doi.org/10.1038/s41467-020-20539-9>), which have been shown to resolve short bursts of activity more effectively than other methods. To our knowledge, superlets provide the highest resolution in both time and frequency compared to wavelets or STFT.

To improve the stability of the estimates, we performed spectral parameterisation on trial-averaged power rather than on individual trials (unlike the approach in Wilson et al., 2022). In contrast, Gyurkovics et al. (2022) who also investigated task-related changes in aperiodic activity, estimated power spectra at the single-trial level, but stabilised their estimates by averaging over 1-second time windows; however, this approach reduced their temporal resolution. We have now clarified this point in the manuscript.

"The authors note in the introduction that 'We hypothesised that the aperiodic slope would be modulated by the processing demands of the n-back task, and that this modulation would vary according to differences in load and stimulus type.'. This type of parametric variation would be a compelling test of the hypothesis, but these analyses only included alpha and beta power (Main text & Figure 4)"

We appreciate the reviewer's comment, but would like to clarify that the comparison between conditions was performed separately for both periodic power and aperiodic parameters. The periodic power analyses included all frequencies from 3 to 50 Hz (or 35 Hz in the case of the second dataset). All factors were included in the linear model (see LMM formula in equation 7 - subsection Methods / Comparisons between experimental conditions), but the figures only include fixed effects that were statistically significant. For example, Figure 7 shows the periodic activity and Figure 9 shows the exponent, with further details provided in other supplementary figures.

"Figure 5 does show some plots with some theta activity, but it is unclear how this representation of periodic activity has anything to do with the major hypothesis that aperiodic slope accounts for task-evoked theta." /.../ In particular, specparam is a multi-step model fitting procedure and it isn't impressively reliable even in ideal conditions (PMID: 38100367, 36094163, 39017780). To achieve the aim stated in the title, abstract, and discussion, the authors would have to first demonstrate the robustness of this technique applied to these data.

We acknowledge these concerns and have taken several steps to clarify the relationship between the aperiodic slope and low-frequency activity, and to assess the robustness of the specparam (FOOOF) approach in our data.

First, we directly compared baseline-corrected activity with periodic and aperiodic components in all three data sets. These analyses showed that low-frequency increases in baseline-corrected signals consistently tracked aperiodic parameters - in particular the aperiodic exponent - rather than periodic theta activity (see Figs 4, S3, S20, S33). Periodic components, on the other hand, were primarily associated with baseline corrected activity in

the alpha and beta bands. The aperiodic exponent also showed negative correlations with high beta/gamma baseline-corrected activity, which is exactly what would be expected in the case of a shift in the aperiodic slope (rather than delta/theta oscillations). See also examples at <https://doi.org/10.1038/s41593-020-00744-x> (Figures 1c-iv) or <https://doi.org/10.1111/ejn.15361> (Figures 3c,d).

Next, because reviewer #1 was concerned that FOOOF might be insensitive to peaks at the edges of the spectrum, we ran a simulation that confirmed this concern. We then applied an alternative phase-based measure of oscillatory activity: the phase-autocorrelation function (pACF; Myrov et al., 2024). This method does not rely on spectral fitting and is sensitive to phase rather than amplitude. Across all datasets, pACF results were in close agreement with periodic estimates from FOOOF and were not correlated with aperiodic parameter estimates (Figs 5, S4, S5, S21, S22, S34, S35).

Taken together, these complementary analyses suggest that the apparent low-frequency (delta, theta) activity observed in the baseline-corrected data is better explained by changes in the aperiodic slope than by true low-frequency oscillations. While we acknowledge the limitations of any single method, the convergence between the techniques increases our confidence in this interpretation.

"How did the authors derive time-varying changes in aperiodic slope and exponent in Figure 6 [now Figure 8]?"

We thank the reviewer for this question. As explained in the Methods section, we first performed a time-frequency decomposition, averaged across trials, and then applied a spectral decomposition to each time point.

"While these methodological details may seem trivial and surmountable, even if successfully addressed the findings would have to be very strong in order to support the rather profound conclusions that the authors made from these analyses, which I consider unsupported at this time:

(a) 'In particular, the similarities observed in the modulation of theta-like activity attributed to aperiodic shifts provide a crucial validation of our conclusions regarding the nature of theta activity and the aperiodic component.'

(b) 'where traditional baseline subtraction can obscure significant neural dynamics by misrepresenting aperiodic activity as theta band oscillatory activity'

(d) 'our findings suggest that theta dynamics, as measured with scalp EEG, are predominantly a result of aperiodic shifts.'

(e) 'a considerable proportion of the theta activity commonly observed in scalp EEG may actually be due to shifts in the aperiodic slope'.

(f) 'It is therefore essential to independently verify whether the observed theta activity is genuinely oscillatory or primarily aperiodic'

[this would be great, but first we need to know that specparam is capable of reliably doing this]."

We believe that our claims are now supported by the aforementioned analyses, namely associations between baseline-corrected time-frequency activity and FOOOF parameters and associations between FOOOF parameters and PACF.

The reviewer found it unclear what low-frequency phase has to do with 1/f spectral changes: 'Finally, our findings challenge the established methodologies and

interpretations of EEG-measured crossfrequency coupling, particularly phase-amplitude coupling'

We thank the reviewer for their comment. To address this concern, we have added further clarification in the Discussion section. Our results are particularly relevant for phase-amplitude coupling (PAC) based on theta, such as theta-gamma coupling. PAC relies on the assumption that there are distinct oscillations at both frequencies. However, if no clear oscillations are present at these frequencies—specifically, if theta oscillations are absent—then the computation of PAC becomes problematic.

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